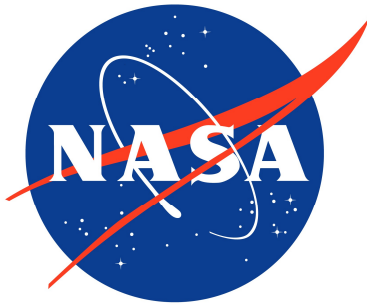


NASA/L'SPACE-2024-T24



Preliminary Design Review
Understanding the character and origin of lunar pole volatiles

Ice Hunters, Team 24

National Aeronautics and
Space Administration

NASA Lucy Student Pipeline
Accelerator and Competency Enabler
2024

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Table of Acronyms

CME	Coronal Mass Ejection
EDS	Electrodynamic Dust Shield
ISRU	In-Situ Resource Utilization
ESDMD	Exploration Systems Development Mission Directorate
NASA	National Aeronautics and Space Administration
LRO	Lunar Reconnaissance Orbiter
PSR	Permanently Shadowed Region
NIRVSS	Near-Infrared Volatiles Spectrometer System
NSS	Neutron Spectrometer System
TRL	Technology Readiness Level
LRO	Lunar Reconnaissance Orbiter
CPU	Central Processing Unit
LEO	Low Earth Orbit
CIGS	Copper Indium Gallium Selenide
IMM	Metamorphic Multi-Junction
RTG	Radioisotope Thermoelectric Generator
ESDMD	Exploration Systems Development Mission Directorate
COTS	Commercial-Off-The-Shelf

RFA	Request for Action
ADV	Advisory
FMEA	Failure Mode and Effect Analysis
CER	Cost Estimating Relationships
NICM	NASA Instrument Cost Model
MCCET	Mission Concept Cost Estimating Tool
ERE	Employee Related Expenses
FY	Fiscal Year
FTE	Full Time Equivalent
GNC	Guidance, Navigation, & Control
GSA	General Services Administration
STEM	Science Technology Engineering Math
SBC	Single Board Computer
MLI	Multiple Layer Insulation
LCS	Long-Wavelength Calibration sensor

LW

Long Wavelength

SW

Short Wavelength

1. Mission Overview

1.1. Mission Statement

The Ice Hunter lunar rover mission concept, supported by NASA's ESDMD and aligned with the Artemis Accords, aims to conduct essential surface measurements and mapping to advance understanding of lunar water distribution and abundance on the PSR of the lunar poles. This mission plans to operate on the lunar surface for a full Earth year, consistent with the Artemis mission framework. The mission's goal is to further the knowledge of the changes in volatile materials and the understanding of the compositional state and distribution of the volatile material it encounters. The mission objectives to achieve this goal are to collect 100g of ice samples from the chosen location, determine the water ice abundance in or near a PSR within the top one meter of regolith, and identify areas of higher ice concentrations with an Artemis III landing site. Furthermore, the chosen location, Haworth, was selected for its PSR in a crater showing promise of ice in previous reconnaissance missions, as well as its low slope and intermediate sunlight, allowing for exploration of its PSR. The two scientific instruments, NSS and NIRVSS, will enable the rover to achieve the mission objectives through the instrument's ability to detect hydrogen within the PSR and then drill for samples within the PSR that the instruments can more finely analyze, allowing ice samples to be analyzed.

1.2. Science Traceability Matrix

The following table will provide the link between the science goals driving the mission and the requirements that the client will set. These two contributing factors will start to root the instruments that will be used to conduct the science experiments that will be performed during the life of the mission. The instrumentation selected will allow the completion of the science goals by allowing for the detection and analysis of samples from the PSR found on the lunar south pole. The goal set is centered around understanding the composition of the PSR. This will allow for a plan defined by objectives, including collecting samples, checking if the samples contain H₂O, and finding a specific area for which the rover can achieve these goals. The rover will be in the Hawthorne region of the south pole and will be using spectroscopy to determine Hydrogen concentration. For example, 400-680 nm wavelengths can be detected to find areas of high hydrogen content. From this, scientists can plot regions of high ice concentration. Finding these regions with greater hydrogen content narrows down the sample collection sites for future manned missions. In line with the Artemis III mission goal to understand the transport, retention, alteration, and loss processes that operate on volatile materials, the rover will detect differences in hydrogen concentration for ground samples versus collected samples. Without disturbing the surface, a spectroscopy reading of a 3 cm² area will be analyzed. Then, the same area will be drilled by the rover, and another spectroscopy reading will be taken of a 100g sample to determine if a sample collection process caused a significant change in hydrogen concentration. If a significant change is detected, there should be additional research on collecting volatile ice on the Moon during manned missions, especially if this ice will be used for vital operations such as oxygen supply and power generation. Certain specifications and parameters will set the standards for the science to be performed and are specified in the table below.

Table 1. Science Traceability Matrix

Science Goals	Science Objectives	Science Measurement Requirements		Instrument Performance Requirements		Predicted Instrument Performance	Instrument	Mission Requirements
		Physical Parameters	Observables					
Understanding the character and origin of lunar poles (Primary) Understand the transport, retention, alteration, and loss processes that operate on volatile materials near and within permanently shadowed regions (NASA, "Artemis III Science.") (Secondary) Determine the compositional state (elemental,	Collect 100g of ice samples at one of the 13 Artemis III landing site locations and determine the composition of the sample	Identify the concentration of Hydrogen wavelengths in a collected sample of lunar regolith	Detect a difference in water concentration from collected samples versus ground samples to predict loss of volatile material using the same 3 cm² area of sample collection,	Wavelength range:	250-600 nm	230-615 nm	Neutron Spectrometer System (NSS)	Vehicle shall traverse to PSR to scan/drill science locations
				Integration time:	1 s	.5 s		
				Sensitivity:	10 ⁻² to below 10 ⁻⁴ w/w	10 ⁻² to below 10 ⁻⁴ w/w		Mission must collect a minimum sample of 100 g of lunar regolith from a depth up to 1 m
				Spatial resolution:	<100 µm	<100 µm		
	Determine the water ice abundance of at least one of the 13 Artemis III landing site locations near or within a PSR within the top 1 meter of	Determine the percent concentration of Hydrogen within the top 1 meter of regolith	Detect Hydrogen wavelengths in the range of 400-680 nm	Wavelength range:	250-600 nm	200-610 nm	Neutron Infrared Volatiles Spectrometer System (NIRVSS)	Vehicle will land within 10 km of PSR
				Integration time:	1 s	1.2 s		
				Sensitivity:	10 ⁻² to below 10 ⁻⁴ w/w	10 ⁻¹ to below 10 ⁻⁵ w/w		

isotopic, mineralogic) and compositional distribution (lateral and with depth) of the volatile component in lunar polar regions (NASA, "Artemis III Science.")	regolith			Spatial resolution:	20 nm	10nm		
	Identify areas of higher ice concentration within an Artemis III landing site	Compare concentrations of hydrogen on a 100 x 100 grid of 1 m ² squares and identify 3 squares with the highest concentration of hydrogen	Detect areas of high ice concentration to predict ice collection sites for future missions	Wavelength range:	400-700 nm	350-750 nm	NSS & NIRVSS	Mission shall scan lunar surface using the two science instruments
				Spatial resolution:	20 nm	10 nm		
				Integration time:	1 s	1 s		
				Sensitivity:	10 ⁻² to 10 ⁻⁴ w/w	10 ⁻² to 10 ⁻⁴ w/w		

1.3. Summary of Mission Location

Upon reviewing the identified candidate regions provided by the Artemis III landing sites catalog, the final decided location for the mission will be the Haworth region. This location has shown promise of potential ice in the Moon's PSRs as previous reconnaissance missions have performed experiments to make these predictions, such as the Clementine spacecraft (Clementine, 2023), the Lunar Prospector (Lunar Prospector, 2024), and the Lunar Reconnaissance Orbiter (LRO, 2024). All three missions have found evidence of ice in the bottom of PSR craters at the Moon's south pole. The location selected has shown to be an optimal landing location due to its exposure to intermittent sunlight near the crater's rims (Heldmann et al, 13). This intermittent sunlight will ensure the successful operation of the spacecraft as all relevant subsystems require the capturing of energy from the sun's light. The coordinates for the location selected are -86.721°N, 338.283°E.

Intermittent sunlight along areas of the selected location will allow the spacecraft to enter PSRs, then exit at marked locations to recharge and regulate temperature as needed (Heldmann et al, 19). The science experiments can be completed within a reasonable range of distance between the PSR and areas of intermittent sunlight without deep concern for the spacecraft's charge. When entering and exiting a PSR, additional research will be completed in order to determine when the spacecraft is entering a shadowed region (not to be confused with a PSR) in order to ensure sufficient charge prior to planned operation cycles (Heldmann et al, 20). The rover will travel to 16 science locations, so intermittent sunlight is vital to the rover's travel and operation at each of these locations to fulfill science objectives.

An additional benefit to the location selected is the multiple regions of flat terrain as constrained by the customer; each location highlight has met the requirement of having a slope $\leq 15^\circ$. Generally, the crater region offers a manageable slope for the craft to navigate and safely carry out science experiments. It also offers proximity to areas with higher elevations resulting in better and faster communication with the LRO (Jia et al). The following images were generated using the JMARS Luna software, which displays the region of interest and specifies locations for potential science experiments. Along these designated locations it can also be seen that sunlight will not be far from these PSRs.

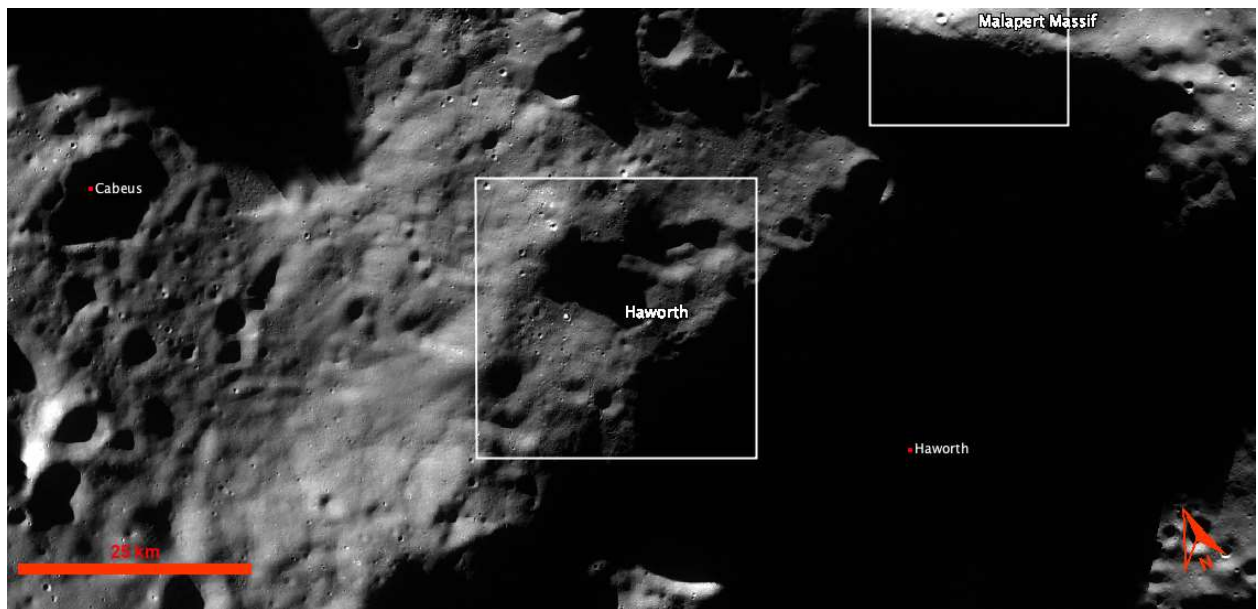


Figure 1.1: Haworth Location

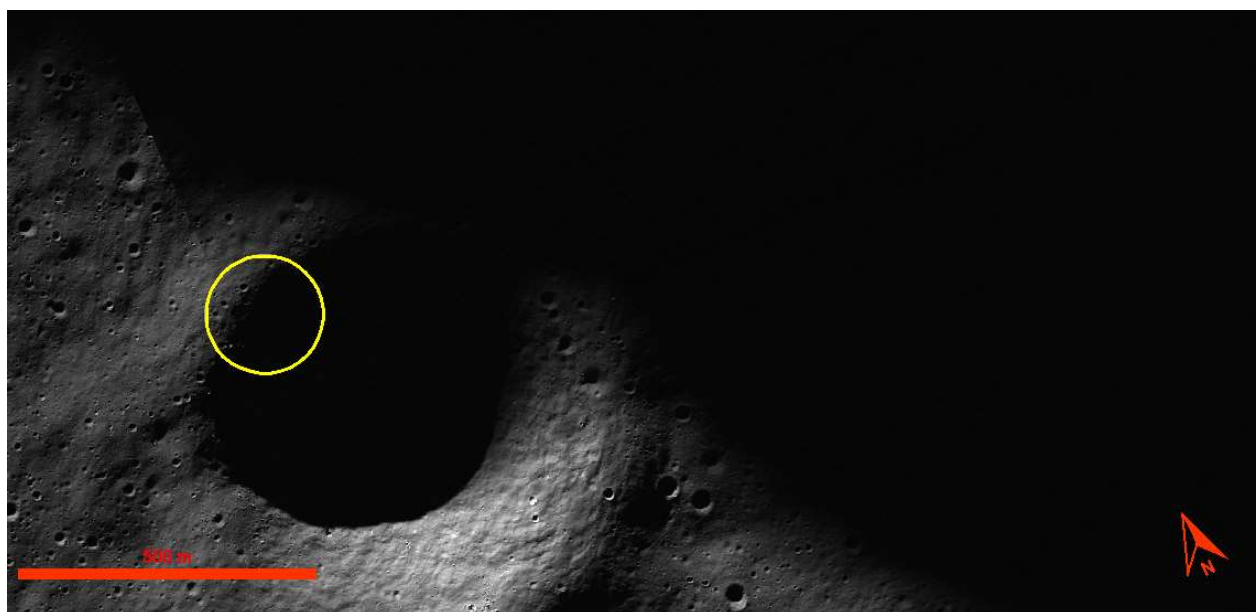


Figure 1.2: Haworth PSR Entry Location

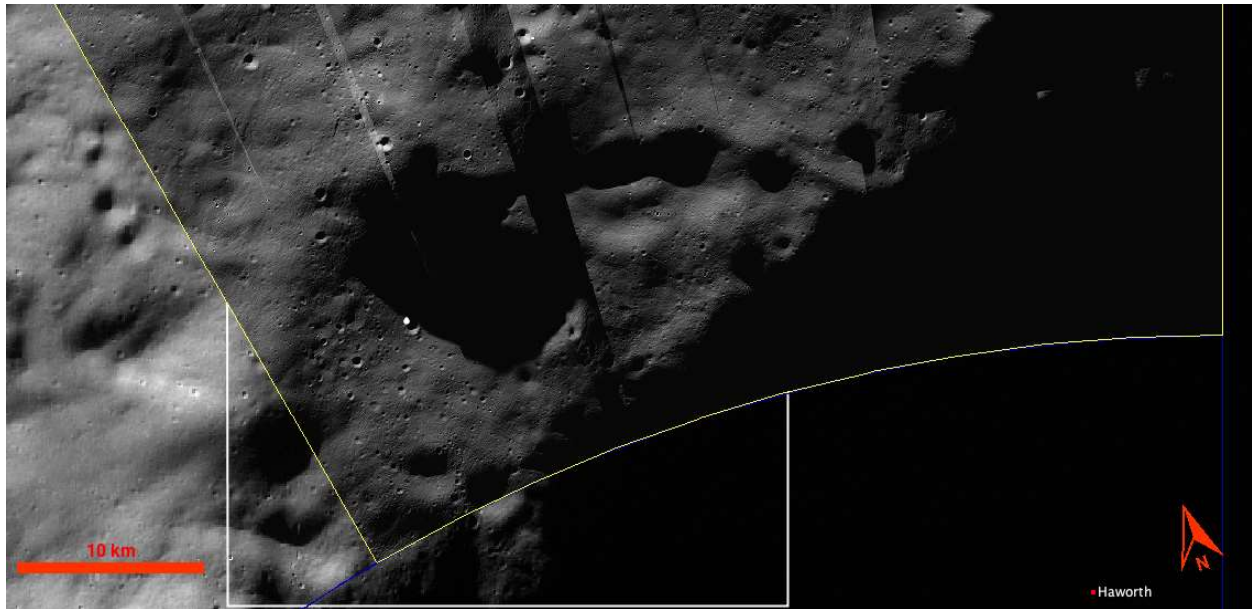


Figure 1.3: Haworth PSR Entry Close Shot

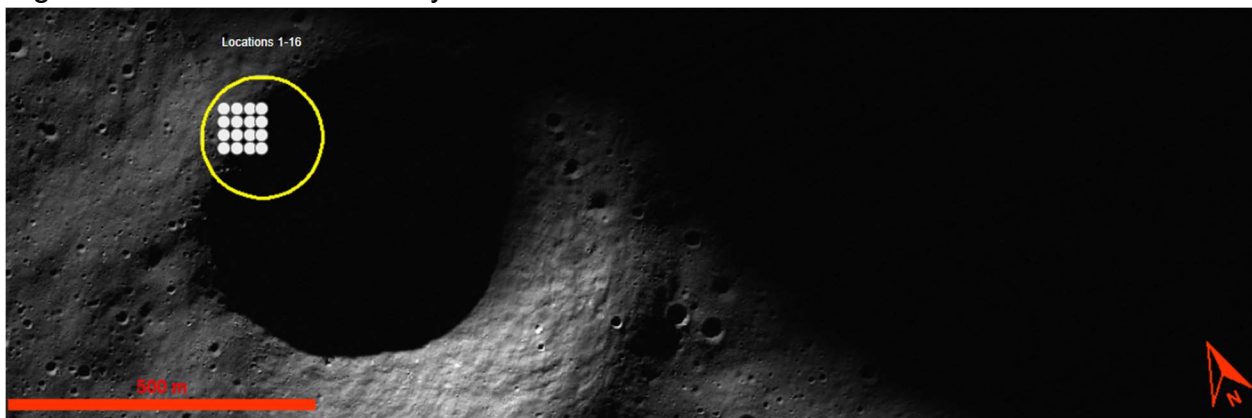


Figure 2: Haworth Location 1-16

1.4. Mission Requirements

As a secondary payload for a discovery class mission, the constraints given by the customer for the Ice Hunter mission are primarily cost, mass and volume. The cost allotted for the Ice Hunter mission shall not exceed \$225 million, which will include materials, science instruments, personnel costs and all related services, but will not include launch or cruise costs. The mass allotted for the Ice Hunter mission shall not exceed 85kg, including vehicle, science instruments, and all systems and subsystems required to accomplish the stated science goals. Finally, the total volume for the Ice Hunter mission shall not exceed a volume Length x Width x Height of 1.5m x 1.5m x 1.5m, though this is not fixed volume measurements, as the vehicle is permitted to expand and exceed these volume constraints post-deployment. Top level requirements for the mission are specified in the table below:

Table 2. Mission Requirements

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Relevant Subsystem	Req met ?
0.1	The system shall survive the lunar environment while collecting data from at least two science locations.	In order to achieve science objectives the system needs to operate/survive	Client	SYS-03	Demonstration	All	Met
0.2	Shall investigate the abundance of near-surface water ice in PSR	Foundational science goal	Client	SYS-01,SYS-03,SYS-05,SYS-06,SYS-07,SYS-09	Test	Payload	Met
SYS-01	The system shall traverse the lunar surface by means of a rover	The rover must be able to travel distances to conduct science objectives	0.1,0.2	SYS-07,SYS-09	Demonstration	Mechanical, Electrical, GNC, Computer hardware	Met
SYS-02	The system shall not exceed a mass of 85 kg	Defined by launch provider	Launch provider	All	Inspection	All	Met
SYS-03	The system shall be able to operate at temperatures ranging from 90.15 K to 379.15 K	The rover must function in extreme temperatures of the lunar pole	0.1	SYS-07	Demonstration	Electrical, Thermal, Computer hardware	Met
SYS-04	The system shall not exceed a volume of L 1.5m x W 1.5m x H 1.5m	Defined by launch provider	Launch provider	All	Inspection	Structural	Met
SYS-05	The system shall collect and store data from the two science instruments on the rover	The data collected will contribute to achieving the science goal	0.1,0.2	SYS-01,SYS-03,SYS-07	Demonstration	Science, Electrical	Met

SYS-06	The system shall communicate with the LRO	After data has been collected and analyzed it will be relayed to earth	0.1,0.2	SYS-07	Demonstration	Electrical, Computer hardware, GNC	Met
SYS-07	The system shall have sufficient power to survive/operate	The rover requires power to perform and communicate all objectives	0.1,0.2	SYS-03	Demonstration	Electrical	Met
SYS-08	The system shall not exceed a cost of \$225M	Defined by contracting client	Contract provider	All	Verification	All	Met
SYS-09	The system shall contain the necessary instruments to support the collection from the scientific instruments	The rover needs the ability to run experiments on the Moon's surface	0.2	SYS-05,SYS-07	Demonstration	Science, Electrical, Computer hardware, Thermal	Met
SYS-10	The system shall not contain a Radioisotope Thermoelectric Generator(RTG)	The rover cannot have any derivative of this concept	Contract provider	All	Verification	All	Met
SYS-11	The system shall be ready for launch by September 1st, 2028	In order to align with the launch provider's secondary payload	Contract provider	All	Verification	All	Met
SYS-12	The system shall not contain radioactive material exceeding a mass of 5g	Limited radioactive materials	Contract provider	All	Verification	All	Met

1.5. Concept of Operations (ConOps)

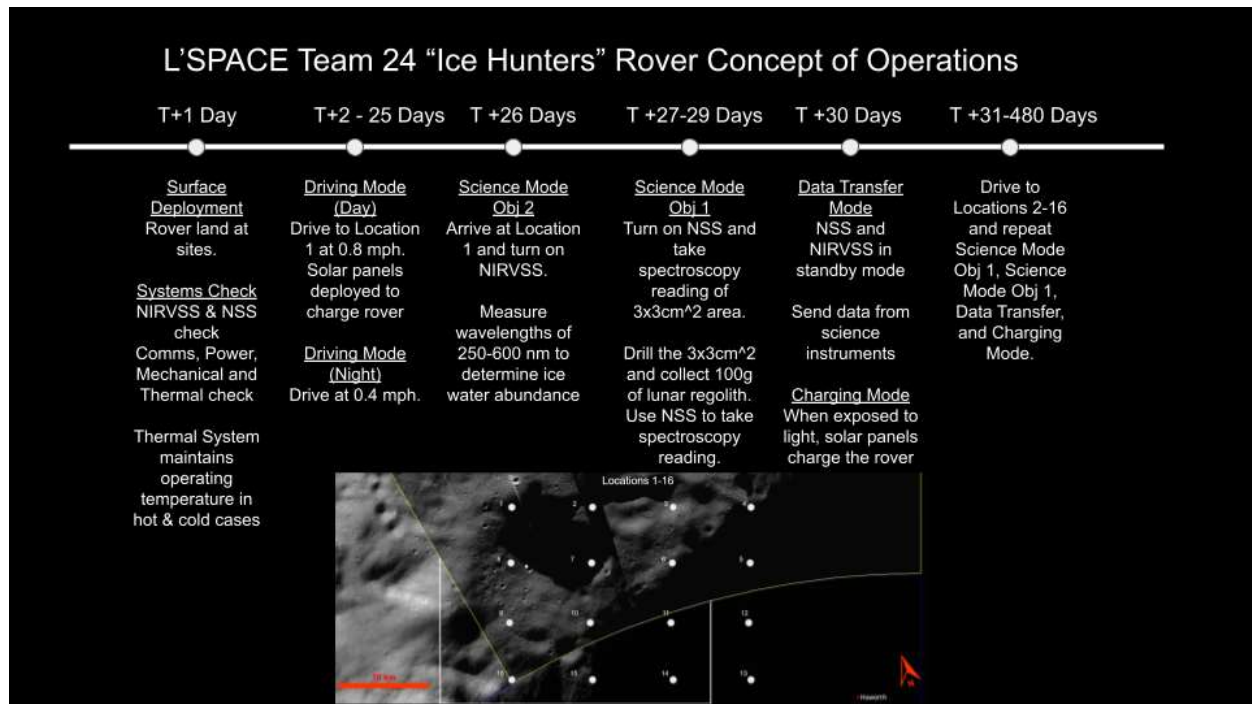


Figure 3: Concept of Operations

The Concept of Operations outlines the rover operation timeline from landing on the Moon to completion of data collection and transfer. At T+1 Day, the Rover will do a total systems check, ensuring science instruments, power, thermal, and communications work. A communication signal will be sent with "subsystem" = "ON" for each subsystem in operation. Throughout the duration of the mission, the thermal system will maintain safe operating temperatures for both hot and cold cases. At T+2-25 Days, the rover will drive to Location 1. Over the duration of the mission, the rover will travel to 16 Science Locations, as shown by the 4x4 dot grid. These locations are a mix of near and in the PSR, to collect enough data for comparison. In Driving Mode (Day), the rover will not be in shadow and be exposed to light. The solar panels will be deployed to charge the batteries and the rover will travel at 0.8 mph. At night, the rover will travel at a slower speed of 0.4 mph to conserve battery and the solar panels will not be deployed. At T+26 days, the rover will begin data collection. NIRVSS will be turned on from standby mode and measure wavelengths of 250-600 nm and will be put in standby after data collection. This will be used to determine ice water abundance for Science Mode Objective Two. At T+27-29 Days, the rover will switch to Science Mode Objective One. The NSS will be turned on and take a spectroscopy reading of 250-600 nm wavelengths of a 3x3cm² area. The same area will be drilled and 100g will be collected as a sample. This NSS will take a multispectral image of 250-600 nm wavelengths of the sample. At T+30 Days, the rover will go into data transfer mode. Data from that location will be sent to scientists for analysis. If the rover is exposed to sunlight, the solar panels will deploy and charge the batteries at this time. At T+31-480

days, the cycle will repeat as the rover collects data at all 16 locations. After data from all locations are sent to Earth, scientists can complete science Objective Three by comparing concentrations of hydrogen on a 30 x 30 km² grid and identify the 3 locations with the highest concentration of ice. At this stage, the mission is complete. The mission should take less than one year to complete. Each science location is about 10 km apart. The rover's average speed is 0.4 mph or 0.644 kmph. Travelling 10 km at this speed takes 15.5 hours. Given the irregular terrain of the Moon, the rover will not travel in a straight line and may have delays, so one day will be allotted for travel. Data collection takes three days and data transmission takes one day. The travel, data collection, and transmission cycle will take approximately 75 days starting from Science Location 1. Adding in the time from surface deployment to Location 1, the mission should take approximately 100 days. However, the rover is designed to operate for 1 year in case of time delay during the mission. Though 16 locations are ideal, the mission will not fail if the rover does not collect and transfer data from all locations. As explain in further detail in section 3.3, the rover shall travel to, collect data from, and transfer data to scientists for at least 2 locations. While visiting only two locations is not ideal, science objectives will be completed. The mission includes 16 locations for better data comparison.

1.6. Vehicle Design Summary

(1 page recommended)

*This section should be an overview of your final designs for the vehicle - focusing on the **overall design, volume, mass, power**. You will go into more detail in later sections; this is just a summary of the overall vehicle design. You should also include an image from your **CAD model(s) of your entire vehicle**.*

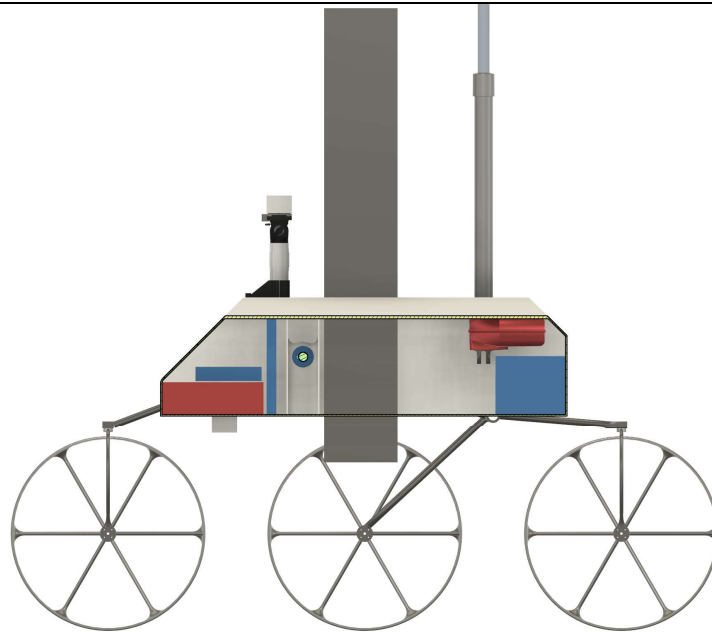


Figure 4: Spacecraft System Overview

The spacecraft takes the form of a six wheeled rover with a rocker-bogie system. The rover also includes two instruments. These instruments, Neutron Spectrometer System and Near Infrared Spectrometer System, will use spectrometry to determine if there are volatile compounds within lunar regolith. The rover design was chosen to perform science in multiple locations throughout the lunar surface. Solar panels will provide power while the rover is not within PSRs and will fold up prior to deployment to meet volume requirements. Power will be stored in batteries to hold energy for use in the PSRs where energy cannot be collected. A drill will be used to expose lunar regolith not already exposed at the surface. Heaters and thermal material will be utilized to ensure the rover maintains a stable and workable temperature. Antennas and radiation hardened computational processors will be utilized to handle data and allow the spacecraft to function and perform its science goals.

Table 3. Total System Mass, Volume, and Power

Mass (kg)	Volume (cm ³)	Power (W)
72.61	502463	432

1.7. Science Instrumentation Summary

The two science instruments onboard the rover are the Neutron Spectrometer System (NSS) and the Near Infrared Volatiles Spectrometer System (NIRVSS). Using two different spectrometers not only lends the mission a greater variety of readings, but also the assurance that if one were to fail, the other would be able to collect most of the same data and the science objectives would still be achieved. The NSS is used for spectroscopy readings on collected samples and will help scientists understand the change in ice content of lunar regolith due to sample collection processes, such as drilling. The drill used in this mission is not a scientific instrument because it is not used to measure data. The NSS takes near surface spectroscopy hydrogen readings and can detect small differences in concentrations. This instrument can measure concentrations as low as 0.5 wt% water-equivalent hydrogen¹. The NSS will also be used with the NIRVSS to complete science objective 3, identifying areas of higher water ice concentration. While the NIRVSS will be the main instrument for mapping, readings from the NSS will verify these results. The NIRVSS will take spectroscopy readings to determine the ice water abundance within the top 1 meter of regolith at 16 science sites. This instrument was chosen because it "... is designed to measure surface and subsurface water, carbon dioxide, and methane. It will also map the surface temperature and changes that occur at the landing site"². Readings from this instrument will not only measure ice abundance, but also the correlating thermal conditions,

¹ Elphic, Richard C. Neutron Spectrometer System

² Colaprete, Anthony. Near infrared volatile spectrometer

reflectance, and elemental composition of regolith. Spectroscopy readings will be taken at varying times of day and night, with extreme temperature differences. This will be the main instrument used when mapping hydrogen concentration across the 16 science sites. The additional data collected by this instrument could help scientists understand how varying environmental properties affect hydrogen concentration in lunar regolith.

1.8. Programmatic Summary

1.8.1. Team Introduction

Matthew Mulderink - Arizona State University (Tempe, AZ)



- Experience in developing a rover that is remotely controlled.
- A business minor, allowing for an understanding of management and organization.
- Experience in working within larger student teams.
- Google App Script experience, which was used when making a program that tracks daily changes in work from team members.

Jeremy Winter - Arizona State University (Tempe, AZ)



- Experience with conducting research on space-related topics and concepts.
- Understanding of how to utilize CAD software to create designs and mock-ups of space technology.

Abhinav Vishnuvajhala - Georgia Tech (Atlanta, Georgia)



- Experience in software development and architecture
- Experience working with upcoming NASA lunar missions at Ames Research Center
- Understanding of budget and scheduling for teams

Alyssa Armstrong - Arizona State University (Tempe, AZ)



- Engineering experience in NX CAD, Thermal Design, rapid prototyping, & technical report writing
- Experience in risk analysis, Gantt charts, requirements, and project management
- Designing instruments to operate in a vacuum and extreme temperature environments

Curtis Sims - Colorado State University (Fort Collins, CO)



- Simple understanding of siemens NX CAD
- Student club and team operational experience as a vice president and a financial officer for engineering based clubs
- Experience designing electrical systems for personal projects as well as for the power system on the rover.
- General understanding of simple construction principles as a general contractor's apprentice.

Enrique Vasquez - University of Texas at El Paso (El Paso, TX)



- A professional understanding of CAD softwares
- Experience with designing a vehicle's differential
- Experience with mechatronics systems

Gavin Knudsen - Utah State University (Logan, UT, USA)



- Technical experience with complicated systems
- Experience designing autonomous robots
- Experience designing outreach materials and events
- Experience designing small satellites

Helen Zhao - Arizona State University (Tempe, AZ)



- Understanding of design and CAD
- Working within team projects
- Research

Jack Young - Arizona State University (Tempe, AZ)



- Experience using JMARS application
- Experience in formal write ups for planetary discovery
- Leadership experience as team lead working in HVAC

Nick Woolsey - University of Colorado Boulder (Boulder, CO)



- Experience in multiple CAD softwares and proficiency in Fusion 360
- Experience in designing multiple robotics projects and participating on robotics teams
- Understanding of engineering fundamentals and principles
- Previous member of NCAS with similar scope and responsibilities

Svar Ajwani - Arizona State University (Tempe, AZ)



- Computer Science major
- Prior experience with software design and architecture
- Prior experience working with a team in a professional setting

1.8.2. Team Management Overview

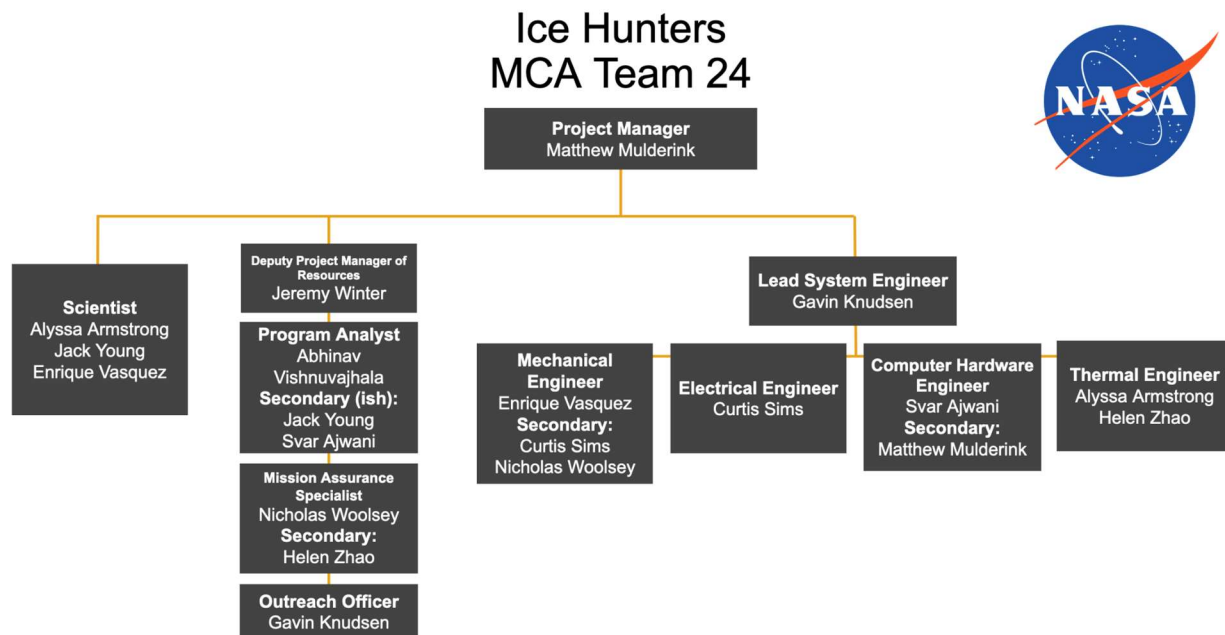


Figure 5: Ice Hunter's Team Structure

The Ice Hunter's team structure is based heavily on individuals taking charge of their primary section alone or with one or two people. This organizational structure was changed from the SRR to the MDR, with promising results but not without downsides. What went well was that compared to when a team member had up to five sections, now team members could become experts in their field and focus on it due to having fewer sections. A benefit that was not predicted was that team members also identified with their roles a lot more, meaning that questions asked on a subject were a lot more likely to get answers. This benefit is because individuals knew that they had the best answer, compared to the previous structure, which had members share their role with five to six members and may have been unsure of themselves compared to their peers in the same role. As predicted, increased trust of the team members meant that most could complete their work without worrying about how a change would affect work already done, as members were doing more work for a particular section.

The new structure posed a challenge: members who struggled with their work went unnoticed. This meant that on the last night of the MDR, a few sections required members from unrelated teams to step up and complete it. This issue stemmed from the difficulty of tracking the progress made in the sections. For the PDR, the solution to the problem was to create a Google App Script that goes through the PDR Google Doc and records the contents of each section in a Google Sheet. This script runs every night, and each section's word count is recorded. This means that sections that do not have progress made for multiple days do not have a change in word count. Also, the Google App Script allows the Project Manager to track 65 sections with statistics instead of just visually. Furthermore, seeing the progress of each section allows for a clear understanding of the progress of each member, meaning that the Project Manager

was able to reach out to those who had not started.

The Ice Hunter's team was reduced by another member after the MDR. This member's primary role was one of the few shared with another person, meaning that the work was redistributed to the member who was also in that role and had more work to do. This member's sections to review were reduced to accommodate the increase in sections to write.

Due to the further reduction in team size, issues members bring up are decided during the weekly meeting or by the Project Manager. The Project Manager's decision usually affirms the belief already given by the members. If not, the issue is resolved during the weekly meetings. Although not all members attend, there has never been a conflict with an actual split among members. The idea behind this decision-making style is to allow members to avoid experiencing hiccups in their work. An example is when subteams ask whether they can change a component. Since most engineering subteams are only one person, this was approved without discussion, with the only request from the Project Manager being to inform all other subteams that this affects. At this point, most members are experts in their roles and need the confidence to make decisions, not have others make decisions for them.

1.8.3. Major Milestones Schedule

As a secondary payload for a discovery class mission, the constraints given by the customer for the Ice Hunter mission are primarily cost, mass and volume. The cost allotted for the Ice Hunter mission shall not exceed \$225 million, which will include materials, science instruments, personnel costs and all related services, but will not include launch of cruise costs. The mass allotted for the Ice Hunter mission shall not exceed 85kg, including vehicle, science instruments, and all systems and subsystems required to accomplish the stated science goals. Finally, the total volume for the Ice Hunter mission shall not exceed a volume Length x Width x Height of 1.5m x 1.5m x 1.5m, though this is not fixed volume measurements, as the vehicle is permitted to expand and exceed these volume constraints post-deployment. Top level requirements for the mission are specified in the table below:

Table 4. Mission Requirements

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Relevant Subsystem	Req met?
0.1	The system shall survive the lunar environment for 2 years.	In order to achieve science objectives the system needs to operate/survive	Client	SYS-03	Demonstration	All	Met
0.2	Shall investigate the abundance of near-surface water ice in PSR	Foundational science goal	Client	SYS-01,SYS-03,SYS-05,SYS-06,SYS-07,SYS-09	Test	Payload	Met
SYS-01	The system shall traverse the lunar surface by means of a rover	The rover must be able to travel distances to conduct science objectives	0.1,0.2	SYS-07,SYS-09	Demonstration	Mechanical, Electrical, GNC, Computer hardware	Met
SYS-02	The system shall not exceed a mass of 85 kg	Defined by launch provider	Launch provider	All	Inspection	All	Met
SYS-03	The system shall be able to operate at temperatures ranging from 90.15 K to 379.15 K	The rover must function in extreme temperatures of the lunar pole	0.1	SYS-07	Demonstration	Electrical, Thermal, Computer hardware	Met
SYS-04	The system shall not exceed a volume of L 1.5m x W 1.5m x H 1.5m	Defined by launch provider	Launch provider	All	Inspection	Structural	Met
SYS-05	The system shall collect and store data from the two science instruments on the rover	The data collected will contribute to achieving the science goal	0.1,0.2	SYS-01,SYS-03,SYS-07	Demonstration	Science, Electrical	Met
SYS-06	The system shall communicate with the	After data has been collected and analyzed it will be relayed to	0.1,0.2	SYS-07	Demonstration	Electrical, Computer hardware, GNC	Met

	LRO	earth					
SYS-07	The system shall have sufficient power to survive/operate	The rover requires power to perform and communicate all objectives	0.1,0.2	SYS-03	Demonstration	Electrical	Met
SYS-08	The system shall not exceed a cost of \$225M	Defined by contracting client	Contract provider	All	Verification	All	Met
SYS-09	The system shall contain the necessary instruments to support the collection from the scientific instruments	The rover needs the ability to run experiments on the Moon's surface	0.2	SYS-05,SYS-07	Demonstration	Science, Electrical, Computer hardware, Thermal	Met
SYS-10	The system shall not contain a Radioisotope Thermoelectric Generator(RTG)	The rover cannot have any derivative of this concept	Contract provider	All	Verification	All	Met
SYS-11	The system shall be ready for launch by September 1st, 2028	In order to align with the launch provider's secondary payload	Contract provider	All	Verification	All	Met
SYS-12	The system shall not contain radioactive material exceeding a mass of 5g	Limited radioactive materials	Contract provider	All	Verification	All	Met

1.8.4. Budget Overview

The Ice Hunters Budget mission has a project cost of \$173,711,430. This total then has a total cost margin of \$20,791,000, meaning that the real cost can total up to \$194,502,430. The budget breakdowns into the following categories: total personnel cost is projected to be \$25,152,201, total travel costs are \$264,525, total outreach costs are \$2,232,787, total direct costs are \$138,835,723, and total MTDC is \$82,561,942. In viewing this data, most of the cost lies in the direct costs, with the personnel budget accounting for a lot of the rest of the budget.

NASA L'SPACE Mission Concept Academy Budget - <i>Ice Hunters (Team 24)</i>							
Mission Phase	Phase C	Phase C	Phase C-D	Phase D	Phase E	Phase F	
Year	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Cumulative Total
FINAL COST CALCULATIONS							
Total F&A	\$ 1,104,727	\$ 2,079,957	\$ 2,102,993	\$ 1,616,717	\$ 237,824	\$ 83,977	\$ 7,226,194
Total Projected Cost	\$ 27,864,301	\$ 47,818,980	\$ 48,880,205	\$ 34,792,822	\$ 9,242,085	\$ 5,113,037	\$ 173,711,430
Total Cost Margin	\$ 4,210,000 15.1%	\$ 6,014,000 12.6%	\$ 6,029,000 12.3%	\$ 3,234,000 9.3%	\$ 694,000 7.5%	\$ 610,000 11.9%	\$ 20,791,000
Total Project Cost	\$ 27,864,301	\$ 47,818,980	\$ 48,880,205	\$ 34,792,822	\$ 9,242,085	\$ 5,113,037	\$ 173,711,430

Figure 6: Final Cost Breakdown

2. Overall Vehicle and System Design

2.1. Spacecraft Overview

The spacecraft will be the culmination of multiple subsystems working as one cohesive unit in order to operate and perform effectively and efficiently throughout the life of the mission. The communication and order of the subsystems shall be integrated to be practical and optimal for all of the spacecraft's operations. The top level requirements for the spacecraft are specified in the table below:

Table 5. Spacecraft Requirements

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Relevant Subsystem	Req met?
0.1	The system shall survive the lunar environment for 2 years.	In order to achieve science objectives the system needs to operate/survive	Client	SYS-03	Demonstration	All	Met
0.2	Shall investigate the abundance of near-surface water ice in PSR	Foundational science goal	Client	SYS-01,SYS-03,SYS-05,SYS-06,SYS-07,SYS-09	Test	Payload	Met
SYS-01	The system shall traverse the lunar surface by means of a rover	The rover must be able to travel distances to conduct science objectives	0.1,0.2	SYS-07,SYS-09	Demonstration	Mechanical, Electrical, GNC, Computer hardware	Met
SYS-02	The system shall not exceed a mass of 85 kg	Defined by launch provider	Launch provider	All	Inspection	All	Met
SYS-03	The system shall be able to operate at temperatures ranging from 90.15 K to 379.15 K	The rover must function in extreme temperatures of the lunar pole	0.1	SYS-07	Demonstration	Electrical, Thermal, Computer hardware	Met
SYS-04	The system shall not exceed a volume of L 1.5m x W 1.5m x H 1.5m	Defined by launch provider	Launch provider	All	Inspection	Structural	Met
SYS-05	The system shall collect and store data from the two science instruments on the rover	The data collected will contribute to achieving the science goal	0.1,0.2	SYS-01,SYS-03,SYS-07	Demonstration	Science, Electrical	Met
SYS-06	The system shall communicate with the	After data has been collected and analyzed it will be relayed to	0.1,0.2	SYS-07	Demonstration	Electrical, Computer hardware, GNC	Met

	LRO	earth					
SYS-07	The system shall have sufficient power to survive/operate	The rover requires power to perform and communicate all objectives	0.1,0.2	SYS-03	Demonstration	Electrical	Met
SYS-08	The system shall not exceed a cost of \$225M	Defined by contracting client	Contract provider	All	Verification	All	Met
SYS-09	The system shall contain the necessary instruments to support the collection from the scientific instruments	The rover needs the ability to run experiments on the Moon's surface	0.2	SYS-05,SYS-07	Demonstration	Science, Electrical, Computer hardware, Thermal	Met
SYS-10	The system shall not contain a Radioisotope Thermoelectric Generator(RTG)	The rover cannot have any derivative of this concept	Contract provider	All	Verification	All	Met
SYS-11	The system shall be ready for launch by September 1st, 2028	In order to align with the launch provider's secondary payload	Contract provider	All	Verification	All	Met
SYS-12	The system shall not contain radioactive material exceeding a mass of 5g	Limited radioactive materials	Contract provider	All	Verification	All	Met

The spacecraft to be designed is a mobile vehicle in the form of a rover. This was decided as the mission's science goals require taking measurements in multiple locations in order to get the most data possible. Many items were considered when designing the spacecraft, initially the envelope of volume and mass were considered as mission requirements will yield a small to medium sized craft in terms of planetary rovers. Science goals and mission requirements were the main drivers of the design and have been considered in this preliminary design. The rover will have a box-shaped body/chassis which will store all the other subsystems and science payloads, the main body will also have some extended sub-bodies that will extend carrying scientific payloads and navigational sensors. Solar arrays will expand to an area that exceeds the volume envelope when fully extended. Lights on the front of the body will also be part of the design when traversing in the PSR. The spacecraft will have 6 wheels to traverse on the lunar surface, the suspension type adopted is the rocker-bogie system which has shown consistent success in similar environments. The spacecraft will primarily be made of aluminum among other low mass materials.

The mechanical subsystem of the spacecraft will provide mobility using the 6 wheeled rocker-bogie system allowing the spacecraft to traverse through lunar regolith. The subsystem will primarily be made of lighter alloys with high strength that can perform in extreme temperatures. The subsystem will provide structure and mobility to the spacecraft to carry all other subsystems and operate for the life of the mission. The mechanical subsystem will have two science instruments mounted to perform the science driving the mission.

The power subsystem generates, stores and distributes energy for all onboard systems that require power such as the CDH subsystem, Science payload and all Mechanical components like motors and actuators. This is done by utilizing the lightweight generation potential of solar panels to recharge the rover's batteries in between periods of testing inside the PSR of the crater. This energy is stored in batteries to power the rover while it is conducting this testing within the PSR and during transition periods between the charging point on the crater rim and the testing location within the crater. The distribution of this energy is done using a solar charge controller, an inverter and a power supply which effectively supplies sufficient power required for the rover at all times.

The Command and Data Handling subsystem acts as the brains of the spacecraft. This subsystem has several specifications that must be fulfilled by the commercial off the shelf (COTS) parts that are being selected. These specifications, discussed further in later sections of this document, include survivability, communication with the rover from Earth, data storage, internal craft communication, weight specifications, and power draw. The pieces of equipment that will uphold these requirements are the central

processing unit (CPU), communication antenna, communication receiver, data bus, storage, and onboard software.

The thermal system will include multiple flexible heaters to keep the internal electronics warm in the frigid environment near and in the PSR. The radiator will be used to dissipate heat generated by the spacecraft to prevent overheating. The radiator will also prevent the heat of the sun from heating the rover above safe temperatures. As a backup, a protective white thermal paint will be applied to the surface of the rover. A thermal switch will be installed to regulate between heating and cooling, depending on the temperature.

The science instrumentation to be used for this particular mission will survey and scan the surface of the Moon using light to determine composition percentages that makeup these sample locations, the instruments used will consist of two high TRL devices including the NSS and NIRVSS instruments. These two scientific payloads will be important in achieving the science goal driving the mission.

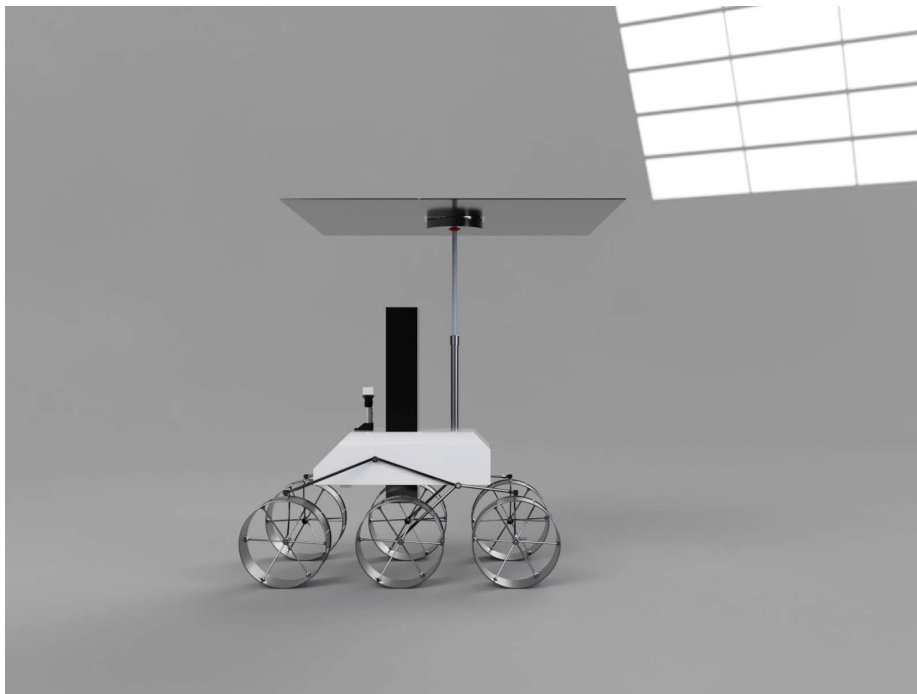


Figure 7: Rover from Side

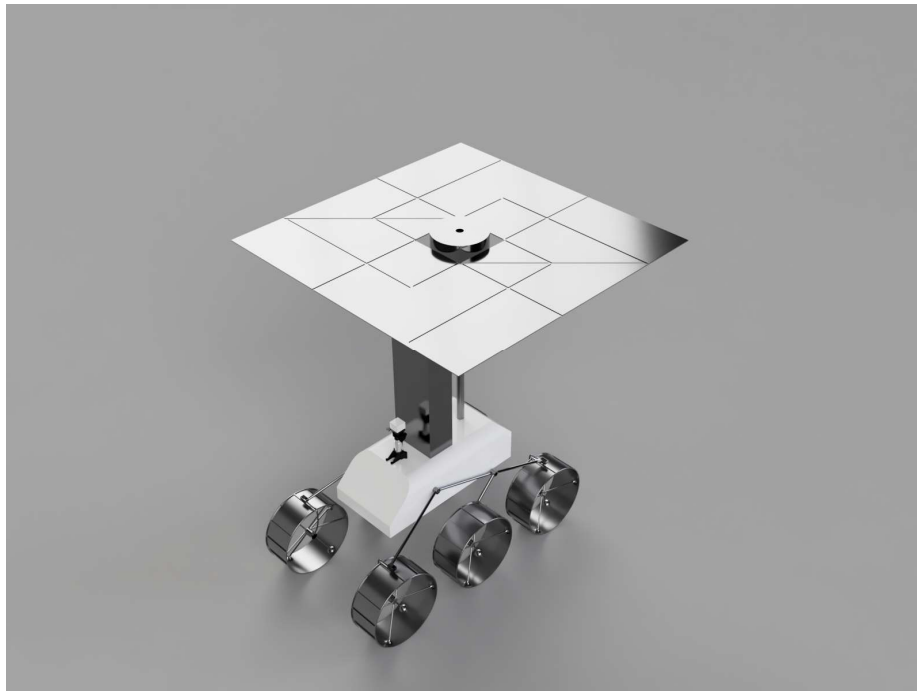


Figure 8: Rover from Eagle Eye



Figure 9: Rover Without Top

Subsystem	Mass (Kg)	Volume (cm ³)	Power (W)
Mechanical	32	486457	39
Power	27	5026	N/A

CDH	7.57	3508	150
Thermal	0.57	5971	212
Payload	5.47	1501	31

Table 6: Total Subsystem Mass, Volume, and Power

2.1.1. Mechanical Subsystem Overview

In the process of developing the mechanical subsystem, many factors contributed to the preliminary design. A combination of Technology Readiness Level (TRL) and practicality are of the highest priorities. The main components to be considered for the subsystem are suspension, materials, wheel design, and science collecting devices.

Research of previous missions has shown how the adopted suspension type of the “rocker-bogie system.”³ has been successful in similar environments, as this technology is flight proven through successful mission operations. The advantage of this component style is that it does not require any separate suspension, allowing for fewer parts, lower mass, and fewer points of failure. This will also allow the load of the body to be distributed equally along each of the vehicle’s six wheels. The rocker-bogie system will ensure safe traversal of the vehicle as it will experience both a rough/rocky and fine/smooth surface during different paths of the Moon’s surface.

Materials to be used in the construction of the subsystem will primarily be of aluminum, but also include titanium, carbon fiber, and plastics. “Since the first aircraft was launched into space, aluminum has been a prominent material used for space structures.”⁴ Aluminum 6061 has proven to be very desired in space applications because of its great strength to mass ratio. Many of the operating satellites, landers, and rovers have shown great success in this material, this will also provide heritage for the primary materials in constructing the bulk of the subsystem’s components. This material has shown to perform well under extreme temperatures as high as 400 K until it starts to lose some of its properties, however as temperatures decrease aluminum not only sustains its properties but even improves these, making this material an excellent candidate when experiencing extended periods in the permanently shadowed regions of the Moon which will yield temperatures as low as 90 K.

The wheel design is imperative to the success of the mission. In order to collect science and power, the vehicle must have capable wheels to traverse successfully through the lunar environment and there have been many variations of redesigning the wheel for different terrain missions. The wheels will endure temperatures from 90 K in shaded regions to 379 K in sunlit areas. Single piece treaded wheels have shown to have fewer points of failure and although they may endure some wear damage,⁵ they still prove to be successful in their function. The wheels are designed to withstand the temperatures and terrain of the lunar pole.

³ Lewin Day, “Rocker bogie suspension.”

⁴ Howard, “Aluminum in spacecraft.”

⁵ NASA, “Rover components.”

The last component to consider will be a drill for collecting samples of regolith at different depths. A top candidate for this function is the TRIDENT⁶ drill. This device can endure the surface of the Moon as it cuts and hammers to fragment hard material. This component will have the lowest technology readiness level. The drill will be a compact size to accommodate the smaller vehicle, and samples can be collected in sizes of 3-5 centimeters for evaluation. All components considered will contribute to the effectiveness of all the other subsystems.

2.1.1.1. Mechanical Subsystem Requirements

The mechanical subsystem comprises four main components: mobility, materials, suspension, and science collection systems. The successful design of these components to work together effectively will be a deciding factor in achieving the science goal driving the mission. Each component has been carefully considered when deciding to move forward with each item, as the first rover mission to the lunar pole increases the complexity and risks involved when the mission is in operation. The approach to each component was decided over trade studies and careful deliberation. Each requirement was methodically determined based on the overall mission driving goal and the environment of the Moon's south pole. Lower level requirements have been considered and evaluated. These requirements are specified in the table below:

Table 7: Mechanical Requirements

⁶ NASA, "VIPER Rover instruments."

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Relevant Subsystem	Req met?
ME-01	The subsystem shall operate/perform under extreme temperatures ranging from 90.15 K to 379.15 K	Achieving science goals relies on the successful operation of the mechanical subsystem	SYS-03	ME-04 ME-06	Test	Thermal	Met
ME-02	The subsystem shall use materials not exceeding a mass of 55 kgs	The mass of mechanical components will be limited	SYS-02	N/A	Inspection	All	Met
ME-03	The subsystem shall not exceed an allotted cost totaling 175 M of all subsystems combined	A finite amount of funds will be allotted for the mechanical subsystem	SYS-08	N/A	Inspection	All	Met
ME-04	The subsystem shall be able to collect energy by means of solar arrays/panels	In order to operate/perform the system needs sufficient power	SYS-07	ME-06	Test	Electrical,CDH, Thermal, Computer hardware	Met
ME-05	The subsystem shall have sufficient volume to carry all payloads	The craft needs sufficient	SYS-04	ME-02	Inspection	All	Met

	required to operate/perform tasks	space to carry all necessary payloads					
ME-06	The subsystem shall have a mobility system to overcome obstacles of the lunar surface such as rough/soft terrain	Science objectives/energy collection rely on the capabilities of the mobility system	SYS-01	ME-04	Test	Electrical,CDH ,Computer hardware	Met
ME-07	The subsystem shall have cameras to scout paths of travel with the least obstacles	Cameras will help assess the location and investigate locations of interest	SYS-01	ME-06	Test	Electrical,Computer hardware	Met
ME-08	The subsystem shall have lights to guide the craft in the PSR	Lights will help navigate the craft in the dark environment	SYS-01	ME-07	Test	Electrical	Met
ME-09	The subsystem shall have sufficient power to operate the mechanical components	To traverse and complete tasks sufficient power is needed	SYS-01	ME-05	Test	Electrical	Met

2.1.1.2. Mechanical Sub-Assembly Overview

Designing the mechanical subsystem yields different sub-assemblies that must work together seamlessly in order to perform to the highest of standards, and because of the specific mission to the South Pole of the Moon, the subassemblies and their respective materials will endure an extreme spectrum of temperatures when in PSRs or collecting energy in sunlit locations. When developing each sub-assembly, heritage and TRL were considered to ensure the success of the system as a whole. Having components that, although haven't been flight tested specifically for this mission's location, have had their success in similar environments lending the idea of effectiveness for different aspects that will make up the different sub-assemblies of the mechanical subsystem. Each component will be broken down into their respective sub-assemblies and discussed from their practicality, TRL, and mechanical properties.

The mechanical subsystem will provide mobility and housing for all relevant subsystems and science payloads. The capabilities of the spacecraft will be imperative in achieving science tasks/goals of the mission. The main items to consider in designing the subsystem are materials, mobility, and dimensions. The materials used for the craft must have a combination of low mass, high strength, and high performance under extreme temperatures. A widely used material with these properties is aluminum among other materials that will be considered in the assembly of the mechanical subsystem. Mobility is important when traversing the harsh terrain of the lunar south pole. A suspension system that can handle a rough and fine surface along with capable wheels such as mesh wheels or a rocker-bogie system will be desirable given their previous success in similar environments. Other obstacles such as increasing slopes and lunar rocks are other situations to expect and design for. Finally the dimensions that make up the craft will be important in meeting the constraints of volume and mass by the client, as well as housing all the relevant subsystems to operate/maintain the mechanical subsystem. There are many components that go into the cumulative operation of the system and having enough space to store them all while providing protection from the harsh environment is key to the mission's success. Achieving the science goal will also help in driving the design of the subsystem as the science tasks may require the craft to be lower to the ground for drilling/scanning regolith. These foundational components of the mechanical subsystem have great importance and will ensure the success of the overall mission.

Suspension Type

The suspension system decided upon is a rocker-bogie style suspension. The main factor in this decision was the high level of heritage that this system has. For example, rocker-bogie suspensions are used in all of NASA's rovers including Curiosity, Spirit,

Opportunity, and Perseverance with a high level of success. With the success of this previously used suspension type, it can be implemented into the spacecraft offering a TRL of 7 since this suspension type has not yet been used specifically on the Moon.

Another key consideration was simplicity and manufacturability. A system that is as simple as possible to keep the amount of potential weak points to a minimum, keep costs down, and ensure the system could be manufactured in a timely manner. A rocker-bogie system offers all of these features. There are no springs or shocks to worry about. The design is simple and easy to implement, and it is reliable. The simplicity of the system means fewer components that could malfunction and fewer mechanical parts that can get clogged with dust, fatigue, or wear. This is a huge benefit considering the rover will be hundreds of thousands of miles away where maintenance is not an option. Keeping the design simple and practical is important in this sub-assembly making up a total of 5 main parts for. There are two lower rocker arms for each side of the suspension and a front rocker for each side as well. There is also a differential gear which regulates each side of the distance traveled by the wheels. This will help when maneuvering or overcoming obstacles.

Considering the terrain of the lunar pole, this suspension type will experience lunar rocks of varying sizes along with fine sand and rough areas. The Moon is a much different environment to Earth and with that come certain challenges to overcome. The Moon has a sandy and dusty surface that could become an issue for traction and sinking. A rocker-bogie suspension has a natural ability to spread the load of the vehicle across all six wheels, which minimizes the maximum ground pressure on any one wheel. This is especially important when traversing soft terrain and ensures maximum traction with all six wheels on the ground at all times.

The combination of a rocker-bogie suspension and wheels as opposed to tracks completely negates the initial worry of overcoming obstacles. Generally, a wheel needs to be twice as tall as the object it is trying to overcome in order to go over it. With a rocker-bogie suspension involved this equation actually reverses and the rover will now be able to scale objects twice as tall as the wheels.

Material

It has been decided that the rover will use a combination of titanium, aluminum, carbon fiber, and plastics to optimize the design and functionality of the rover while keeping the weight to a minimum. Different materials will be used on different components of the rover depending on their needs and specifications. Initially, magnesium was considered as a construction material because of its incredible weight-saving potential, but its high reactivity posed an extreme risk to the mission and was consequently eliminated from consideration. Notable reactions include corrosion: magnesium is highly reactive in the

presence of moisture, which is a primary concern for the mission. Flammability: magnesium is highly flammable and can burn intensely. This is a significant concern, particularly during the launch phase when materials are exposed to high temperatures. Special coatings and treatments are often used to mitigate this risk. While magnesium has good thermal conductivity, it also has a relatively low melting point compared to other metals like titanium or aluminum. This can be a disadvantage in certain high-temperature applications, especially on the Moon where temperatures can reach 379.15 K. The primary material used in the construction of the subassemblies is Aluminum 6000 series, offering lower mass compared to other metals and excellent mechanical properties to perform in extreme hot and cold temperatures. This material type will provide a high TRL of 7 as it is a widely used material in many space applications.

Along with alloys, other materials will also contribute to the overall subassemblies, many of which have also been used on previous missions to other worlds. Heritage and high TRL components only go so far because the environment they travel to will always differ in composition, temperatures, and terrain. The Moon has virtually no atmosphere and is a much smaller body compared to earth resulting in a fraction of the gravity experienced on Earth.

The materials selected will also undergo analysis and simulations to ensure that the mechanical properties of each material type is maintained under different temperatures expected on the Moon as well as loads experienced by the mechanical system and all other subsequent subsystems. This will determine if any sub-assemblies need further adjustments or material changes through the maturation process of the mechanical subsystem.

Wheel design

Designing wheels for space applications can be challenging. Rovers on the Moon will experience extremely low temperatures which will cause traditional tires like what was used on Earth to shatter like glass. Because the design and material of the wheels have been carefully considered, maintaining mechanical properties in the harsh temperatures of the Moon are of the utmost importance. The wheel design is imperative to the success of the mission. In order to collect science and power, the vehicle must have capable wheels to traverse successfully through the lunar environment. There have been many variations of redesigning the wheel for different terrain missions. The wheels will endure temperatures from 90 K in shaded regions to 379 K in sunlit areas. Single piece treaded wheels have shown to have fewer points of failure and although they may endure some wear damage,⁷ they still prove to be successful in their function. The wheels will be fabricated out of a single block of aluminum to reduce failure points. This

⁷ NASA, "Rover components."

will likely be done at a designated NASA facility. The wheels are designed to withstand the temperatures and terrain of the lunar pole.

Science collection

The mechanical subsystem will support the drill device that will contribute in achieving the science goal, this device will be integrated into the main body of the rover in order to dig at a depth of up to 0.75 meter of the lunar regolith. A top candidate for this function is the TRIDENT⁸ drill. This device can endure the surface of the Moon as it cuts and hammers to fragment hard material. This component will have the lowest technology readiness level. The drill will be a compact size to accommodate the smaller vehicle, and samples can be collected in sizes of 3-5 centimeters for evaluation. These drilling sessions at different sites will provide insightful data to the composition of the samples produced using this drill. This device will be integrated into the center of the mechanical system's chassis. Lights and cameras will accommodate the drill in the multiple drilling points mapped out for the mission.

The following table will display the specifications of each sub-assembly totals making up each mechanical subsystem component. This will provide mass, volume, and power draw.

Table 8: Mechanical Subsystem Components

Component	Mass [kg]	Volume [m]	Max Power Draw [W]
Chassis	13	1 x .5 x .25	10
Suspension	1	.8 x .3 x .05	2
Wheels	6.05	.349	24
Drill	12	.145 x .025 x .126	3

2.1.1.3. Mechanical Subsystem Recovery and Redundancy Plans

The mechanical subsystem designed will have multiple points to ensure the successful operation of the vehicle given unforeseen circumstances such as obstacles in the terrain and multiple wheel design where failures can be mitigated by

⁸ NASA, "VIPER Rover instruments."

the suspension and drivetrain of the spacecraft. The structure design will also have multiple layers for thermal and strength redundancy. The mechanical subsystem will include the two science instruments to scan the lunar surface offering recovery if one system happens to be inoperable. This also includes multiple cameras and multiple lights as part of the navigating/monitoring recovery.

The mobility component of the mechanical subsystem offers recovery on the basis of multiple wheels with their respective motors to drive each wheel forward. In the case of a failed/damaged wheel or motor the rover will still be able to traverse the surface, however this could slow down previously determined speeds when all wheels/motors are at 100%. Although speeds may decline this component offers multiple points of redundancy so long as at least one wheel from each side can be powered to drive the spacecraft forward. This component will be of the utmost importance as traversing between science locations and sun rich areas to charge solar panels which will regulate operating temperatures and power all of the vehicle's components. This ensures a capable spacecraft to survive and carry out all the tasks required to accomplish the science goal driving the mission.

In the case of the component housing/science payload deck there will not be any recovery or redundancy as this component is too large in mass and dimensions to duplicate, rather a multiple layer insulation (MLI) can be implemented to allow some form of redundancy for this component, this will be low in mass and cost. This MLI will help in protecting all other subsystems on the spacecraft.

The next component to consider is the energy hub. This is also a component that will be much too large in mass and dimensions to duplicate, so no recovery or redundancy is in place should anything go wrong. This means that it is critical that the solar array can deploy out of the energy hub and unfold properly to be able to function efficiently. Given the high degree of importance for this component to be successful, it can be determined that it will be safe to move forward without any recovery or redundancy plans being that the devices that make up the energy hub will be of the highest product quality, and it only has to work one time. Once the spacecraft has landed on the Moon, it will only need to deploy one time. The reason the energy hub cannot be landed with the panels already deployed is because the total area that the solar panels expand to would exceed the allowed volume constraint set by the launch provider.

As a part of the mechanical subsystem, another device to acknowledge is the onboard drill. Although this device has no recovery or redundancy due to the overall mass and dimensions, there are other things to consider. For this particular component it is safe to continue as instruments on the payload deck will still be able to assess the composition of the volatiles on the surface and slight subsurface of the

Moon without having to drill at exceeding depths. The drill is important in contributing to the overall science goal of the mission and will be integrated into the body of the component housing. Ensuring the drill is operable and efficient will be imperative to mission success.

In the case of navigating the lunar terrain, there are means of recovery and redundancy in this area as multiple cameras/lights will be mounted on both the mast of the rover and beneath the rover to monitor the drilling process. There will be a total of 4 cameras and a total of 4 lights as part of the navigation component of the mechanical subsystem. Each pair of lights and cameras will be mounted in the areas providing recovery and redundancy in the case where one of each pair of lights or cameras happen to malfunction or become damaged by other unforeseen circumstances. Navigation and drill monitoring will be important to reduce risks when the rover is traversing paths between drill locations and energy collecting locations.

Considering all of the components that make up the mechanical subsystem there is recovery and redundancy on three of the five major components. The other two components have been determined to be safe to move forward with as their risk of failing is determined to be low. The majority of the components will offer recovery and redundancy ensuring a capable rover given undetermined failures or malfunctions of the mechanical subsystem.

2.1.1.4. Mechanical Subsystem Manufacturing and Procurement Plans

The subsystem will primarily consist of the mobility system, component housing/science payload deck, energy collecting, and navigating. Each component will have its own process of design and development. A combination of COTS and in-house manufacturing will contribute to the development of the mechanical components. It is crucial that these components can perform to the best of their abilities but to also ensure that it can also be done given the time constraint to be ready for launch. The manufacturing and procurement plans must be feasible to align with all of the constraints designated by the client. As a result the COTS and in-house manufacturing was decided due to the uniqueness of the mission to the lunar south pole while also taking into consideration heritage products that can provide positive results while saving time and money.

The mobility system of the mechanical subsystem begins with the wheels, and because the wheels used on a vehicle depend on the environment, there aren't many COTS options available, so this is one of the components where engineers can use NASA resources to manufacture this part of the system. All six wheels would be machined

from a single block of aluminum at a designated NASA facility. This will ensure a reliable product from an in-house facility such as the NASA Glenn Research Center, where a rich history of “reinventing the wheel⁹” has taken place. These wheels will be powered using a COTS product, 6 EC32 brushless motors¹⁰ provided by Maxon, which is a reputable company in business for over 60 years. This company has also used similar motors that have been successful with spacecrafts on the surface of Mars. This third party product provides a reliable field tested component with great savings in fabrication and testing. The last component of the mobility system is the suspension. The suspension type for the system is a rocker-bogie style. This will not yield a COTS product as the dimensions of the spacecraft will be unique in size, therefore NASA facilities will be used to develop and manufacture the suspension component. Many factors will contribute to the development of the suspension component such as the extreme spectrum of temperatures that lie on the surface of the Moon’s south pole, as well as the weight of the overall spacecraft and the loads it will endure given the lower force of gravity on the Moon’s surface. The expected times for development for manufacturing the mobility system can be from 9-12 months.

The component housing/science payload deck will allow instruments to be mounted while maintaining housing temperatures for all other subsystems at normal operating conditions. This component is important to the life of the other subsystems as it will provide safe stowage of all other components, which will need to survive the harsh conditions of the Moon. Given the unique size of the vehicle due to client constraints, this component will be manufactured in-house using the material Aluminum 6061, a proven and widely used material in the aerospace industry due to its desirable properties. This component will have sufficient volume to store all the necessary subsystems and can be fabricated and calibrated at a designated NASA facility. The estimated time to design and produce this component is anywhere from 6-9 months time.

To power the different subsystems an energy collecting system is required to collect energy. This will be done using COTS parts, an EC32 brushless motor from Maxxon, which can be used to deploy a solar array along with a LA40P actuator¹¹ from Linak, and multiple coreless servo motors from BETU¹². Together these components can raise and aim solar arrays in a given direction for the most effective energy capture given different positions and locations on the lunar surface. Linak is a family-owned business and leader in the industry for over 100 years providing exceptional products. The solar

⁹ NASA, “NASA Glenn Research Center.”

¹⁰ Maxon, “High precise drive system.”

¹¹ Segments, “Compact and robust electrical linear actuator.”

¹² Betu, “Coreless servo motor.”

array will be folded up in the energy housing of the payload deck and after landing it will deploy out of the energy housing and expand to an area of 2.25 m² providing sufficient area to collect energy from the Sun's rays to power all of the subsystems. The importance of the solar array deployment is crucial to mission success. Collecting energy to power not only the mechanical system but all the other subsystems will determine the operation and survival of the entire spacecraft. These products will all be contracted which will yield lower costs and time in developing these components of the energy collecting mechanism. The time to produce these products is estimated to be several weeks as these products are easily available.

Another component that will contribute to achieving the science tasks and goal is by means of a drill. This drill will be specifically made for this mission given the default dimensions of the contractor's device currently exceeds the volume constraints provided by the client. As a result of a scaled down version of this component, the depth capabilities will also be scaled down by ~25% reaching depths of up to .75 m, still giving sufficient depths to determine composition underneath the surface of PSRs. The device to be used will be the TRIDENT drill¹³This is a COTS product that will be contracted through Honeybee Robotics, providing innovative products since 1983. This particular drill has been designed specifically for this mission type, with plans to go on future trips to the lunar surface. The company has adapted well to aggressive schedules and can produce quality products estimated to be anywhere from 3 to 6 months time. Although this product has the lowest TRL of the other components, this product is convenient having already been produced and tested to perform in conditions specifically for the Moon's surface. This will save time and money on designing and fabricating this component by other means.

The last component to consider for the mechanical subsystem is navigating. This will require on-site imaging of the landscape. This can be accomplished by multiple cameras and lights for navigating and/or science location reconnaissance. The main cameras will be mounted on a Leo Rover Universal Camera Mast style¹⁴. The two mastcams will be with The Imaging Source Color Camera¹⁵ allowing imaging or videos from Teleskop service. This company has established a positive reputation in imaging devices for different applications. These will allow the safe navigation of the spacecraft through the PSR environment. These cameras will also monitor the drilling process under the chassis of the vehicle. With proper imaging of the lunar terrain, safe paths can be scouted to travel on with the least obstacles. Reducing risks is key to mission success so the importance of these images will be extremely important in mitigating the

¹³ Honeybee Robotics, "Drill products."

¹⁴ Leo Rover, "Universal camera mast."

¹⁵ Teleskop-express, "The Imaging Source."

risks associated with the environment. The lights and cameras will also provide a view from below the rover to monitor the progress of the drilling tasks. Finding the best areas for drilling is important and when in the complete darkness of the permanently shadowed regions lighting and imaging are of great importance in completing the driving mission goal. The products mentioned are easily available and the procurement times of these items will be hasty.

All of these components that make up the mechanical subsystem have been methodically decided using trade studies, research, and discussion. The manufacturing and procurement plans were determined based on the uniqueness of the mission destination and the launch date determined by the provider. With the plans stated above, results will yield a time efficient product while being conservative as possible with costs. The total time for all the components to be manufactured or procured is estimated to be anywhere from 9-12 months. This aligns with the timelines dedicated to the phase associated with the fabrication of these components. Delivering the mechanical subsystem on schedule will provide sufficient time for successive phases to prepare for launch.

2.1.1.5. Mechanical Subsystem Verification Plans

As the development of the mechanical subsystem matures, it is important to acknowledge Verification and Validation (V&V), ensuring that the components designed and fabricated meet the requirements of the uniqueness of the mission type. Things to be considered are environmental factors, temperatures, and potential obstacles. By developing a preliminary verification plan for each requirement of the subsystem, validation can be established to move forward safely and confidently.

Table 9. Mechanical Requirements

Req #	Requirement Summary	Verification Method	Rationale for Method	Preliminary Verification Plan
MEV-01	The operate and survive the components of the mechanical subsystem must withstand temperatures ranging 90.15 K to 379.15 K	Test	To prepare for the extreme temperatures on the Moon the subsystem needs to handle a spectrum of these temperatures	Due to the unique temperatures of the Moon, materials of each of the components that make up the mechanical subsystem will be tested for hot and cold extremes
MEV-02	To align with mass constraints the mechanical subsystem will have an allotted amount of mass	Inspection	Launch provider has allotted a finite amount of mass	Inspections will occur throughout the development of the mechanical subsystem to ensure mass constraints are not exceeded
MEV-03	To align with cost constraints the mechanical subsystem will have an allotted amount of funds	Inspection	Client has allotted a finite amount of funding for the mechanical subsystem	Inspections will occur throughout the development of the mechanical subsystem to ensure cost constraints are not exceeded
MEV-04	In order to be operable the mechanical subsystem is responsible for deploying and collecting energy to power the mechanical and other subsystems	Test	Ensuring the energy hub will deploy successfully extensive testing will be performed	Testing of the components and all other subcomponents associated with the energy hub will be tested many times over to ensure the successful deployment of the solar array

MEV-05	The mechanical subsystem is responsible for housing and protecting all the other subsystems of which volume cannot exceed 1.5 x 1.5 x 1.5 [m]	Inspection	Launch provider has allotted a finite amount of volume	Inspections will occur throughout the development of the mechanical subsystem to ensure the volume constraint is not exceeded
MEV-06	To travel through the lunar surface between science locations and charging locations the mechanical subsystem is responsible for having a capable mobility system	Test	To have a reliable mobility system testing of the wheels and suspension will occur on lunar like terrain	Testing of the wheels and suspension will take place simulating environments the rover will likely encounter on the lunar pole
MEV-07	While traveling from one location to another location imaging will assist in assessing safe paths of travel as well as drill monitoring	Test	Ensuring the safest paths along between tasks and charging testing of the cameras will be completed	Testing of individual cameras positioned on the rover will occur to ensure the safety of the rover
MEV-08	While traveling in the permanently shadowed regions lights on the mechanical subsystem will prove helpful in mitigating obstacles as well as	Test	Ensuring the safest paths between tasks and charging testing of the lights will be completed	Testing of individual lights positioned on the rover will occur to ensure the safety of the rover

	monitoring drilling tasks			
MEV-09	The mechanical subsystem needs sufficient power to operate after landing on the Moon	Test	Ensuring enough power can be collected to power all of the subsystems testing of the available power and power draw from each component will be tested	Testing of the power collected and the power drawn from each individual component will ensure the proper operation and survival of the rover

2.1.2. Power Subsystem Overview

The Power Subsystem and its subassemblies work together to generate and provide power to the rover's systems that require energy at any point after disconnection from the delivery vehicle. These include the mechanical sub system, CDH sub system, and the science payload as well as any other parts outside of these systems that need to be powered such as lighting or heating elements. In this section the power draw in watts of each subsystem and its individual components will be evaluated to ascertain a maximum power draw for the rover in order to calculate the amount of energy storage needed, operational time period, and necessary power generation. The total power needs of these systems are 236 watts with the largest total power draw coming from the Mechanical Subsystem.

For the Mechanical Subsystem on this rover there are three separate categories of components that draw power after the rover lands on the lunar surface. These categories contain; the hub motors of which there are six Maxon EC 32 flat motors, linear actuators of which there are two LINAK LA-12 linear actuators, and rotary actuators or servos of which there are three BETU 45kg Coreless Servo Motors. The electrical rating of the hub motors is 42.6mA at 24V for each motor, resulting in an individual power draw of 1.1W per motor. Thus, for all motors there is a total possible current draw of 255.6mA and power draw of 6.6W. For the linear actuators the power draw is 1A and 24V under full drain each. This gives a power draw of 24W each for a total of 48W. Last, for the rotary actuators and servo's, each has a power draw of 4A and 8.3V which when calculated yields a power draw of 16.8W each for a total of 50.4W when all three servo's are combined. The total power draw for the mechanical components in this section is approximately 103.7W at peak power draw assuming all components are under load and operating at the same time.

Moving on to the CDH system, there are several necessary components that draw power. First is, the motherboard/CPU which when powered draws a maximum of 6.7A and 5V summing up to just under 35W at maximum power draw, one important note for the power subsystem is that this component will likely never experience this amount of power draw unless transmitting and receiving information, nonetheless it will be assumed that the power draw is at a constant maximum for the sake of margin and unforeseen computational loads. Next, both the transmitter and receiver array draw approximately 55W each for a total of 110W and a total of 145 watts for the system. The amount of power draw for the transmitter and receiver will be assessed at this level for one hour before and after data collection as it is unlikely that the transmitter and receiver will be needed at all times for navigation to and from testing locations. For the memory that the CDH system will use the motherboard/CPU has allocated space and power draw for this part but it is worth calculating as if it were by itself to ensure that it does not impact the function of the power system. This component draws very little power with only a 30mA and 3.3V draw totaling to approximately 0.12W in read/write mode for 8MB of nand flash memory. Knowing this, it can be assumed that the power draw while in read/write mode will not exceed the allotted power draw that the

motherboard/CPU SBC has allotted for in its own max power draw and is therefore not required in the final power draw amount for needed battery capacity.

Finally, the Science Payload has three pieces of equipment that need to be powered during the data collection period and instrument functionality testing periods of the mission. First is the NIRVSS which uses a maximum of 29.6W of power when in use, second is the NSS which uses 1.6W of power when in use, and finally the TRIDENT drill which uses a maximum of 5 watts of power when in use. These three parts use a combined 36.2W of power for the entirety of the Science Payload System. These instruments take a minimal amount of time to complete their tasks when the rover reaches the bottom of the Haworth crater with the trident drill likely taking the longest period of time to gather material and therefore likely using the most power. The power draw for this system will be evaluated as in use at maximum power draw for 1 hour in order to maintain margin for failed tests and more tests than originally planned per test site. As the total power draw for this system is low and the amount of time that these systems will be on is also very low, the amount of power drained from the batteries for this system is less compared to other systems.

As the total power draw for all systems is approximately 236.1 watts, the runtime for the rover at this max power draw is 10 hours which is considerably shorter than the maximum possible runtime as there will be long periods of time especially during travel where the power draw is much less giving a longer runtime. To manage the power draw from systems that are not being actively used will turn off each system to allow for an increase in total runtime. For the traverse phase cycle, the only systems that will be running are the 6 drive motors in the mechanical system and the SBC in the CDH system. The max power draw for these two components will be approximately 41.6 watts. Using only these two categories of equipment the rover's runtime increases from 10 hours to 58 hours giving more time to complete mission tasks when the rover reaches the crater bottom. Once the rover has reached the bottom, the battery should have discharged to 65% of its total capacity and the rover will begin the data collection phase. To reduce the total power consumption during this period, this phase will take place when the rover is stationary and the rover will not be transmitting or receiving information at this time. This operational phase cycle will involve using the NSS and NIRVSS which when combined use 31.2 watts of power. This combined with the SBC which uses 35 watts will use approximately 66.2 watts at peak operation. Therefore at peak operation for one hour the battery should drain to approximately 1 percent with a wide margin, to 63% of the total charge capacity. The next phase of the concept of operations is data transmission. During this period of time the downsized data transmitter and receiver will be switched on and the data collected will be transferred to earth. The both the transmitter and receiver will use 55w of energy for a total of 110 watts while in use during maximum transmission. The actual power use may be lower than this because the amount of data that will need to be moved will likely be less than what is needed and will therefore result in a lower power draw. Given a 1 hour transmission period this should result in a drain of approximately 1% with margin on the batteries when compared to full charge leaving the charge of the batteries at the end of this stage at 63%. Finally the next steps of this process are to move to a point where the rover can recharge its batteries and repeat the previous steps. This may

require a charge of up to 35 percent to move between the crater and crater rim, leaving a remaining charge of up to 28% for both overhead and the possibility of completing multiple regolith tests in one traverse from crater rim to test location.

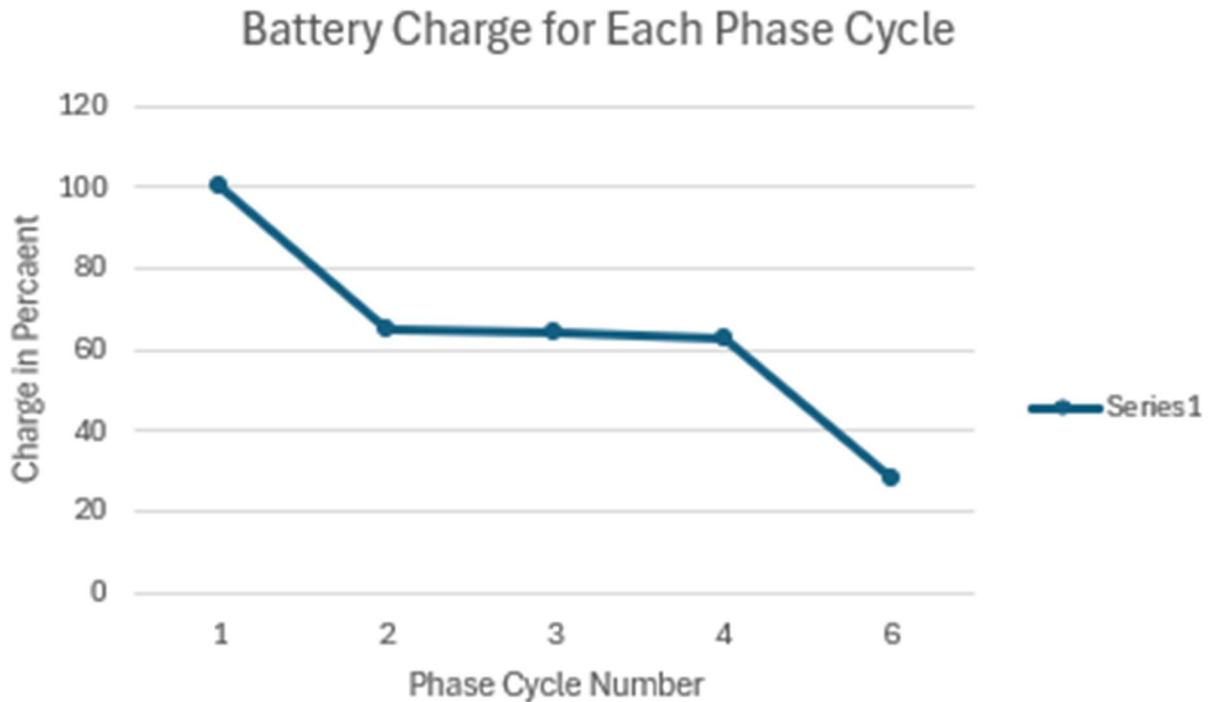


Figure 9.1: Battery Charge of Each Phase

2.1.2.1. Power Subsystem Requirements

The following table defines the requirements needed for the power subsystem. These requirements are based on and derived from the mission requirements and objectives which outline the goals and scope of the mission. By understanding the needs of the other subsystems and the payload instruments themselves, the team gets an understanding of the needs the power subsystem must fulfill.

Table 10. Power Requirements

Requirement Number	Requirement Description	Rationale	Parent Req	Child Req	Verification Method	Relevant Subsystem	Requirement met?
PS-01	The Power Subsystem shall be able to withstand the temperature range on the lunar surface. (90.15 K to 379.15 K)	The system must be able to work in the operating conditions on the Moon.	SYS-03		Test	Thermal Control Subsystem	Yes
PS-02	The Power Subsystem shall be able to withstand the environmental radiation on the surface of the Moon. (1,369 microsieverts per day)	The system must be able to function on the surface of the Moon for the lifetime of the mission	PS-06 PS-07		Test and Specifications	Power subsystem	Untested
PS-03	The Power Subsystem shall be able to supply all power requirements of the spacecraft between solar charging intervals. (1 week hold)	The system must be able to ensure that the science goals of the craft can be fulfilled and therefore must be able to supply all instruments with power for up to 1 week	SYS-07		Test and calculations	Power generation and storage	Untested

PS-04	The Power Subsystem shall weigh no more than 20 percent of the total allotted kilograms. (17.1 kg)	The system must not encroach on the maximum weight requirement for the whole craft	SYS-02		Measurement	N/A	Up for Revision
PS-05	The Power Subsystem shall require no more volume than 20 percent of the total allotted craft volume. (0.675m ³)	The system must not encroach on the volume requirements of the spacecraft	SYS-04		Measurement	N/A	Yes
PS-06	The Power Subsystem shall be no less than 75 percent efficient.	The system must not restrict the operation of the craft excessively or to a point where it draws more power than the rest of the craft.	SYS-07	PS-03	Test	Thermal Control Subsystem	Untested
PS-07	The power subsystem shall be able to generate and supply 275 watts at peak rover operation.	The power subsystem must be able to supply the craft with power at peak operation	N/A	PS-03	Calculation and test	Computer Design Hardware, Mechanical Subsystem, Science Payload Subsystem	Yes

2.1.2.2. Power Sub-Assembly Overview

In this section, the Power System of the rover and its design will be explained in detail to discuss the component choices as well as the advantages and disadvantages of each in order to sufficiently supply the craft with power at peak loads and at non peak loads in order to conserve battery in permanently shadowed regions. By doing this, this section hopes to ascertain the capabilities of the craft and discuss steps to mitigate any foreseen and unforeseen issues or inherent disadvantages within the power subsystem. The first step in this process is referencing information about the power usage and generation capabilities of the craft that was discussed in section 2.1.2 of this document. The team will begin by looking at the power draw first for the Mechanical System, then Command and Data Handling system, and finally the Science Payload.

For the Mechanical System on this craft as said previously, there are six hub motors which have a total power draw of 6.6 watts, two linear actuators which have a total power draw of 48 watts, and three rotary actuators or servos which have a total power draw of 50.4 watts. This gives the Mechanical system of this craft a total power draw of 103.7 watts at peak operation. For the CDH system, as discussed previously there is one SBC which draws approximately 35 watts of power at peak computational load. There is one transmitter and one receiver which have a combined 110 watt power draw at peak transmission and reception loads. And finally 500GB of flash memory integrated into the SBC that will use the SBC's dedicated power. This system totals to about 145 watts at peak load for the Power System. Finally the Science and Payload system has three components. The NIRVSS which draws 29.6 watts of power, the NSS which draw's 1.6 watts of power and finally the TRIDENT drill for sample collection which draws a maximum of 5 watts of power while in use (Paulsen, 2018). Totalling the max power draw for the Science Payload team have a maximum power draw of 36.2 watts and totalling the maximum power draw for each subsystem have a maximum power draw of 236.1 watts.

Next the team will discuss the components of the Power System. The team will begin with the power generation, continue with power storage and end with power distribution for each system. To generate power on this craft, this rover will use 2.25 square meters of RocketLab ZTJ triple junction solar cells. This solar cell has several attributes that separate it from other solar cells that RocketLab can supply. Firstly, The ZTJ solar cell is not the most advanced solar cell that RocketLab supplies, but it was selected because its power generation capabilities are comparable to their most advanced options with a maximum 29.5% BOL efficiency compared to 31.3% for their most advanced cells such as the Z4J+ and its stated better performance in low light environments (RocketLab, 2024). Second, The ZTJ solar cell has extensive flight heritage (RocketLab, 2024) giving it a higher certified TRL than the more advanced Z4J+ Solar cell. Finally, the ZTJ solar cell has a specified and proven solar cell performance degradation of less than 2% for 2000 cycles (RocketLab, 2024) which more advanced and theoretically "better" solar cells like the Z4J+ do not. This is important because although the current mission lifecycle does not include an extensive and time intensive mapping of lunar ice deposits, the Power System and specifically this component will be able to handle extended operations on the Moon if there is an

increase in scope. All of this is important when considering the lunar environment that the solar cells will be in because each of these attributes allow for the solar cell to generate power in low light environments and maintain its performance moving in and out of notoriously cold PSR's.

Moving on from the factors that led to choosing this solar cell the team will discuss the setup and power generation specifications of the final, installed solar cell array. The surface area of the solar panels on this rover will be 2.25 square meters or 1.5 x 1.5 meters when unfolded. This will provide the power subsystem of the craft with approximately 400W of peak power which will allow the craft to charge the batteries and operate the rover at max power simultaneously. This solar cell array will be backed with a backplate outlined in section 2.1.1.2. By the Mechanical Engineering Team and mounted to the top of the rover in a folded position for flight. Each cell will be placed connected to a main power stem allowing each cell to function independently to increase the redundancy so each cell does not rely on other cells to function, as if the cells were in a series setup. There will effectively be two solar panels that function independently of each other and will be individually connected to one of the two array power regulators on the PDU. Therefore, the performance and resilience to environmental factors that this cell provides in this setup allows the following components Power System to effectively supply power to the rest of the rover's subsystems.

Next, to store power on the craft the team will use twelve Eaglepicher 55 Ah lithium ion battery cells. These cells have several attributes that allow the Power System to effectively supply power to the craft for the period of time that the rover is within the PSR. This battery has several attributes that make it an excellent choice for this rover compared to different batteries that Eaglepicher and other manufacturers provide. Most importantly this solar cell has one of the highest energy densities when compared to other battery types such as nickel cadmium or nickel hydrogen batteries at 343 Wh/L compared to about 150 Wh/L for nickel cadmium or 60 Wh/L for nickel hydrogen types. This allows the battery pack to take up less space in the rover. On top of this, while some batteries are lighter and have a higher specific energy than this chosen battery, the specific energy for this lithium ion battery is 134 Wh/kg compared to 55-75 Wh/kg for Nickel Hydrogen batteries giving it a higher power to weight ratio than some other forms of batteries. This information is important because of the environment that the Power system will be operating in, for long periods of time the batteries must supply power to the rover's subsystems while the rover is within the PSR. Considering other environmental factors such as the temperature, because this battery is using a unique low temperature formula its temperature rating has a wider and colder band than traditional lithium ion batteries giving the battery pack a low operating temperature of -20 degrees celsius. Which while not as favorable as the low operating temperature of Nickel Hydrogen batteries is plenty considering the advantages of this battery cell. To wrap the environmental considerations of this component, the TRL for this component is 8 because of its proven function in space-like environments.

To utilize this battery correctly and most efficiently, they will be arranged in a parallel circuit with each of the 12 batteries being individually connected to one of the 60A

battery ports and to the main negative line from the PDU. This setup will run one battery which has a max output of 440 amps at 3.6 volts through each port. This limits the amount of peak power that each battery can provide and also provides the highest redundancy as multiple batteries can fail and as long as one remains working, the craft can power itself albeit not for very long. These battery ports are internally connected to both the integrated solar charge controller and battery regulator inside of the PDU allowing the batteries to be charged and discharged through the PDU from the solar cell array. This component of the assembly weighs approximately 1.69 kg each and when all 12 are compiled this part of the Power subsystem weighs 20kg.

The final component that will be discussed in the power subsystem is the AAC Clyde Space STARBUCK-MINI, PDU. This component has a TRL of 9 after being installed on several satellites currently in orbit and combines several functions that would otherwise be performed by multiple parts of the same system. The components it replaces are the Solar Charge Controller, Battery Voltage Regulator, and a multitude of voltage converters for specific voltages. There are several advantages to combining all of these parts into one, Primarily it makes the system much more simple in terms of assembly and general construction but there are also several environmental advantages. Firstly, the singular component is much easier to protect from the physical environment as it is easier to seal from lunar regolith that could short circuit certain pieces and it is also easier to conduct temperature control on because it is a singular piece with a singular temperature requirement.. This would be much harder if the separate components were used because there are more points of entry and more complexity. Second, because this component is a singular piece that does not contain any components that have unnecessary functions, there are less points of failure. This is important in terms of overcoming the environmental factors that the component and system will experience because it creates less possible areas where this component can fail due to environmental factors such as temperature, physical debris, or mechanical failure for each physical connection point in the system.

When it is in use, the STARBUCK-MINI will be set in between the solar array, and battery pack functioning as both a solar charge regulator and battery voltage regulator to effectively manage when the batteries are charging and supplying power to the craft. The solar cells and batteries will be connected to the PDU as previously mentioned. Once these components are connected the last physical connection to make a working power system is to connect individual ports on the PDU to the other systems that require power. This step will start with the CDH system. There are three components of the system that directly require power from the PDU. First is the SBC which will require 6.7A at 5 volts which will be connected to one high power (10A) output. This will be followed by the transmitter and receiver which both require 10A at 5v and will therefore each be connected to two more high power outputs on the STARBUCK-MINI. Moving on to the mechanical subsystem there are three types of components that require power. The first are the 6 hub motors which require a maximum of 1A at 24 volts which will be supplied by 6 of the 12 primary nominal output ports on the PDU which can supply up to 3 amps at 24 volts. For the 2 linear actuators which require 1A at 24 volts which will be supplied by 2 more primary nominal output ports on the PDU. To wrap up the mechanical system, the 2 rotary actuators which require 4A at 8 volts at peak

operation. These will be connected to two high power outlets individually. Finally for the science payload, the NIRVSS, NSS, and TRIDENT drill will be connected to one high power output, one secondary bus output, and one secondary bus output respectively. Once all components are connected, this component will be housed inside the structure of the rover.

Some of the attributes of these components help mitigate certain environmental factors but do not solve them. To further increase the reliability of these components and the system as a whole, there are several steps that will be taken to ensure that failures from environmental factors are properly mitigated. The primary physical point of failure for the PDU will be at each power connection point, because of the high chance that the power cables come disconnected while the rover is in flight or on the lunar surface, each of these power cables will be secured to the PDU using screws for the built in securement system. Furthermore, to decrease the possibility of overheating for this component, its temperature will be maintained by the Thermal system to ensure that its temperature stay's in the -30 to 60 degree celsius range. For the solar array there are several points of failure mainly from physical impacts or degradation of the solar cells themselves. To ensure that the solar arrays are not impacted and damaged by lunar regolith and to decrease the max torque on the solar array's mount during movement. they will be stored in the stowed position while the rover is moving. In addition to keeping the solar array's stowed while moving, the solar arrays will be stowed while the rover is not charging to decrease degradation of the solar cells from both light and cosmic radiation. For the Batteries that are onboard the rover, the long term degradation can mostly be ignored for the duration of the mission as over 2000 cycles must be completed before the cells reach 80% capacity and the possibility of stranding the rover in a PSR where it cannot charge increases drastically. However, to decrease this degradation in case a longer mission cycle is approved, the batteries will be maintained at a constant temperature within the operating range of -20 to 60 degrees celsius and will be kept from draining completely in order to avoid damage to the batteries.

These components were selected in part because of their flight history with each having a high Technology Readiness Level or TRL. This is important especially for the Power System because without all components functioning the rover does not function and because many components like the batteries are heavy and may make up as much as 20% of the weight of the craft it is difficult to introduce redundant components to decrease the risk involved. This selection process began with selecting the solar cells which are RocketLab's ZTJ solar cells that have flown on over 1100 missions to date (RocketLab, 2024). This displays the flight heritage of this component and gives its TRL which is level 9. Continuing on with the selection of the battery cells, the battery cells selected were the Eaglepicher 55 Ah lithium ion battery cells. These cells at a minimum were designed and intended for use in the aerospace industry (Eaglepicher, 2024) and were therefore tested and displayed ability to perform in space environments. As Eaglepicher is a manufacturer that has a flight history with NASA and this cell is one of their few cells designed for space, the likelihood that it has flown in the past is high which therefore earns it at minimum a technology readiness level of 8 and at a maximum at TRL of 9 as it has been tested and demonstrated its performance in a space like environment. The PDU is similar. The PDU that was selected is the

STARBUCK-MINI from AAC Clyde Space, This component is a space-grade power supply designed for small-sat use in low Earth orbit. Because of its use case and the design/testing process that it went through it automatically has a TRL of 8 but because it has recently flown on the Intuitive Machines 2024 lunar mission (AAC Clyde Space) it has a TRL of 9 in an environment that almost exactly matches the environment this rover will be entering. Therefore, the overall TRL of this system sits at a level of 8 with the Eagle

2.1.2.3. Power Subsystem Recovery and Redundancy Plans

For the Power System the consequences regarding the failure of one component are high and because of this the likelihood that the rover will fail part or all of its mission within the required mission timeframe is also high. Because of the high risks, the ability for the system to recover itself and continue to function or the selection of redundant components must be carefully considered in order to ensure that the failure or partial failure of one or more parts of the system does not result in a total failure of the rover. In this section of the report the recovery and redundancy features of the rover will be discussed along with their benefits and drawbacks.

To ensure that the Power system for the rover is reliable before considering making components, components that meet or exceed the required environmental tolerances must be chosen. This involves considerations like, temperature, cycle degradation, radiation tolerance, and mechanical wear on components as well as the systems overall TRL. Starting with the ZTJ solar cells, the Electrical Engineering team selected these components because of their increased performance in low light conditions as well as its resilience to low temperature cycling which would be experienced moving in or out of PSR's and physical failure which can be caused by debris that are kicked up into the solar panel by the wheels. (RocketLab, 2024). These considerations combined with the flight proven TRL of 9 for this component make it a reliable choice before considering altering the design of the craft to make the ZTJ solar cell more redundant and the craft more recoverable. This consideration involves other components as well like the Eaglepicher 55 Ah lithium ion battery cell. The Electrical Engineering team selected this component because of its reliability given by its large working temperature range and large cycle life compared to the number of cycles for the mission (Eaglepicher, 2024). These considerations combined with a flight proven TRL of 9 make this component reliable before the Electrical engineering team needs to consider its redundancy. Finally, the final chosen component which ties these two together and therefore needed special consideration is the AAC Clyde Space STARBUCK-MINI power distribution unit. This component performs several tasks as both the solar charge controller, voltage converter and the power distribution unit, and therefore must be more reliable in the first place. Some of the considerations that the Electrical Engineering team made in the selection of this component were, designed purpose,

Ideally the Power System would include multiple redundant components to ensure its reliability and ability to recover if a fault in the system occurs. All of the components in the Power System on this rover are critical to the mission and should they fail before mission completion they would result in mission failure. As some of the components are more prone to failure but also have high weight like the batteries and PDU this makes it harder to improve the reliability by adding components in parallel. Therefore to improve

the reliability of this system there are several steps the team should take first to avoid adding weight. For components that are heavier it will be more advantageous to select components that are overbuilt for the job that they are tasked with. Combining both of these strategies will improve the reliability, redundancy and self recovery of the craft. This was used for each of the parts in the Power system. First the PDU, STARBUCK-MINI was selected because of its high TRL and much higher power capacity than what will be used on the craft. For these components.

2.1.2.4. Power Subsystem Manufacturing and Procurement Plans

The Power Subsystem as specified by the system requirement PS-03 must supply the craft with power for the duration of the mission. To do this, the Power Subsystem will use several COTS parts to fulfill its role in supplying power to the rover. These parts and components are; Solar cells in array, Batteries in array, Solar Charge Controller, a Converter, and finally a Power Supply Unit. In this section, the Electrical Engineering team will go over the manufacturing and procurement for each of these components in reference to the entire system.

The solar cells in the array that were chosen are the ZTJ triple junction solar cell from Rocket Labs. These solar cells were chosen from this company because of Rocket Labs' proven reliability in the satellite market with over 1100 installations powered by their technology in orbit as well as product testing through a thermal cycle before delivery. The Electrical Engineering team decided to go with a COTS triple junction solar cell as the development of an in-house triple-junction cell would have taken more time and money than using a COTS component. The lead time for this component is around 18 months and is the longest expected lead time. In case this component cannot be used, The Sparkwing solar panel from Sparkwing Space will be used in its stead. This company was chosen as a backup because it has over 40 years of history supplying satellite solar panels and its product has flown on over 85 missions while also maintaining a similar amount of power supply for the craft.

Moving on, the batteries that are currently specified in the Power Subsystem are the Eaglepicher 55 Ah lithium ion battery cells. These batteries were chosen because of Eaglepicher's legacy as a satellite battery manufacturer with over 3 billion installed cell hours in space and because they were designed with a space focused use case in mind, creating specifications that align with the craft's needs. Eaglepicher also has a history with NASA as it has provided batteries for missions like Mars Pathfinder and Mars Odyssey further increasing its credibility (Eaglepicher, 2024). The lead time for this product will be around 5 months as industry standard for a custom or low volume battery. The backup component that will be used if this component cannot be delivered

is the Optimus-80 from AAC Clyde Space. This company has a history of providing the UK Space Agency with components and has supplied components for, and built several satellites with their components on board that are currently operating in orbit like the YMIR-1 (AAC Clyde Space, 2024).

The next part in the system is the solar controller. For this, the Electrical Engineering team chose the AIMS Power 10 AMP Solar Charge Controller. This part and company were chosen for several reasons initially but will likely be changed in place of a more space dedicated component. This component initially fits the thermal, environmental and power needs of the system, however, this solar power converter was not designed by a company in the aerospace business and while reputable in the solar business is not yet proven in space. Therefore, in the future this component will be replaced with an all in one power supply and solar controller like the STARBUCK-MINI from AAC Clyde Space which currently has several operational cubesats in orbit and has also worked closely with the UK space agency. As this part replaces the next two components as well, the backup part supplier and lead time for this component will be discussed at the end of the justification section.

The next component that the Electrical Engineering team chose for the Power Subsystem was the DC to AC inverter. The EE team decided on the 24V to 120V inverter made by and sourced from inverter.com. This fulfills several requirements such as the power conversion and input/output voltage requirement, but is lacking in several aspects such as TRL. On top of this, the company's lack of experience making products for the space industry, the products unproven reliability at low temperatures and the reputability of the company decrease the confidence the Electrical Engineering team has in this component. For these reasons this component will also be replaced by the STARBUCK-MINI from AAC Clyde Space as it fulfills the requirements and is produced by a more reputable and proven company.

Finally, the power supply that the system is using to deliver power is the MeanWell SPV-300 series AC/DC Power supply. This power supply fits all of the power requirements of the system but the selected company does not meet NASA expectations as it has not constructed any components for space and therefore there are several points where this product could fail such as vibration from launch, radiation exposure, and temperature resistance which would need to be tested. To remedy the extended certification process for this part the EE team will likely also switch this power supply in lieu of the STARBUCK-MINI from AAC Clyde Space as this company is reputable in the aerospace industry. This component has a lead time of 9-14 months however if this component cannot be sourced, there are two backup parts from different suppliers, the first is the SuperNova PCDU from Bradford Space which makes an almost identical product but may force a redesign of the battery pack as it only takes in power at 28 volts from this source. This company has a similar background to AAC Clyde Space and has experience similar to that of it as well. The second option for the previous three parts is to use the previously selected options. This is not a good option in terms of TRL and reliability but provides a nearly guaranteed delivery as they are all widely available commercial parts.

2.1.2.5. Power Subsystem Verification Plans

Table 10.1. Power Verification Plans

Req #	Requirement Summary	Verification Method	Rationale for Method	Preliminary Verification Plan
PS-01	The Power System shall be able to withstand the temperature range on the lunar surface. (90.15 K to 379.15 K)	Demonstration	The Power System should be able to work in the controlled temperature range for all sensitive components, and for all nonsensitive components it should be able to work outside of the controlled temperature range	PS-01 will be demonstrated by operating the system in a space-like environment at operating temperatures to verify this requirement. This will give the team a baseline for the system's performance at certain temperatures during the mission. A preliminary version of this demonstration will test its operation at three different temperatures, one at the lowest controlled operating temperature, one at a medium operating temperature, and one at the highest operating temperature for the system. these tests will involve the system charging, discharging, and operating at low and high battery levels. A failure of this demonstration will be a power drop that is below the required 275-watt supply for continuous peak operation.

PS-02	The Power System shall be able to withstand the environmental radiation on the surface of the Moon. (1,369 microsieverts per day)	Demonstration	The Power System should be able to withstand the environmental radiation found on the surface of the moon to ensure that all components of the system will be able to function for the duration of the mission.	PS-02 will be demonstrated by operating the system in a space-like environment at background radiation levels to verify that it will function in the mission environment. This will give the team a baseline for the system's performance at certain temperatures during the mission. A preliminary version of this demonstration will test its operation at three different radiation levels in series, one at the radiation level on Earth, one at the radiation level in transit, and one at the radiation level on the Moon. These tests will involve the system charging, discharging, and operating at low and high battery levels. A failure of this demonstration will be considered a power drop that is below the required 275-watt supply for continuous peak operation. The different tests in this demonstration will be performed one after the other to replicate the actual absorbed radiation level during the mission.
PS-03	The Power System shall be able to supply all power requirements of the spacecraft between solar charging intervals in standby mode. (1 week hold)	Analysis	The Power System should be able to keep the craft operational in PSR's for an extended period of time in order to ensure that the rover is able to complete the mission	PS-03 will be proven through an analysis of the power draw during standby mode with only the computer on. This will be calculated by finding the nominal power draw of the SBC that will be used in this rover and dividing by the watt hours of the rover's battery pack. This will give a time in hours which will then be divided by 24 hours in a day to give the final time in days that the rover can remain in standby mode without charging. A failure of this test will be considered a time to discharge the batteries from 100% to 10% which is less than a week.

PS-04	The Power System shall weigh no more than 20 percent of the total allotted kilograms. (17.1 kg)	Analysis	The Power System should not diminish the amount of weight that other systems are allowed to use in order to avoid weakening system too far making the reliability of the craft less	PS-04 will be proven by weighing the assembled system in its entirety including proper gauge cables to ensure that it does not reduce the allotted weight of other systems. A failure of this test will be considered any weight that is above 20% of 85 kg's or 17 kg without an exception from the team. if an exception is granted by the team this weight may be increased
PS-05	The Power System shall require no more volume than 20 percent of the total allotted craft volume. (0.675m ³)	Analysis	The Power System should not diminish the amount of space that other systems are using too far to ensure that the capabilities of other systems are not diminished to a point where they weaken the ability of the craft to function and uphold the mission	PS-05 will be proven by calculating the total volume of the assembled system in its entirety including proper gauge cables to ensure that it does not reduce the allotted volume of other systems. A failure of this test will be considered any weight that is above 20% of 2.25 cubic meters or 0.45 cubic meters without an exception from the team. if an exception is granted by the team this volume may be increased
PS-06	The Power System shall be no less than 75 percent efficient.	Analysis	The Power System should maintain as high of an efficiency as possible in order to ensure that it is not wasting power	PS-06 will be proven by analyzing the stated efficiencies of each component, notably the batteries and power supply to ensure that the amount of power generated by the solar panels is enough to effectively charge the rover during its mission cycle. A failure of this verification will be considered any combined efficiency that is less than 75% efficient.
PS-07	The power System shall be able to generate and supply 275 watts at peak rover operation.	Test	The Power System should be able to supply power to all of the components that require power simultaneously in order to ensure that it can complete tasks in a timely manner	PS-07 will be proven by conducting a test on the entire assembled Power System including electromagnetic interference testing, and thermal vacuum testing to ensure that the thermal and electromagnetic environment experienced on the Power System is within tolerance and will not cause the system to fail. A failure of

				this test will be considered any drop in power below the 275-watt threshold and any decreasing trend in performance that results in a drop in power below 275 watts before the end of the mission lifecycle.
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2.1.3. CDH Subsystem Overview

The Command and Data Handling subsystem is essentially the brains/nervous system of the spacecraft. This subsystem has several specifications that must be fulfilled by the commercial off the shelf (COTS) parts that are being selected. These specifications are found in section 1.5.5.1 of this report and include: survivability, communication with the rover from Earth, data storage, internal craft communication, weight specifications, and power draw. The pieces of equipment that will uphold these requirements are the central processing unit (CPU), communication antenna, communication receiver, data bus, storage, and onboard software. The final selection of these components must meet the list of requirements found in section 1.5.5.1 but also must be able to operate on the lunar south pole.

The components being used are as follows: Single-Board Computer: BAE Systems RAD5545, Transmitter and Receiver: BAE Systems Ka-band Multi Orbit LEO Optimized Antenna, Memory: 2GB Flash Storage, Data Bus: SpaceWire, Software: Linux OS.

The CDH subsystem interacts with every other subsystem and guides the spacecraft in its data collection. The CPU is the brain of the CDH subsystem, and is responsible for executing software commands and managing the data flow. The data bus is the device that exchanges data between the different sub-assemblies. The antenna is composed of a transmitter and a receiver. The transmitter will send data to the Lunar Reconnaissance Orbiter (LRO), which then sends it to Earth. The receiver will do the opposite by way of acquiring data sent from the LRO. The memory will store data and software instructions for the CPU to access at any time during the spacecraft's operation. The software consists of the different commands to be sent to the different subsystems and the science instruments in order to make the rover traverse the lunar surface, sustain operation, and properly send and receive data.

Data Flow

All of the rover's instruments and subsystems send data collected and operation statuses to the CPU via the data buses and software, and will receive

commands/instructions from the flight software. The CPU processes said data and writes it to the memory. The CPU will send data to the transmitter in order to send information to the LRO, and the receiver will send information to the CPU for processing.

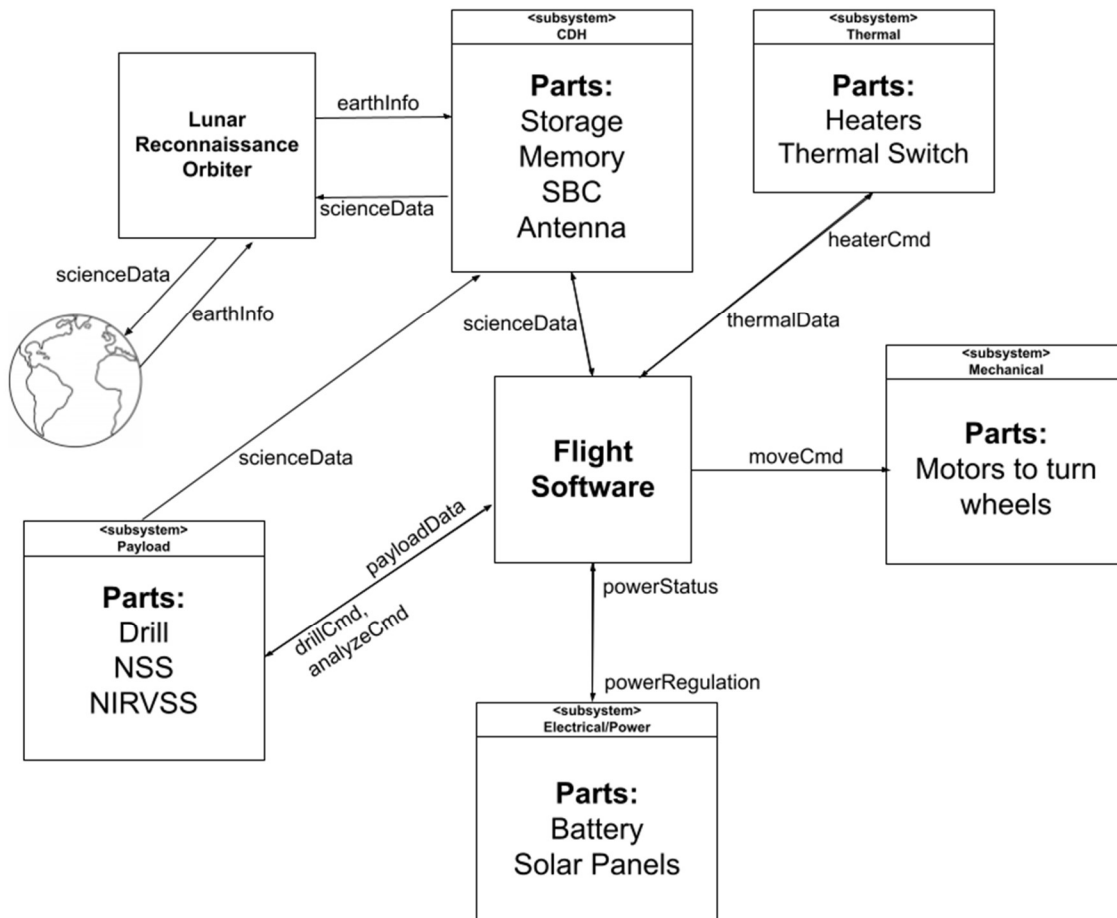


Figure 10: CDH Outline

2.1.3.1. CDH Subsystem Requirements

The command and data handling subsystem requirements are based on the ability for the various components needed from the command and data handling subsystem and the components and instruments from the other subsystems working together to send information in and out of the processor, storage, and the chosen communication method. The ability for the command and data handling subsystem creates its own requirements, as the components that create the ability to communicate need the ability to do so.

Table 11. CDH Subsystem Requirements

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Relevant Subsystem	Req met?
CDH-01	The CDH subsystem shall be able to withstand the lunar pole temperatures range of 90.15 K to 379.15 K	Temperature range is needed for operations to perform	TCS-1.2	N/A	Test	Thermal Control System	Met
CDH-02	The CDH subsystem shall be able to send communications to the primary mission orbiter.	Data must be available to NASA for observations	SYS-06	N/A	Test	Command & Data Handling	Met
CDH-03	The CDH subsystem shall be able to receive communications from a primary mission orbiter.	Rover must be able to be commanded by NASA when needed	SYS-06	CDH-03.1	Test	Command & Data Handling	Met
CDH-04	The CDH subsystem shall be able to store data and communications received	Allows for computation to happen	SYS-09	CDH-04.1	Test	Command & Data Handling	Met
CDH-05	The CDH subsystem shall be able to compute data onboard	Allows for data to be properly sent and reliable actions	SYS-09	N/A	Test	Command & Data Handling	Met

CDH-06	The CDH subsystem shall be able to use software to manage the various operations of the rover	Allows for all systems to work as intended	SYS-01	N/A	Test	Command & Data Handling	Met
CDH-07	The subsystems shall have a mass of no greater than 85 kgs.	System needs to be light enough to guarantee harmony with the other systems.	SYS-02	N/A	Inspection	System	Not Met
CDH-08	The CDH subsystem shall use no greater than 275 watts.	System needs to split power among the systems	PS-07	N/A	Test	Payload, Command & Data Handling	Not Met
CDH-02.1	The CDH subsystem shall be able to send and receive communications at a distance of 100km	Orbiter will be within this radius.	CDH-02	N/A	Test	Command & Data Handling	Met
CDH-04.1	The CDH subsystem shall be equipped with a storage device capable of performance that will not delay the rover	Storage must be large enough for all of the information.	CDH-04	N/A	Test	Command & Data Handling	Met
CDH-06.1	The CDH subsystem shall be able to act autonomously when communication is not	Allows for missions to PSR	0.1	N/A	Test	Command & Data Handling	Met

	available						
CDH-06.2	The CDH subsystem shall be able to send and receive communications to and from the other subsystems	CDH is responsible for making sure the other subsystems are functioning properly	CDH-06	N/A	Test	Command & Data Handling	Met

2.1.3.2. CDH Sub-Assembly Overview

Single-Board-Computer

The component chosen for this important part of the craft will be the RAD5545 CPU made by BAE Systems. This CPU was selected because it satisfies multiple roles in the CDH subsystem, such as the motherboard, CPU, and graphics card, decreasing the complexity of the assembly and physical failure points of the system. The RAD5545 is a single-board computer (SBC) with a combined system on a chip, effectively combining the motherboard, processor, and graphics card together which decreases the number of parts in the system and also simplifies the power supply for these components as it only needs one 5V connection to power the board rather than 3 separate power connection points for each of the subcomponents. On top of this, the power consumption for this board is below 30 watts at peak performance, out-matching other SBCs for a low power draw.

Besides the physical simplifications that make this board advantageous, it also outperforms in several other key areas. The first of which is its radiation hardness, while the RAD5545 may not be as radiation tolerant as other SBCs on the market, it far exceeds the radiation requirements for the environment on the Moon, with a specified radiation tolerance of 100,000 rads which exceeds the 0.139 rads of radiation received on the surface of the Moon each day and the total number of rad's received for 40 days which falls significantly below the 100,000 rad limit received for the mission timeline found in section 1.7.3.

There are several other characteristics that make this SBC advantageous compared to other options. This CPU operates at higher computation/clock speeds around 250 MHz compared to other SBCs which offers the advantage of faster processing times and quicker response times when processing data. In some ways this is not advantageous as higher clock speeds can increase the chances or opportunity for a bit to flip from a 0 to a 1 in a SEU as previously discussed, however depending on the error rate observed the CPU can be clocked down to decrease this error or kept at a higher rate to process more information faster offering more flexibility for the team depending on the amount of radiation hardening and processing power needed.

Data Bus

Outside of the CPU, an important component for internal data transfer and communication is the data bus. This component which makes up the wiring between the subcomponents of this system is necessary for the components to communicate with each other and operate effectively. For this purpose, the CDH team selected Space Wire as the data bus. There are several advantages to using space wire. Primarily the

radiation hardness that space wire provides is significantly greater than other options and supports a much better fault-tolerant network than other data bus options like MIL-STD-1553 wire or regular ethernet cable. This is advantageous to the use case because the lower error rates that space wire provides because of its greater radiation hardness allow the system to function with less error as a whole, thus increasing the clarity of data transfer and decreasing operating errors. Outside of this, the total power consumption of the space wire system is significant at around 50mW per driver/receiver pair compared to ethernet—which can use up to 30W for a driver/receiver pair. This makes space wire advantageous to the use case as it consumes less energy and therefore can support the same operations in a role where power consumption should be kept to a minimum.

Antenna (Transmitter and Receiver)

There are several other components that are necessary for the rover to uphold its requirements outlined in the SU 2024 mission task and in the CDH requirements section 1.5.5.1 of this document. Two important components that are necessary for communication to and from the rover are the communications antenna and the communications receiver. The communication antenna that was selected by the CDH team was the BAE Systems Ka-band Multi Orbit LEO Optimized Transmitter in a half-size array of 1 transmit antenna. This antenna was selected primarily because it has high rates of data transfer over the Ka-band and uses the same radio band as the LRO. The power consumption and weight for this transmission antenna are approximately 55 watts for transmission or reception at peak operation and 2.2 kg. This fits the needs of this mission well as it has a low weight and power draw compared to other options. On top of this, the transmitter is electronically steerable.

The second component that was selected for the communications array was the receiver, which the CDH subteam selected as the BAE Systems Ka-band Multi Orbit LEO Optimized Receiver in a $\frac{1}{3}$ size array of 2 receiving subarrays. These communication arrays were selected in the $\frac{1}{2}$ and $\frac{1}{3}$ sizes because the COTS components are size-optimized for LEO, coming in a large array for communication at distances of up to 800 km, thus exceeding the requirements for communication with the LRO which resides at 50 km above the lunar surface. Additionally, this array size fulfills both the transmitter and receiver requirements for the rover in section 1.5.5.1.

Memory

The last hardware piece of the CDH subsystem is the memory. For this system, the memory architecture that the CDH team chose was flash storage because of its low power requirements, superior memory stability, and vibration resistance in space environments. This makes flash memory a great option for the CDH subsystem and for

this rover. After deciding on the memory architecture the CDH team decided that the rover would have 500GB of data storage for images, sample and test information, and the operating system. This is because the craft will need to store information between transmission intervals with the LRO. This amount of storage will allow the rover to continue functioning when it is not in contact with the LRO and store information that is needed onboard at all times. On top of this, flash storage is not heavy, space-consuming, or energy-intensive therefore the memory size of the storage module will impact the craft very slightly. This means that the memory size can be much larger than what is actually needed and not drastically impact the weight or volume requirements for the system.

Software

The Operating System will be Linux, a well-established OS in space applications, such as the SpaceX Satellite network (Tung). On top of this, it is a commonly utilized OS in a multitude of non-space operations. Thus, there is a widely supported support network for both the Linux platform and related hardware components. Outside of this, Linux is very streamlined, making it much more power-efficient than similar Operating Systems while also maintaining its versatility, as opposed to more dedicated space operating systems. For instance, Real-Time Executive for Military Systems (RTEMS) is known as an operating system designed for spaceflight applications but is more strictly dedicated to rocket control. Because of these reasons, the CDH team chose Linux as the operating system for the rover.

TRL

To conclude the overview of the CDH sub-assemblies the team will discuss the TRL of all subcomponents and the whole system in conjunction with the weight, volume, and power requirements and specifications of the whole system. For the subcomponents of the CDH system, the team will start from the highest TRL and proceed to the lowest to discuss the strengths and weaknesses of the system. Starting off, the SBC motherboard and CPU has a TRL of 5 as it has been designed for use in a space environment and tested on Earth in a space-like environment. The 2GB of flash memory has been used in space extensively over the years and has therefore been flight-proven in a space environment. This earns it a TRL of 6. Moving on, the receiver, transmitter, and data bus have all been built for and tested in space environments. This earns the antenna a TRL of 6, and the data bus a TRL of 7, as SpaceWire has been used quite extensively in spacecrafts such as Gaia and ExoMars. Linux OS has been widely used in space applications, giving it a TRL of 7. Overall, this subsystem when all of its components have been combined will have a TRL of 5 as all of the components have been tested or proven in a space-like environment but the whole system has not been tested similarly,

therefore to improve the system's TRL, testing must be done in a lunar-pole-like environment.

Table 12: CDH Components

Component	Weight (kg)	Volume (L*W*H) (cm)	Power Requirements (watts)	TRL
Single Board Computer	0.24 kg	18.5 cm x 18 cm x 4.5 cm	30W max	5
Transmitter	7kg T+R	(10 cm x 20 x 5cm) T+R	55W upload when sending data to LRO	6
Receiver		(10 cm x 20 x 5 cm) T+R	65W download when receiving data from LRO	6
Data Bus	0.03 kg/m	N/A	N/A	7
Storage	0.03 kg	5 x 2.5 x 1 cm ³	Write: 0.0067 W Read: 0.0027 W Off: 0 W	6
Software	N/A	N/A	Low	7
Total:	7.57 kg	3507.8 cm ³	150.0094 W	5

2.1.3.3. CDH Subsystem Recovery and Redundancy Plans

The main computers can utilize error checking and error correcting algorithms to ensure proper computation and data handling. Similarly, duplicate computational systems can be used to ensure redundancy in data handling and processing. Systems, such as those indicated in the trade studies, can also be chosen for radiation hardness which will mitigate risk further. Communication issues can be mitigated by utilizing antennas which have been rated for space as well as incorporating systems that will attempt to make contact should communications be temporarily suspended. The size, weight, and cost of the antenna and transponder makes it unrealistic to include redundant components, so those will be documented as Single Point Failures should any malfunction occur.

2.1.3.4. CDH Subsystem Manufacturing and Procurement Plans

SBC

The single-board computer chosen is the BAE Systems RAD5545 Radiation-Hardened Processor. The supplier will be contractor L3Harris Technologies. L3Harris Technologies is a historically reliable supplier for space-grade CPUs, as NASA's perseverance rover is currently operating using a BAE CPU supplied by them. If procurement of the RAD5545 is not possible, the team's backup will be a RAD750 which is still compatible with all of the other components. An exact estimate of manufacturing time was not found, but according to multiple sources, radiation-hardened CPUs take a significant amount of time to be manufactured. In this case, it seems as if the RAD5545 will take around 3 months to procure. This is a safe estimation based on quote time, international shipping and potential delays, along with estimated lead times for the RAD5545's predecessor, the RAD750.

Data Bus

The data bus chosen is SpaceWire's data buses. The component will be purchased COTS, and the primary supplier will be STAR-Dundee. STAR-Dundee seems the most reliable out of the SpaceWire suppliers researched, and they possessed a handful of documentation regarding SpaceWire. The backup supplier is Cobham Advanced Electronic Solutions. Standard products' lead times are estimated to be around 4-12 weeks. Thus, 12 weeks is the baseline estimate for procuring a SpaceWire data bus.

Antenna (transmitter + receiver)

The antenna (transmitter + receiver) will be the BAE Systems Ka-band multi-orbit LEO optimized antenna. The supplier/COTS vendor for the antenna will be ViaSat. The backup supplier will be L3Harris Technologies. BAE systems have proven to be trustworthy and create well-performing radiation-hardened equipment. The estimated lead time when compared to similar antennas from different vendors is approximately 3 months.

Flash memory

The storage chosen is 2GB flash memory supplied by the contractor Phison USA. This company manufactured the storage on the Perseverance rover¹⁶. The company should be able to provide it completely radiation-hardened, and the team will be responsible for testing. The backup will be to purchase the memory from L3Harris technologies, as they are also proven to be trustworthy. The lead time for this was especially difficult to find, but it should not take longer than 3 months.

¹⁶ Mellor, "Fresh Details on the Flash That Sits inside Mars Perseverance Rover."

Software

The software being used is Linux OS. Linux is a commonly used operating system among different space technologies, such as the SpaceX Satellite network. NASA will need to program this software in-house. The estimated lead time for this is however long it takes the developers to program it. According to the National Academies Press it can take “up to 28 months”, which is almost the entirety of Phase C.

Note: the lead times are fairly rough estimates. It was very difficult to find credible sources online that could give lead times for even other models of each component. However, since they are all COTS, three months should be a generous estimate for shipping and delivery.

2.1.3.5. CDH System Verification Plans

The CDH system plans on verifying its components by testing them in order to be sure that they have operated on the moon. This is important as communication needs to occur when possible, as the rover will be in and out of the PSR.

Table 13. CDH Verification Matrix

Req #	Requirement Summary	Verification Method	Rationale for Method	Preliminary Verification Plan
CDH-01	Withstand lunar pole temperatures	Test	Ensure CDH subsystem can withstand lunar pole temperatures	Environmental Test, Ensure CDH subsystem can withstand lunar pole temperatures. Equipment: Thermal chamber, Temperature control system, Data logging equipment.
CDH-02	Send communications to LRO	Test	Verify transmitter function in lunar environment	Functional Test, Verify communication capability with primary mission orbiter. Equipment: Communication test equipment, Primary mission orbiter simulator, Data logging equipment.
CDH-03	Receive communications from LRO	Test	Verify receiver function in lunar environment	Functional Test, Verify communication capability with primary mission orbiter. Equipment: Communication test equipment, Primary mission orbiter simulator, Data logging equipment.
CDH-04	Store data and communications	Test	Measure storage speed	Performance Test, Measure data storage capacity and speed. Equipment: Data storage test equipment, Data logging equipment, Timing equipment.
CDH-05	Compute data onboard	Test	Measure computing speed and accuracy	Performance Test, Measure computing speed and accuracy. Equipment: Computing test equipment, Data logging equipment, Timing equipment.
CDH-06	Utilize software to oversee rover functionality	Test	Verify software functionality through testing	Software Test, Verify software functionality. Equipment: Software testing tools, Rover simulator, Data logging equipment.
CDH-07	Total mass of system not to exceed 85kg	Test	Verify subsystem mass is adherent to constraints	Inspection, Visually inspect subsystem mass. Equipment: Scale, Measuring tape, Data logging equipment.

CDH-08	CDH subsystem power draw $\leq$ 275 watts	Test	Measure power draw	Performance Test, Measure power consumption. Equipment: Power meter, Data logging equipment, Timing equipment.
CDH-02.1	Send and receive communications at a distance of 100km or less	Test	Verify communication at distances specified by mission task	See CDH-02
CDH-04.1	Storage device shall not commit errors or delays	Test	Measure storage accuracy	See CDH-04
CDH-06.1	Act autonomously when communication is unavailable	Test	Verify autonomous functioning	See CDH-06
CDH-06.2	Send and receive communications to/from the other rover subsystems	Test	Verify communication with other subsystems	See CDH-06

2.1.4. Thermal Control Subsystem Overview

The Thermal Control Systems (TCS) include two main controls: Passive Thermal Control and Active Thermal Control. Principles of the Passive Thermal Control is to prevent heat from entering and isolate heat from the spacecraft and the other components within the system. As stated in 7.2, the design methods include: (1) material selection is a critical aspect that meets thermal performance to regulate the spacecraft in harsh lunar conditions over the mission duration. (2) spacecraft orientation is needed during different mission phases and placed to withstand deep space. (3) thermal interfaces facilitate the transfer of heat from the relationship between the different parts of the spacecraft to ensure well functioning and stability. (4) circuit board design considerations to keep the heat away from the board.

Material selection is chosen to meet the critical and important requirements to ensure the best thermal performance for the spacecraft. Materials selected must manage and regulate temperature throughout the duration of the mission in the lunar environment.

Table 7-2: SmallSat Passive Thermal Technologies

Passive Technology	Description
Sprayable Thermal Control Coatings	Specialized liquid coatings applied to the spacecraft surfaces to manage and regulate the ratio of absorptivity and emissivity for optimal energy balance and thermal performance.
Films, Tapes, and MLI	Materials used to modify spacecraft surface property to manage and regulate temperature.
Thermal Straps	Provide a conductive link between a heat source and thermal sink to conductively transfer heat
Thermal Contact Conductance and Bolted Joint Conductance	Heat transfer by “contact” conductance between two surfaces pressed together by uniform pressure can be varied by using interface filler materials and conductive gaskets.

Sunshields	Deployed material that reduces the amount of incident solar flux on a spacecraft.
Thermal Louvers	Controls heat transfer between spacecraft surfaces.
Deployable Radiators	Dedicated surface for dissipating excess heat via radiative heat transfer.
Phase Change Materials	Energy is absorbed as material within a metal compartment changes phase because of exposure to a heat source.
Thermal Switches	An electromechanical device which controls the flow of electrical current in response to temperature change.

Figure 11: Table of SmallSat Passive Thermal Technologies

The components that were selected include the following: Sierra Nevada Thin Plate Heat Switch, Sprayable Thermal Control Coating (Socomore A276 Polyurethane) and Q-Rad Deployable Radiator. In these components, lithium-ion batteries are considered as they operate between -20 Celcuis to 60 Celcuis.

The Active Thermal Control system provides temperature regulation throughout all the different systems. However, unlike the passive systems, which rely heavily on thermal properties, the active system manages the temperatures of other systems.

Table 7-10: SmallSat Active Thermal Technologies

Active Technology	(Description)
Electrical Heaters	Space heaters use an electrical-resistance element in between two sheets of flexible electrically insulating material.
Cryocoolers	Miniature refrigeration system designed to cool components to extremely low temperatures.
Thermoelectric Coolers (TEC)	miniature solid-state heat pumps which provide localized cooling via the Peltier effect, which is cooling resulting from passing electric current through a junction formed by two dissimilar metals
Fluid Loops	System that circulates a working fluid through the spacecraft to a heat sink.

Figure 12: Table of SmallSat Active Thermal Technologies

The active technologies chosen include the following: Omega's KHLVA, PLM-Series,

Lockheed Martin Space's MICRO1-3 and the Laird Technologies' HighTemp ETX Series.

The three main modes of heat transfer are conduction, convection, and radiation. Due to the lack of atmosphere on the Moon, convection is not included on the diagram.

Assume steady state.

In the following diagrams Q_{cond1} is the conduction between the lunar surface and the rover's left wheels. Q_{cond2} is the conduction between the lunar surface and the rover's right wheels.

In the Hot Case:

$$Q_{\text{net in}} = Q_{\text{net out}}$$

$$Q_{\text{sun}} + Q_{\text{cond1}} + Q_{\text{cond2}} + Q_{\text{rad_moon}} = Q_{\text{rad_rover}} + Q_{\text{coolers_rover}}$$

Hot Case

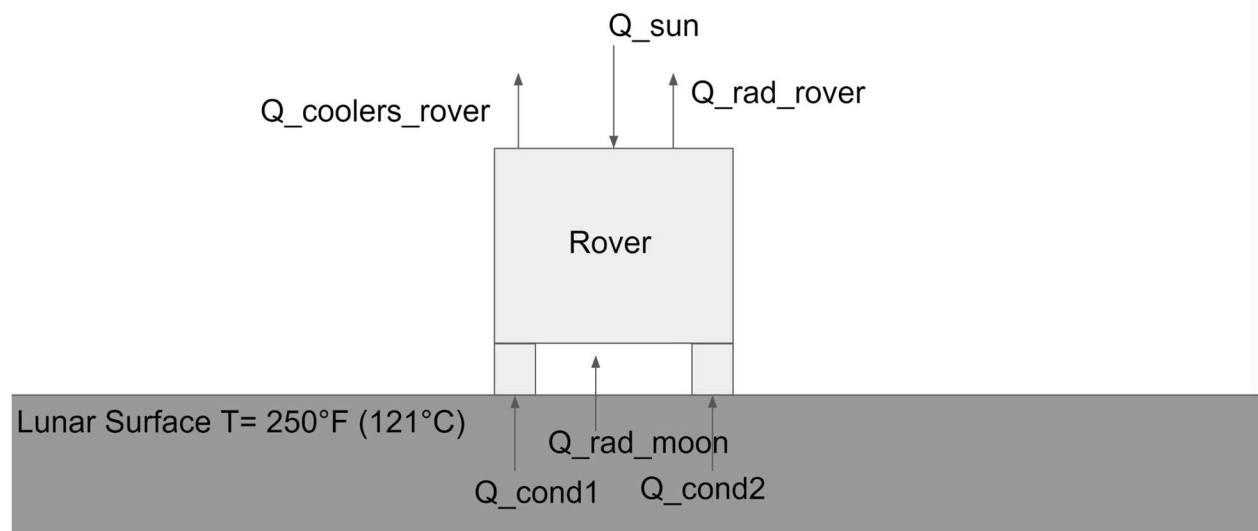


Figure 13: Thermal Diagram of the Hot Case

In the Cool Case:

$$Q_{\text{net out}} = Q_{\text{net in}} + Q_{\text{gen_net}}$$

$$Q_{\text{rad_rover}} + Q_{\text{cond1}} + Q_{\text{cond2}} = Q_{\text{gen_rover}}$$

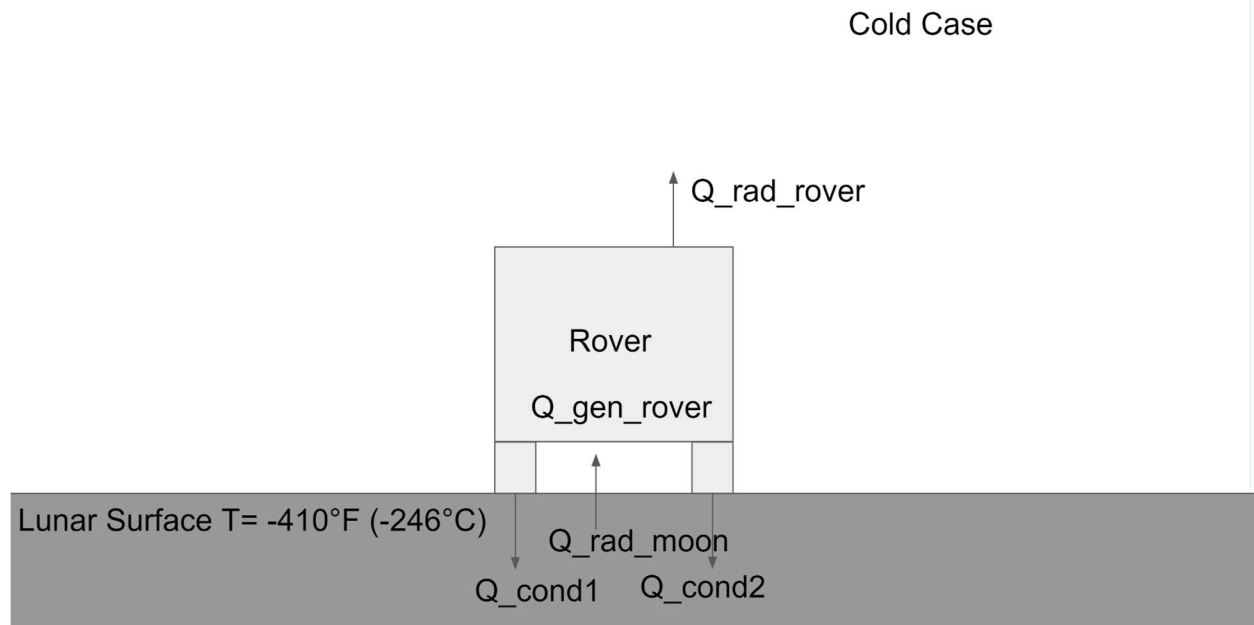


Figure 14: Thermal Diagram of the Cold Case

Note: $Q_{\text{gen_rover}}$ is heat generated by the thermal system, not heat from the environment.

2.1.4.1. Thermal Control Subsystem Requirements

The thermal management subsystem requirements aim to create an operating temperature window in which the rover will operate, allowing for an understanding of how all subsystems will behave in the thermal environment of the lunar pole and PSR, meaning that decisions regarding the rover can be made accordingly. The ability to regulate the rover's internal temperature despite changes externally is at the core of what the thermal management subsystem defines:

Table 14. Thermal Control System Reqs (TCS)

Thermal Control System Reqs (TCS)	Requirement	Rationale	Parent Req	Child Req	Verification method	Relevant Subsystem	Req met?
TCS-1	The system shall maintain an internal temperature of 20 ± 15 degrees Celcius at all times until the end of the mission.	Ensure systems are able to properly function.	MR-1	TCS-1.1, TCS-1.2	Test	Thermal Control System	Met
TCS-1.1	The system shall maintain the defined	Ensures system	TCS-1	N/A	Test	Thermal Control	Met

	internal temperature range when at max and minimum power.	functions properly at power extremes.				System	
TCS-1.2	The system shall maintain the defined internal temperature range in the coldest and hottest lunar surface temperatures (90.15 K to 379.15 K)	Ensures system functions properly at temperature extremes.	TCS-1	N/A	Test	Thermal Control System	Met
TCS-2	The system shall not cost greater than the amount of money allocated to the thermal system.	Ensures there is enough money to design, build, and test the system.	TCS-1	N/A	Analysis	Thermal Control System	Met
TCS-3	The system shall have a thermal model that shows net heat transfer equal to zero.	Ensures the system accounts for all forms of heat transfer throughout the design process.	TCS-1	TCS-3.1 TCS-3.2	Inspection	Thermal Control System	Met
TCS-3.1	The thermal model shall include the main forms of heat transfer in space, including conduction and radiation.	Ensures the system accounts for all forms of heat transfer throughout the design process.	TCS-3	N/A	Inspection	Thermal Control System	Met
TCS-3.2	The thermal diagram shall model extreme hot and cold cases.	The hot extreme will be at the hottest surface temperature and full power of components. The cold extreme will be at the coldest	TCS-3	N/A	Inspection	Thermal Control System	Met

		surface temperature with the system at low power. This ensures the thermal system accounts for the worst extremes.					
TCS-4	Exposed exterior components such as solar panels shall survive extreme lunar surface temperatures from 90.15 K to 379.15 K throughout the duration of the mission.	Ensures components not contained within controlled temperature in the instrument can survive extreme temperatures.	TCS-1	N/A	Analysis	Thermal Control System	Met
TCS-4	The internal thermal system shall not exceed internal structural dimensions.	Ensures the thermal system fits within volume constraints.	TCS-1	N/A	Analysis	Thermal Control System	Met

2.1.4.2. Thermal Control Sub-Assembly Overview

The thermal subsystem regulates the internal temperature of the rover to protect internal electronics from extreme lunar surface temperatures. The main components to mitigate extreme temperature fluctuation in the rover are a heater, a radiator, and a thermal switch. The heater will keep electronics from getting too cold in PSR and the radiator will keep the internals cool when in sunlight. The thermal switch automatically regulates the thermal system between temperature extremes by switching on the heater if the temperature is too low. The thermal switch is a passive device, allowing the system to adjust to the surroundings automatically. In addition to these components, a sprayable thermal control coating can be applied to the surface as a passive temperature regulator.

The heater chosen for the rover is the Omega KHLVA, PLM-Series. It comes in a variety of sizes ranging from 1-5 inches in Length and Width. All heaters are 0.01 in

thick and can be mounted on curved surfaces due to their flexibility. These heaters can operate in -57 to 232°C without adhesive and -40 to 149°C with adhesive.

The Diaphragm Thin Plate is the recommended device that allows the rover to switch a heat conduction path of strong and weak thermal coupling. Sierra Nevada develops Diaphragm Thin Plates that have a max conduction of 607 W/m²/°C and the minimum conduction of 7.1 W/m²/°C. The size is also relatively small, which becomes relevant to the design of the spacecraft itself. Compared to the alternative options of the thermal switches which are— a Passive Thermal Control Heat Switch and the HP Thin Plate, the Passive Thermal Control Heat Switch has a low max and minimum conductivity, however, the HP Thin Plate comes extremely close to a match in the trade studies. Both the Diaphragm Thin Plate and the HP Thin Plate have similar conductivity and volume.

Radiators within lunar exploration provide heat rejection for rovers or spacecraft for the experience through the lunar harsh atmosphere and environment. Active Thermal Architecture (Utah State University) allows flexibility of the design with the factors of versatility with the high performance deployable MFPL radiator. Unlike the Q-Rad and the Triangular Fins radiators, the Active Thermal Architecture's max TRL is higher with a wide range of operating temperature range and has is relevantly small.

A sprayable thermal coating is an additional option to aid the surface of the rover in radiating heat. While this paint can be expensive, its ability to passively radiate heat from the spacecraft helps protect the internal electronics from hot surface temperatures. Based on the trade study, the Socomore A276 Polyurethane white paint is the best option, mainly due to its lower cost. It was also rated as an easy to apply paint, which will shorten the manufacturing time.

2.1.4.3. Thermal Control Subsystem Recovery and Redundancy Plans

The thermal system is highly important for the transportation through the lunar's atmosphere— the temperature, the lunar dust and the harsh environment overall. The risk can lead to the system to overheat causing damage to other subcomponents to the spacecraft itself.

The thermal system can include multiple heaters for redundant heating. The heater chosen from the trade study is less than \$100, flexible, and small, making it an easy redundancy measure to implement. If one heater fails, the others should keep the internal components warm enough to function. To reject heat from the surface of the rover, both a radiator and thermal coating are used in the thermal system. The thermal control coating acts as a redundant backup in case it fails or is inefficient, though this is not likely.

2.1.4.4. Thermal Control Subsystem Manufacturing and Procurement Plans

The thermal system will be using the following components: heaters, radiators, a thermal switch, and a protective thermal coating. The heater used is an Omega KHLVA, PLM-Series. This is a flexible heater that comes in a variety of sizes. The chosen model is PLM-112/2-P heater which is 1 inch in width and 12 inches in length. The order time is 1 week shipped from India to the United States. The team chose Omega as the supplier because the website offers detailed information about the heater's shipping time, cost, and size. The shipping time can range from 1 week to 11 weeks, so the team chose a heater with the shortest shipping time.

The radiator chosen was the Q-Rad Deployable Radiator from redwire space, based in Jacksonville, Florida. The company has great heritage, with involvement in missions such as Artemis I and Mars 2020. This component is off the shelf, however, shipping time is unclear despite extensive research on both the company and third-party websites.

The Sierra Nevada Thin Plate Heat Switch passively controls the thermal system. It is situated between the electronics and radiator and increases conductance when electronics become too hot to dissipate heat into the radiator. Sierra Nevada Space Technologies was used because its components were low-cost and self-regulating. There are locations all across the US, which will decrease shipping time. The website does not have detailed information on manufacturing and shipping time, however, it can be estimated to take about a month. The components are manufactured and shipping time across the United States takes about a month at most.

The thermal protective coating chosen is the Socomore Aeroglaze A276 Polyurethane Coating in white. The manufacturer, Ellsworth Adhesives, ships the paint in 1-2 business weeks. The paint is already made so there is no manufacturing time. Buying the paint directly from Socomore is another option, however, the shipping times are unclear on the website.

The MICRO1-3 from Lockheed Martin Space satellites are known for their advanced technology and capability to support different lunar missions. Cryocoolers are specialized devices used to cool components to extremely low temperatures, but the MICRO1-3 generates less noise, maintaining a stable and low temperature that helps the spacecraft from temperature drift. Small standard models are \$10,000 - \$15,0000 from Lockheed.

Laird Technologies UTX Series (interface materials designed to provide heat transfer) are engineered to perform across a wide temperature range— typically up to 150°C or more. Depending on getting a unit of thermal pads or a roll can range from \$10-100 per unit.

2.1.4.5. Thermal Control Subsystem Verification Plans

Table 15. Thermal Control Verification

Req #	Requirement Summary	Verification Method	Rationale for Method	Preliminary Verification Plan
TCS-1	The system shall maintain an internal temperature of 20 ± 15 degrees Celcius at all times until the end of the mission.	Test	Putting the spacecraft in a simulated environment to see if thermal system adapts to changing temperatures.	Thermal Vacuum Chamber Testing (TVAC)
TCS-1.1	The system shall maintain the defined internal temperature range when at max and minimum power.	Test	Operate the spacecraft on minimum and maximum power using a prototype to simulate the thermal system.	Testbed in TVAC
TCS-1.2	The system shall maintain the defined internal temperature range in the coldest and hottest lunar surface temperatures (90.15 K to 379.15 K)	Test	Putting the spacecraft in a simulated environment to see if thermal system adapts to Extreme hot and cold temperature cases.	TVAC
TCS-2	The system shall not cost greater than the amount of money allocated to the thermal system.	Demonstration	If system costs significantly more than other subsystems, reevaluate components.	Budgeting sheet
TCS-3	The system shall have a thermal model that shows net heat transfer equal to zero.	Analysis	Using thermal laws, a diagram showing all the modes of heat transfer in the hot and cold cases.	Heat Flow Diagrams
TCS-3.1	The thermal model shall include the	Demonstration	Hot and cold case thermal diagrams include the forms of	Heat Flow Diagrams

	main forms of heat transfer in space, including conduction and radiation.		heat transfer.	
TCS-3.2	The thermal diagram shall model extreme hot and cold cases.	Demonstration	Hot and cold cases shall be shown on the the diagram.	Heat Flow Diagrams
TCS-4	Exposed exterior components such as solar panels shall survive extreme lunar surface temperatures from 90.15 K to 379.15 K throughout the duration of the mission.	Inspection	View the data sheet of the solar panels to see survival temperatures.	Component Data Sheet
TCS-4	The internal thermal system shall not exceed internal structural dimensions.	Analysis	In NX, import step files of each thermal component and ensure it fits with all other system components.	NX Assembly

2.1.5. Payload Subsystem Overview

There are two science instruments being used on the mission: an NSS and an NIRVSS.

The NSS detects the amount of hydrogen by volume through wavelengths within a 100 g sample of Lunar regolith. The NSS has a wavelength range of 230 - 615 nm, an integration time of 0.5 seconds, sensitivity of 10^{-2} to below 10^{-4} w/w, and a spatial resolution of $< 100 \mu\text{m}$. The instrument has two gas-proportional counter (GPC) sensors filled with helium to characterize the local thermal ($< 0.3 \text{ eV}$) and epithermal (0.3 eV to 1 keV) neutron environment. One GPC is surrounded in a layer of 0.63 mm thick cadmium to absorb any thermal neutrons. The difference between the exposed and Cadmium wrapped sensor measurements provides the thermal neutron rate. This rate gives information on the concentration of hydrogen present. During an ideal neutron detection case, the proton and triton lose all of their kinetic energy within the Helium gas, which produces a signal whose magnitude is proportional to the energy deposited in the helium gas. The sensors then pick up this signal and utilize this along with the thermal neutron rate to provide necessary information to accurately evaluate the amount of hydrogen within the 100 g sample of Lunar regolith.

The NSS will be in use while the rover is actively looking for trace amounts of hydrogen within the lunar surface and powered off when traversing to designated areas on the lunar landscape to prevent unnecessary power consumption. To achieve this, the NSS requires a power system supplying $28 \pm 6 \text{ V}$ during operation and 1.6 W of power, as well as a data processing system that can store information on all samples. The instrument weighs 1.9 kg . The operating temperature requirements of the NSS shall not exceed 60° C or go below -30° C , requiring a heating system and a cooling radiator system to maintain proper operating temperatures.

The NIRVSS detects the chemical composition of a sample of suspected hydrogen filled lunar regolith, whether that be water, hydroxyl, hydrogen atoms, etc. The NIRVSS has two optical engines, the NIRVSS SW and NIRVSS LW. The NIRVSS SW covers $\sim 1600\text{-}2400 \text{ nm}$ with a $\sim 9 \text{ nm}$ sampling, whereas the NIRVSS LW covers $\sim 2300\text{-}3400 \text{ nm}$ with a $\sim 12 \text{ nm}$ sampling. Each optical engine is attached to a fiber optic cable with the other end of each cable connected to the bracket assembly that is designed to view the surface underneath the RP rover. Also located on the bracket assembly, and essential for spectrometer operation in shadowed regions on the Moon, is the IR illumination source, located in the lamp assembly at the end of a Winston cone that directs the output energy toward the surface. The illumination source is an IR-emitter consisting of a tungsten filament located inside an industry standard TO-8 package. The TO-8 package has an internal gold plated parabolic reflector and contains an inert gas sealed inside by a sapphire window. The camera used on the NIRVSS consists of a 2048×2048 pixel complementary metal-oxide semiconductor sensor. The LCS consists of three thermopile detectors sealed with Xe gas inside an industry standard TO-5 package. The package contains an aperture with a ~ 60 degree FOV that is covered by various windows and filters. Filters isolating three wavelengths, 7.9, 10.3,

and 12.33 m, were selected to provide sensitivity to the 200-350 K range. Mounted on the front of the bracket assembly, the LCS FOVs cover the region underneath the rover that is illuminated by the lamp assembly and viewed by the drill observation camera, SW, and LW spectrometers¹⁷.

The NIRVSS will be turned off until a sample of lunar regolith has been taken. Once the sample is taken, the instrument will power on and perform its scanning of the sample. A power system will provide 29.6 W to ensure proper functioning. The instrument weighs 3.57 kg. There is a data processing system that will store all information gathered on all samples from the 16 different drilling locations. The operating temperature requirements of the NIRVSS shall not exceed 60° C or go below -30° C, requiring a heating system and a cooling radiator system to maintain proper operating temperatures.

2.1.5.1. Science Instrumentation Requirements

Table 16. Science Instrument Requirements

¹⁷ NASA, “Near-Infrared Monitoring of Volatiles in Frozen Lunar Simulants While Drilling”.

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Relevant Subsystem	Req met?
PAY-01	The subsystem shall operate/perform under extreme temperatures ranging from 90.15 K to 379.15 K	Achieving science goals relies on the successful operation of the science instrumentation systems	0.1, SYS-03, SYS-07	N/A	test	Thermal	met
PAY-02	The subsystem shall be mounted on the spacecraft where the line of sight is clear of any obstructions	To collect data efficiently the instruments will need an open view of the lunar surface	0.2	N/A	inspection	Mechanical	met
PAY-03	The subsystem shall have sufficient power to operate	In order to run the instruments power is required	SYS-07	PAY-3.1, PAY-3.2	test	Electrical	met
PAY-3.1	The NSS shall have 1.6 W of power to operate	Required power for hydrogen detection beneath the surface	SYS-07, PAY-03	N/A	test	Electrical	met
PAY-3.2	The NIRVSS shall have 29.6 W of power to operate	Required power for analyzing of lunar samples	SYS-07, PAY-03	N/A	test	Electrical	met
PAY-04	The NIRVSS SW shall cover ~1600-2400 nm with a ~9 nm sampling	Necessary parameter for proper hydrogen detection	SYS-05, SYS-09, PAY-08	N/A	test	Hardware, Mechanical	met
PAY-05	The NIRVSS LW shall cover ~2300-3400 nm with a ~12 nm sampling	Necessary parameter for proper hydrogen detection	SYS-05, SYS-09	N/A	test	Hardware, Mechanical	met
PAY-06	The subsystem shall be able to collect 100 g up to 1 m depth of lunar regolith	To interpret the abundance of volatiles under the surface, beneath surface	0.2, SYS-05, SYS-09	PAY-08	test	Mechanical	met

		samples will contribute					
PAY-07	The subsystem shall store all collected data for relay to the LRO	After samples and data has been analyzed, it must be relayed back to earth for analysis	SYS-05,SYS-06	PAY-08	test	Electrical, Computer, Hardware, CDH	met
PAY-08	The NSS shall be able to detect 5% hydrogen volume within a 100 g sample up to 1 m depth of lunar regolith	From initial constraints, hydrogen must be traceable up to 1 m below the surface	0.2, SYS-05,SYS-09, PAY-06, PAY-07	PAY-04	test	Hardware, Mechanical	met
PAY-09	The NIRVSS camera optic must be positioned near the sampling drill	Proper positioning assures accurate analysis of samples	SYS-05, SYS-09	N/A	inspection	Mechanical, Hardware	met
PAY-10	The NIRVSS shall be able to detect for trace amounts of hydrogen within the lunar sample	Accurate analysis of hydrogen content is crucial to success of science objectives		PAY-04, PAY-05	test	Hardware, Electrical	met

2.1.5.2. Payload Subsystem Recovery and Redundancy Plans

The Payload subsystem is composed of the NSS and the NIRVSS instruments and the relevant assemblies that are required for full operation. Due to the high importance of both instruments in the science objectives, there is no foreseeable reason for either to go offline. They work together, the NSS first and the NIRVSS second, for a total of sixteen different sampling sites. The high cost of the NSS and NIRVSS eliminate the possibility of adding a backup of each piece of equipment to the rover. The added weight would affect not only the presented launch parameters, but also the mobility and effectiveness of the rover once on the lunar surface.

In order to mitigate the disastrous effects of a payload subsystem failure, extra care will be taken in the design, building, and testing of the subsystem. Once installed on the rover, special reinforcements will be constructed to foolproof the integration of the instruments with the rover. A protective grate will be placed over the NSS which is positioned near the ground to detect hydrogen concentrations, protecting the instrument from unexpected terrain such as sudden drops and large boulders. The wiring harnesses for both instruments will be doubled in high risk areas to avoid any loss of power during operation in the harsh lunar environment. In the case that one is exposed to temperatures outside operating range or is cut due to excessive off road driving, there will be a backup to supply and receive necessary data and power.

2.1.5.3. Payload Subsystem Manufacturing and Procurement Plans

The science instruments selected for the mission's tasks include two instruments, which will be COTS instruments because of their high TRL and success on similar missions.

The first instrument selected to achieve the scientific goal is the Neutron Spectrometer System (NSS), which will be contracted through the company Astrobotic. This company has been entrusted to develop technology for the Moon to Mars plan along with the Jet Propulsion Laboratory and Arizona State University¹⁸. The estimated lead time including manufacturing and shipping for this is 3-4 months.

The second instrument of the two instrument's max payload will be the Near Infrared Volatiles Spectrometer System (NIRVSS), which will also be contracted through Astrobotic to develop this science instrument. This instrument was also chosen to be onboard VIPER; however, the mission was recently canceled. This was also a mission to the south pole to investigate lunar volatiles. One option would be to reuse VIPER's

¹⁸ Mitchell, Audra. "Astrobotic Awarded

NIRVSS¹⁹. Shipping time would be 2-3 weeks. Alternatively, Astrobotics would manufacture and ship a new NIRVSS, which could take up to 3 months.

The team is aware these instruments were originally designed to be used in the Peregrine 1 Mission for NASA's Commercial Lunar Payload Services (CLPS). Though this mission had a successful launch, the lander lost control due to a propellant leak. However, the Peregrine mission was also a lunar mission, so the instruments were made and tested for the same environment that they will be used for in this mission. There are no backup suppliers for these specific instruments as they were designed by Astrobotics. Alternative spectrometers can be found. However, this would cause a delay in the schedule.

2.1.5.4. Payload Subsystem Verification Plans

Table 17. Payload Subsystem Verification Matrix

¹⁹ "TO20A (Viper) Payloads - NASA Science."

Req #	Requirement Summary	Verification Method	Rationale for Method	Preliminary Verification Plan
PAYV-01	To operate and survive the components of the payload subsystem must withstand temperatures ranging 90.15 K to 379.15 K	test	To prepare for the extreme temperatures on the Moon the subsystem needs to handle a spectrum of these temperatures	Due to the unique temperatures of the Moon, materials of each of the components that make up the payload subsystem will be tested for hot and cold extremes
PAYV-02	To align with mass constraints the payload subsystem will have an allotted amount of mass	inspection	Launch provider has allotted a finite amount of mass	Inspections will occur throughout the development of the payload subsystem to ensure mass constraints are not exceeded
PAYV-03	To align with cost constraints the payload subsystem will have an allotted amount of funds	inspection	Client has allotted a finite amount of funding for the payload subsystem	Inspections will occur throughout the development of the payload subsystem to ensure cost constraints are not exceeded
PAYV-04	The NSS and NIRVSS systems must work cohesively with necessary Power, CDH, and Thermal subsystems	test	To achieve science goals, all payload systems must be fully operational	Inspections will occur throughout development ensuring parameters are met for both instruments. A comprehensive test will take place upon completion of construction of rover
PAYV-05	Ensure NSS can properly detect 5% volume of hydrogen in a 100 g sample	test	Required amount of hydrogen for the power consumption and time taken to be worth it to drill out and inspect samples	Several rounds of tests will be conducted when the instrument is finished, each time with different amounts of hydrogen present to ensure proper detection

PAYV-06	Ensure NIRVSS can detect proper characteristics of hydrogen concentration from sample	test	Required function for the power consumption and time taken to be worth it to drill out and inspect samples	Several rounds of tests will be conducted when the instrument is finished, each time with different hydrogen forms to ensure proper detection
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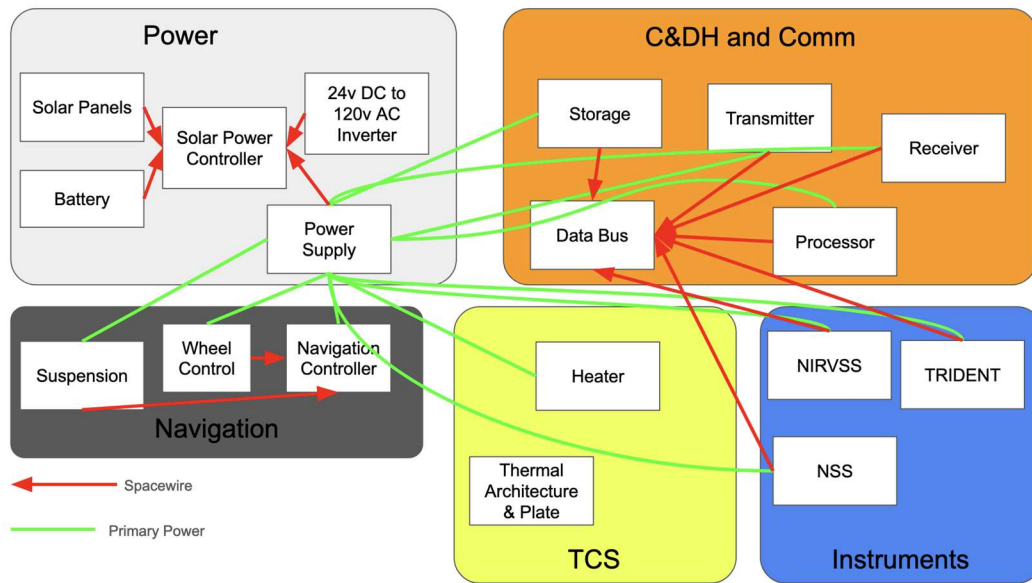
2.2. Interface Control

The interface between payloads is based around communicating only the necessary data from each subsystem. The data handling will communicate to all other subsystems, and all subsystems will communicate to the data handling subsystem. This creates the basis for how the communication will work, as most communications will happen this way. Furthermore, power is given to all other subsystems, though this should be obvious because all subsystems need power for at least one component or instrument. This structure creates a simple way of communicating, meaning that everything runs in and out of data handling, and power flows out of that subsystem.

Table 18. N² Diagram

Payload	↗ instrument info				
⊖ instrument commands	Data Handling	↘ instrument data, rover telemetry	↘ location to go to	↘ needs of subsystems temperature	↘ when to get more power
	⊖ comms to rover	Communications with LRO			
	⊖ telemetry		Navigation and Motion Control		
⊖ when samples are collected, specific thermal	⊖ current temperature			Thermal	
⊖ Power for instruments	⊖ Power for computation and processing	⊖ Power for transceiver	⊖ Power for motors and attitude determination sensors	⊖ Power for sensors and heating elements	Power

Figure 15. Block Diagram



3. Science Mission Plan

3.1. Science Objectives

The first science objective is to collect 100g of ice samples at one of the 13 Artemis III landing site locations and determine the composition of the sample. This science objective was chosen in alignment with the Artemis III goal to understand the changes volatile materials undergo when they are altered and transported. Specifically, the purpose of this objective is to determine if there is any loss of water during the sample collection process. To achieve this, the Neutron Spectrometer System (NSS) will be used to take a spectroscopy reading of a 3 cm x 3 cm area to determine the percent hydrogen. This instrument was selected because of its accurate measurements for unclosed, small areas. Then, a drill will be used to break up the regolith, as it would be for a 100g sample collection. After that, the NSS will take another reading of the same area and record the percent of hydrogen in the sample. The two readings will be compared to determine if the sample collection process has a significant effect on the hydrogen percentage and ice water abundance in lunar regolith in or near the PSR. This process will be repeated for all 16 science locations.

The second science objective is to determine the water ice abundance of at least one of the 13 Artemis III landing site locations near or within a PSR within the top 1 meter of regolith. This objective was provided to the team as a secondary objective. Using the Near Infrared Volatiles Spectrometer System (NIRVSS) at 16 science locations, wavelengths of 400-680 nm will be measured to determine the hydrogen percentage which can be translated to ice water abundance by scientists. This instrument was chosen because it can measure surface and subsurface water²⁰.

The third science objective is to identify areas of higher ice concentration within an Artemis III landing site. The processes and instruments used to achieve the previous objectives will result in ice water concentrations at 16 different science locations. The NIRVSS will also track environmental properties such as temperature and elevation to better understand the data. The ice water concentrations will be plotted in a 4x4 color coded grid to illustrate areas with the highest ice water abundance. Future manned missions to the Moon can use this information to strategically plan sample collections or to gather large quantities of ice to be used for power, oxygen, and drinking water.

3.2. Experimental Logic, Approach, and Method of Investigation

Once the rover is deployed on the surface of the Moon (T+1 Day), the rover will run a "life check" for all subsystems. This begins with sending a signal back to scientists if the computer is on. Then, the computer will do preliminary checks for each subsystem. For thermal, the temperature of the inside will be checked to ensure that all internal components are within the operational temperature range. The power system will ensure all components are receiving power and solar panels shall be deployed to

²⁰ Colaprete, Anthony. Near infrared volatile

test solar charging. Both the NSS and NIRVSS will take spectroscopy data at the landing site to ensure functionality and calibrate the instruments. The mechanical system will check the drill functionality, along with the driving controls, to ensure the rover can move in any direction. After completing the systems life check, the communication system shall send a report back to scientists on Earth, with a detailed report. Each subsystem will have a status of ONLINE, OFFLINE, or ERROR. Any system with an error will include a report detailing the issue, such as an offline heater. In the case of an error, redundancy and recovery plans shall be implemented to prevent mission failure. This concludes the system life procedure. The following steps will be repeated for 16 science locations in the Hawthorn region. The 16 locations are marked in white.

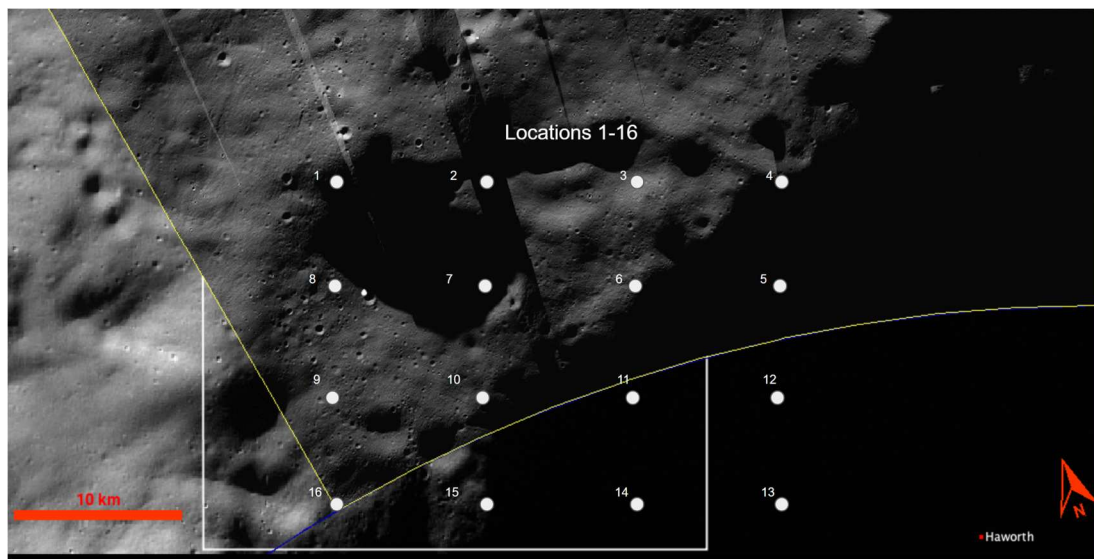


Figure 16: Hawthorn Locations

From T+2-25 Days, the rover goes into Driving Mode. Depending on the science location, terrain, and amount of daylight the rover has access to, the travel time will change. In Driving Mode (Day), the rover will be exposed to light and the solar panels can recharge the battery. The rover will drive at a higher speed. When the rover is in Driving Mode (Night) the rover will drive at a lower speed to conserve power. The thermal system shall use more heaters to keep temperatures within the operational temperatures despite extremely low atmospheric temperatures. The mechanical system shall take a route to avoid large boulders, steep inclines, and other obstacles. During the driving time, the GPS system will keep track of the location and the computer system will periodically send life checks to the satellite. When the rover arrives at the science location, NIRVSS will be turned on and run its calibration procedure. After, it will take multispectral images to measure wavelengths of 250-600 nm to determine ice water abundance for Science Mode Objective Two. After the data is stored in the hard drive, NIRVSS will be put in standby mode. The focus is data collection for Science Objective One. The NSS will be turned on from standby mode and run its calibration procedure and take a spectroscopy reading of 250-600 nm wavelengths of a 3x3cm² area. While this is ongoing, the drill will be turned on. The

rover will position itself so the drill is directly over the same 3x3cm² area. The area will be drilled as if to collect 100g of lunar regolith samples. There was originally a plan to include a sample collection procedure for later missions to recover collected sample capsules. However, the team reviewed mission requirements and science objectives and determined that sample collection was not necessary. Treating the lunar regolith with a sample collection process to observe changes in hydrogen composition would be sufficient to accomplish mission goals. Sample collection would require additional mechanisms such as a sealable capsule and scoop that were out of scope for this mission. Returning to the drilled sample, the rover would reposition itself so the NSS is in the same position as it was when reading the undrilled sample. The spectroscopy process will be repeated and the data will be stored in the hard drive. All data saved to the hard drive will automatically be titled with the following template "DAY_TIME_LOCATION_TEST TYPE_TEST NUMBER_INSTRUMENT NAME". NSS will be put in standby and the rover will return to a sunlit region to recharge and send data back to scientists in Data Transfer Mode. When the rover has completely recharged, it will repeat these steps for all 16 science locations. Throughout the mission, life checks will be sent back to Earth when a satellite is in range. After data from all locations are sent to Earth, data from NSS and NIRVSS will be used to complete science objective 3 because the instrument measures environmental factors such as temperature and surface reflectance. NSS data will be used to verify these readings. NIRVSS data will be used to compare concentrations of hydrogen on a 40 x 40 km² grid. The three locations with the highest concentration of hydrogen will be identified as the locations with the highest concentration of lunar ice volatiles. At this stage, the mission is complete.

3.3. Payload Success Criteria

The NSS and NIRVSS work together to collect data from 16 science locations. However, the mission will not fail if the rover fails to collect data from all 16 locations. Science objectives 1 and 2 need one location at the very least in order to accomplish the respective objectives' goals. Science objective three aims to compare data from locations to determine the one with the highest ice water abundance. The more science locations the rover visits, the more data to compare. While not preferred, visiting only 2 locations would provide enough data for comparison. The payloads must collect data from at least 2 locations for the mission to be a complete success. The following table is an overview of the instruments and criteria needed to achieve all three science goals.

Table 19. Payload Criteria

	Instruments	
Success Criteria	NSS	NIRVSS
Detect wavelengths 250-700 nm	X	X
Collect temperature data		X

Collect reflectivity data		X
Can measure as low as 0.5 wt% water from H wavelengths	X	

The Neutron Spectrometer System (NSS) fulfills half of the payload criteria. It can detect 250-700 nm wavelengths, like the NIRVSS. What differentiates it from the other spectrometers is it can detect small traces of water, as low as 0.5 weight percent water from Hydrogen wavelengths. “The instrument works by making measurements to determine the local thermal (<0.3 eV) and epithermal (0.3 eV to 1 keV) neutron flux, which can be used to determine the local hydrogen content.”²¹ If the other instrument were to fail, the mission would still succeed as long as the rover reaches at least two science sites. Again, while it is not ideal, these two sites would still help scientists determine a location with higher ice water content.

The Near Infrared Volatile Spectrometer System NIRVSS fulfills three of four payload criteria. It can measure 250-700 nm wavelengths, though not to the same level of precision as the NSS. What makes this instrument unique is its ability to measure environmental data, such as reflectivity and temperature. These two criteria are valuable in creating context behind the spectroscopy data. If this instrument were to fail, the NSS would still be able to collect all the data required from at least two science locations. However, it would be difficult to verify the data and understand the environmental factors that affect it.

3.4. Testing and Calibration Measurements

Calibration methods for the Neutron Spectrometer System (NSS) have been developed in the lab setting in preparation for the VIPER mission. The lab test simulated flight testing by using a neutron-moderating pile (NMP) of reactor grade graphite to create planetary-like neutron spectra.²² An NMP is required for lab testing because cosmic-ray protons are deflected by Earth’s magnetic field. The Moon does not have a significant magnetic field so these cosmic rays do reach the surface of the Moon. The same lab procedure can be used with actual cosmic ray protons instead of simulated ones and the lunar data can be compared with the lab data. For this procedure, the NSS is positioned 12 inches above the surface. In the lab, “The NSS measured count rate versus time for each configuration, along with sample spectra... These measurements provide an indication of typical NSS measurements. For example, moving from configuration #1 to #4 is analogous to moving from hydrogen-poor to hydrogen-rich (>100s of ppm H) materials.”²³ Data collected on the Moon will be calibrated with the benchmark data from the lab. Additionally, a white calibration panel can be placed in range of the NSS. The white panel will have a known spectral graph, and the in-situ spectral image taken by the NSS of the panel will be used to

²¹ Elphic, Richard C. Neutron Spectrometer System

²² Peplowski, P. N., R. C. Elphic

²³ Peplowski, P. N., R. C. Elphic

offset the spectral curve to ensure the data accurately represents the regolith's composition.

The NIRVSS can be calibrated by taking a spectral image of a white calibration panel with a known composition. This panel will be attached to the lander at a point where the NIRVSS can take an image. The experimental spectral curve will be compared with the known spectral curve to calibrate the instrument. The NIRVSS can also be calibrated by taking a spectral image of the same area the NSS took a spectral calibration image. The spectral curve shall be compared using a computer algorithm. While these are two different instruments, the composition measured from a common target should be the same. A secondary calibration method can be introduced as well for NIRVSS temperature reading. A Long-wavelength Calibration Sensor (LCS) can be used to document surface temperature²⁴ to calibrate the NIRVSS temperature readings.

3.5. Precision and Accuracy of Instrumentation

The VIPER mission wrote a technical report called VIPER: PATHFINDING IN-SITU RESOURCE UTILIZATION, which includes a slideshow outlining the specs of the NSS and NIRVSS.

The Neutron Spectrometer System (NSS) has an accuracy of 5-10% absolute.²⁵ The data collected should be assumed have a margin of error of 10% or +/- 5%. This error difference is most important to consider when comparing the spectroscopy data for science objective 1. The spectroscopy data taken for an undrilled sample will have a margin of error of +/- 5% and the drilled sample will also have a +/- 5% margin of error. This means that if the detected hydrogen percent difference between these two samples is less than 10%, the difference is not significant. Any difference less than 10% could be instrumental error.

The Near InfraRed Volatiles Spectrometer System (NIRVSS) has an accuracy of radiance measure less than 25%²⁶ and it can be assumed the spectroscopy accuracy is similar to NSS, which is 5-10% absolute. The NIRVSS is the primary instrument for science objective 3, so this margin of error of 10% or +/- 5% should also be considered when analyzing the data. Science objective 3 aims to determine areas of greater lunar ice by comparing hydrogen concentrations between science locations. Similar to the logic for science objective 2, only locations with greater than 10% hydrogen difference should be deemed to have a difference in ice abundance.

3.6. Expected Data & Analysis

The NSS uses neutron spectra to measure thermal flux. The NSS operates by measuring and displaying data in the form of EJ versus Energy graphs. To create these

²⁴ Roush, Ted L., Anthony

²⁵ Andrews, Daniel. "VIPER

²⁶ Andrews, Daniel. "VIPER

graphs, the “NSS measures pulse height spectra that are used to derive counting rates in the bare and Cd-wrapped neutron sensors.”²⁷ From there, “...the local thermal (<0.3 eV) and epithermal (0.3 eV to 1 keV) neutron flux, [which] can be used to determine the local hydrogen content.”²⁸ Neutrons will reflect energy differently based on the composition of the material and the velocity and angle at which the neutrons hit and interact with the material. The process of translating the flux to hydrogen abundance “...requires iterative comparison between measured and simulated neutron fluxes. The simulations include detector response information, specifically how likely a neutron is to produce a signal in the sensors as a function of its energy and sensor-incidence angle.”²⁹ As shown in Figure 17, simulated data shows the difference between the resulting neutron flux of a lunar sample (right) versus a radioactive neutron source (left), which is often used as a baseline for testing neutron sensing instruments.

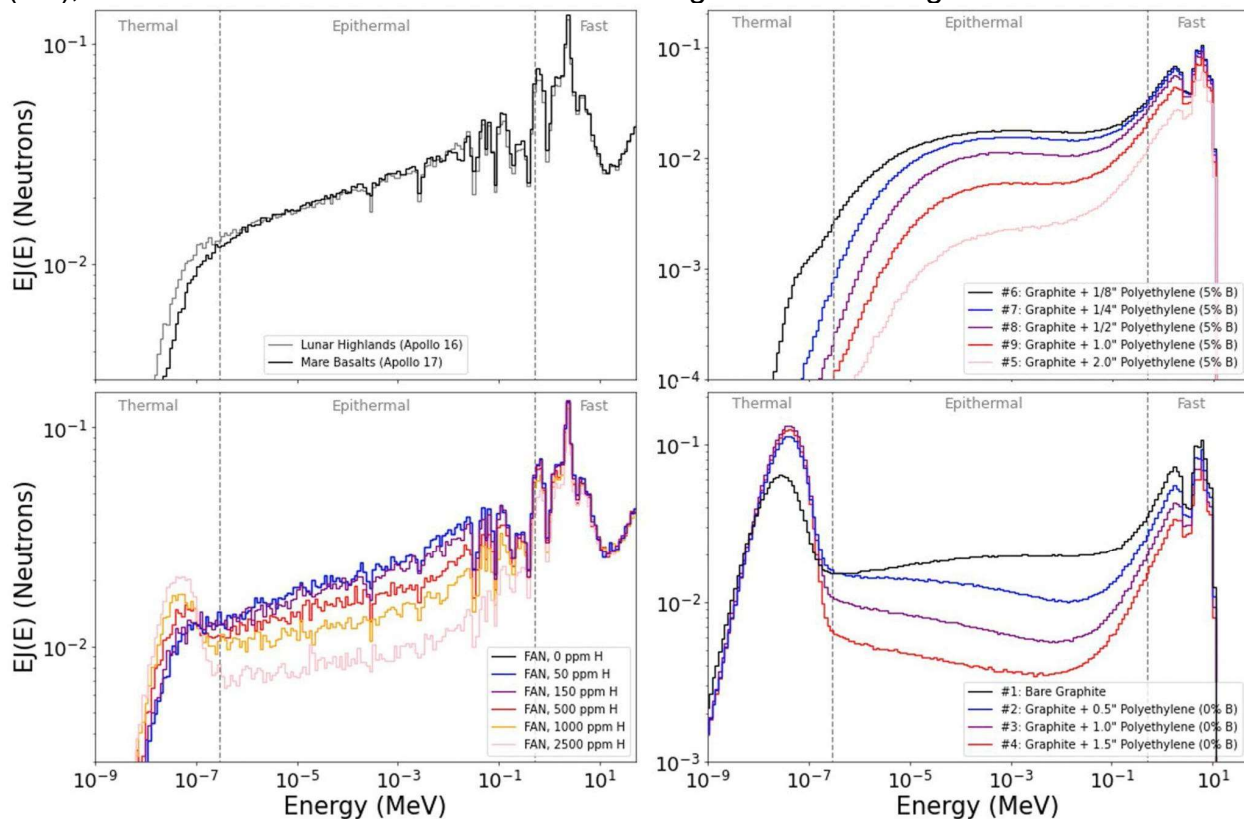


Figure 17: Geant4-simulated neutron emission spectra for simulations of cosmic-ray-induced neutron production on the Moon (right) and from the APL neutron pile with an AmBe source (left).³⁰

From these graphs, the detector response will be compared with neutron fluxes to create Counts versus Time graphs. These show change in epithermal neutron rate. As shown in Figure 18, the Cd-wrapped sensor of the NSS measures a decrease in

²⁷ Peplowski, P. N., R. C. Elphic

²⁸ Elphic, Richard C. Neutron Spectrometer System

²⁹ Peplowski, P. N., R. C. Elphic

³⁰ Peplowski, P. N. Calibration of Nasa's Neutron Spectrometer System

epithermal neutron rate over time. A high Count represents a hydrogen lacking region whereas a low Count corresponds to a hydrogen abundance.³¹

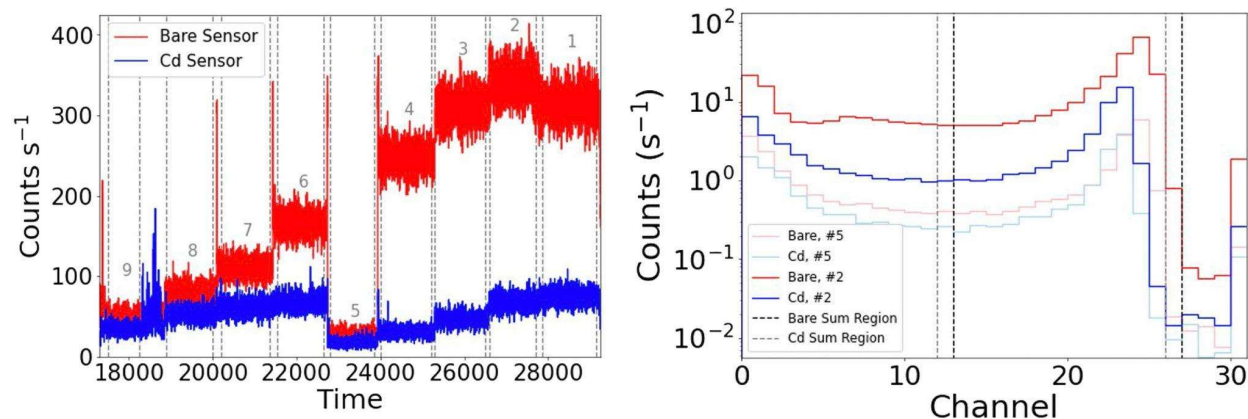


Figure 19: (Left) Time-series data of the bare- and Cd-covered sensors for each pile configuration (numbered; see legend).

Figure 20: (Right) Sample spectra from two of the pile configurations, with dashed lines indicating the summing region used to derive the neutron capture peak rates.³²

This relationship will be used to compare hydrogen content of a sample before and after drilling. If there is a greater than 10 percent difference in epithermal neutron rate from before drilling the sample to after drilling the sample, it shall be determined that this sample collection process has an effect on the ice water abundance in lunar regolith. While the data collected from this instrument will also be used in science objective 3, determining areas of high ice concentration across the different science locations. Each science location will be visited at a certain time. When observing drops in the Counts versus Time graph, it can be noted that locations corresponding with these drops have higher hydrogen and ice water abundance, given that these differences are greater than 10%. This will be used to verify the science objective 3 data taken by the NIRVSS.

The NIRVSS uses data from absorption, emission, and reflectance data to analyze the composition, thermal, and physical properties of a sample. NIRVSS spectroscopy analyzes the change in energy of electrons instead of protons. When an electron absorbs energy from light, it will jump different energy levels based on the wavelength of light it absorbs. The JWST uses spectroscopy to determine the compositions of bodies lightyears away, with hydrogen being the most abundant element in the universe. According to the JWST website, “Electrons can also lose energy and drop down to lower energy levels. When an electron drops down between levels, it emits photons with the same amount of energy—the same wavelength—that it would need to absorb in order to move up between those same levels. This is why hydrogen’s emission spectrum is the inverse of its absorption spectrum, with emission lines at 410 nm (violet), 434 nm (blue), 486 nm (blue-green), and 656 nm (red). The highest energy

³¹ Peplowski, P. N. Calibration of Nasa’s Neutron Spectrometer System

³² Peplowski, P. N. Calibration of Nasa’s Neutron Spectrometer System

and shortest wavelength light is given off by the electrons that fall the farthest.”³³ By observing either absorption or emission versus wavelength graphs, the composition of a sample can be determined. Absorption is observed by dips in the graph, as shown in Figure 21, a spectroscopy graph from JWST of a bright star called Altair. Scientists can observe hydrogen content just from looking at the graph, because each element behaves differently when exposed to certain wavelengths of light. This is shown in Figure 22, where sodium, nitrogen, hydrogen, and oxygen absorb or emit unique patterns of light.

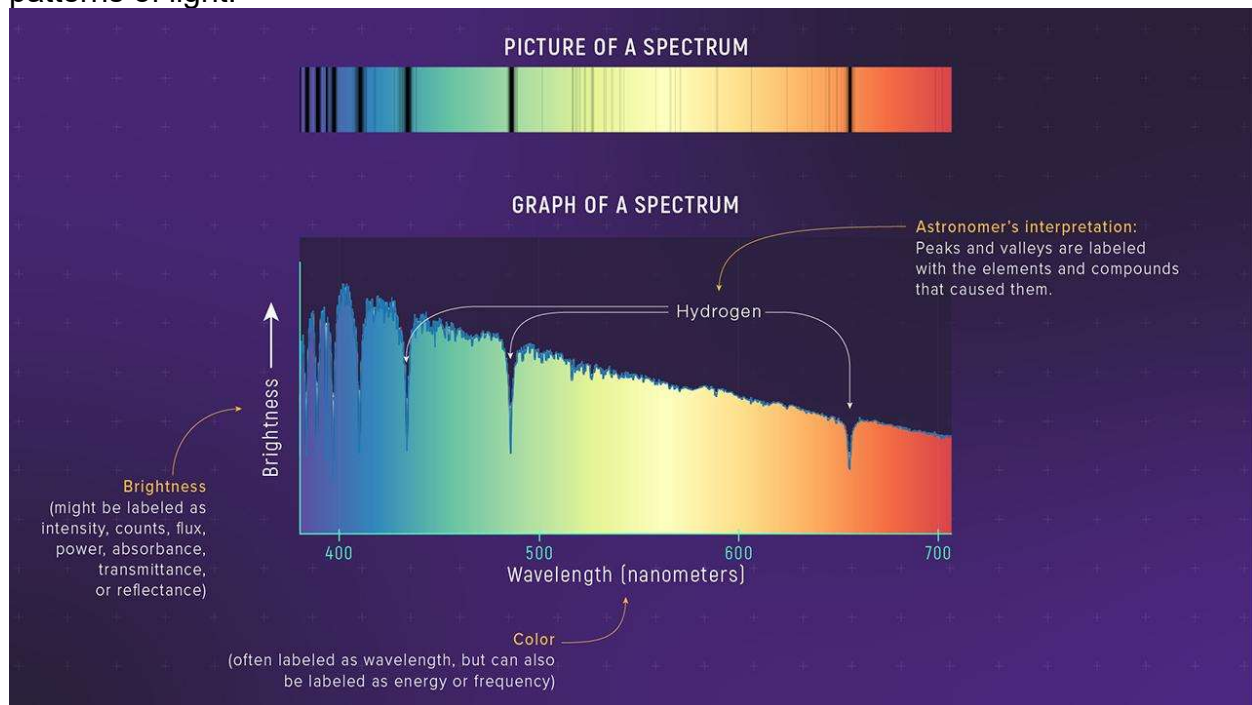


Figure 21: Spectroscopy Graph of Star Altair from JWST³⁴

³³ Spectroscopy 101 Webb

³⁴ Spectroscopy 101 Webb

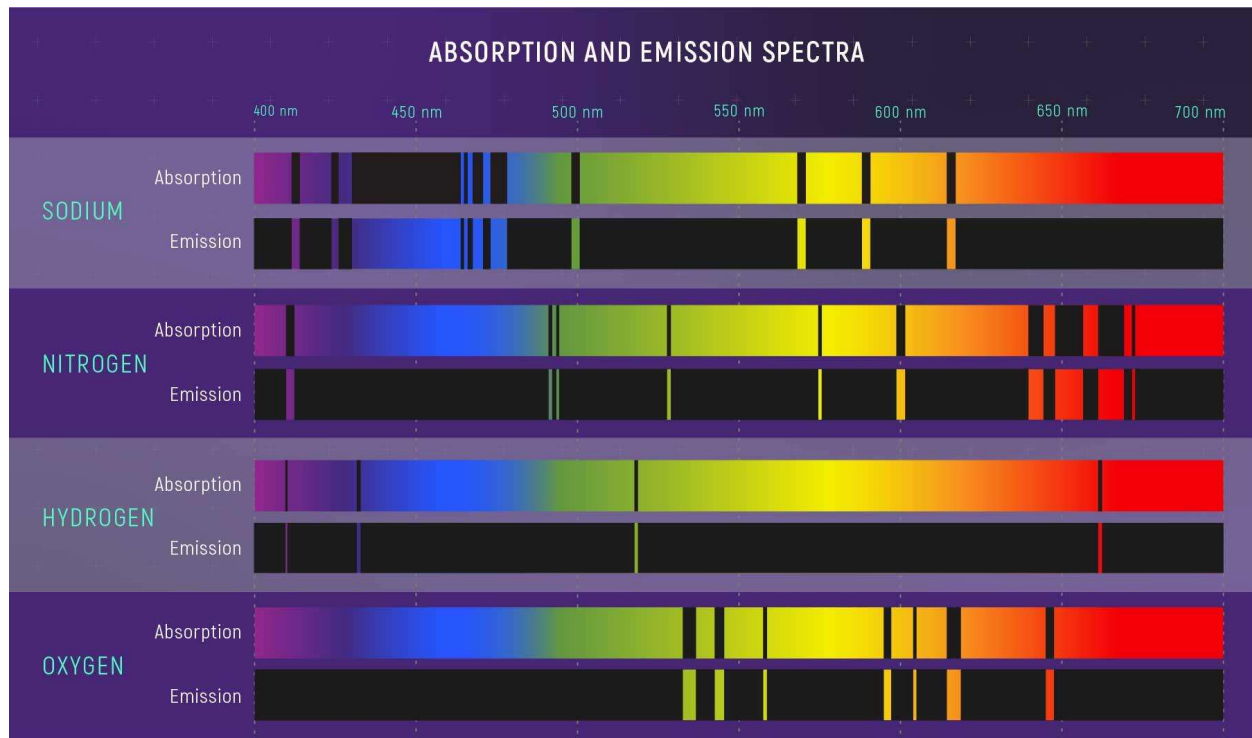


Figure 22: Absorption and Emission Spectra from JWST³⁵

Simulated testing with the NIRVSS has already been conducted. As shown in Figure 23 different, two different icy regolith simulants were measured by the NIRVSS. The unique changes in the curves over different wavelengths represents the different elements present in the sample.

³⁵ Spectroscopy 101 Webb

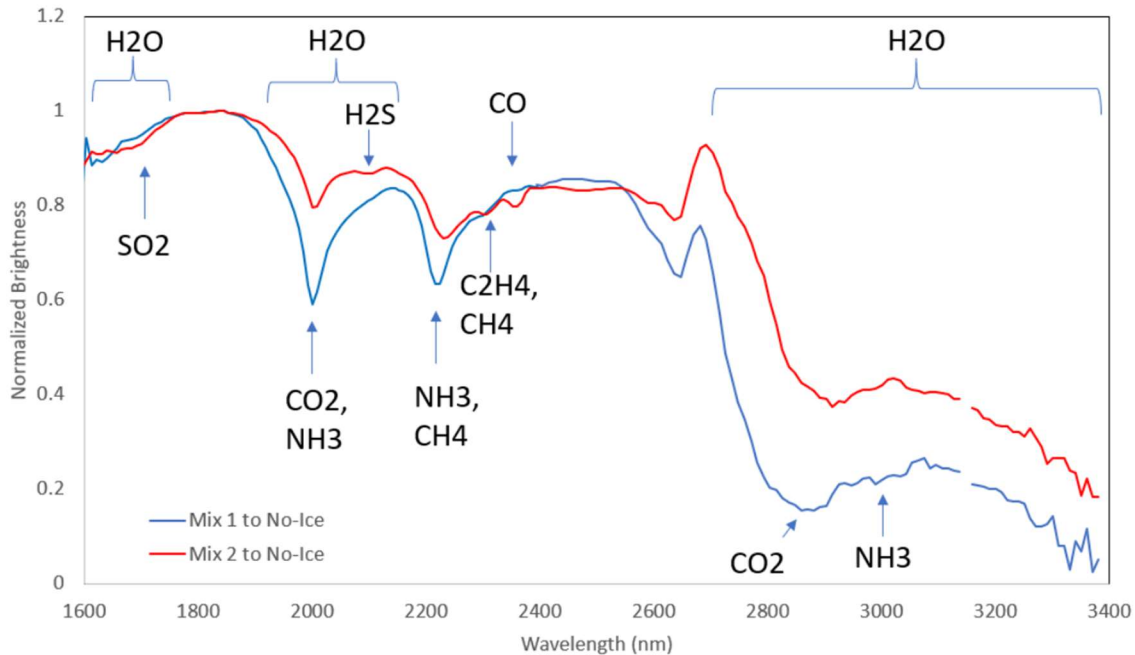


Figure 23: Initial NIRVSS Spectral Data on the two versions of OPRFLCROSS icy regolith simulant³⁶

While this test focuses on detecting H₂O in the samples, the NIRVSS will be used to detect hydrogen wavelengths in the range of 400-680 nm in the top 1 meter of regolith. This data will be used for science objective 2, determining the percent concentration of hydrogen within the top 1 meter. To account for experimental error, percent hydrogen should include +/- 5%. From there, ice water abundance can be calculated. The NIRVSS is the primary instrument for science objective 3. Along with absorbance, emission, and reflectivity data, the NIRVSS records the temperature throughout the lunar day and night. This additional data will provide context to the water concentrations found across the different science locations. Not only will areas of high ice water concentrations be found, but scientists will have an in depth understanding of how the lunar time of day affects this concentration. Locations with a greater than 10% hydrogen concentration shall be defined as a significant difference. Data from the NSS will validate the results from NIRVSS.

³⁶ Roux, Vincent Testing NIRVSS

4. Mission Risk Management

4.1. Safety and Hazard Overview

To effectively manage risks, the team has implemented several risk management strategies that begin with risk identification in research and development, where all members contribute to potential risks. Identified risks are categorized and labeled through the use of a risk chart and risk matrix, which helps in prioritizing based on likelihood and consequence. The chart also depicts the team's approach and if there are mitigation tactics in place, the risk is being watched, or the risk is acceptable. A trend analysis is also updated showing how the status of each risk changes over time and whether they are increasing, decreasing, or remain unchanged.

The highest priority risks are determined from their combined likelihood and consequence ranking and move on to the FMEA for further analysis. Using the FMEA chart the team can systematically quantify and prioritize risks based on their probability of occurrence and potential impact. Prevention tactics and further steps are then provided to mitigate these severe risks.

Continuous reevaluation is a critical component of the team's risk management. This involves regularly considering risks and updating the charts to incorporate new information and adjusting mitigation strategies as needed. Clear communication ensures that all team members are aware of the risks and the steps being taken to manage them. With these strategies in place, the team can prevent magnification of risks and their associated issues, allocate resources more effectively, and maintain the mission schedule.

4.1.1. Risk Analysis

Each subsystem has its respective risks. With that, there are also external mission risks that cannot be prevented. These risks must be accounted for and addressed because if left unmanaged, issues can compound, causing potential mission failure. The risks themselves need to be categorized and labeled, allowing the team to understand the level of severity that each risk poses. Each risk is quantified by likelihood and consequence. This way, proper resources can be committed to each issue, and the risk can be fully understood and mitigated.

The team must continuously reevaluate risks to consider additions and stay on top of potential issues as the team gains more information and knowledge. This allows the team to remain diligent and mitigate further issues. This is a part of the team's CRM strategy and helps RIDM as well.

Table 20. Risk Summary

PDR Risk Summary							
ID	Summary	L	C	Trend	Approach	Risk Statement	Status
1	Mechanical	2	5	↑	M - redundancy with 6 wheels, better wheel design	Given the terrain may be rocky and harsh, the rover's wheels could potentially fail or become damaged, impeding the rover's movement or immobilizing it, potentially halting the mission.	Wheels were remodeled thinner to save weight
2	Mechanical	2	5	↓	R - Research deployment methods and implement a durable design	Given the mechanical and electrical intricacies of the solar array's deployment mechanism, the arrays may fail to deploy, reducing available power, and causing a premature mission failure.	An alternative actuator has been chosen and some deployment procedures have been researched
3	Power	3	5	→	M - High-Capacity Batteries with 31 hours or reliable runtime. Optimized route planning minimizing time in PSRs. Energy conservation modes to limit non-essential activities	Given the rover's exposure to prolonged shadow periods, solar cells could be kept in a PSR too long, depleting the power supply, and causing mission failure.	No updates
4	Power	2	2	→	M - Panels can be folded and stowed away in the rover's chassis	Given the presence of lunar dust, dust may accumulate on solar panels, impeding recharging capabilities, decreasing the efficiency of the solar system.	CAD model has been updated to reflect folded panel design

5	CDH	3	5	→	M - Suitable radiation hardened components have been chosen for CDH	Given the high radiation levels in space, radiation may cause damage to the semiconductor, leading to corrupted data or component, instrument, or vehicle loss or damage, and mission failure.	Components exceed necessary radiation protection levels, although the camera and motors do not have radiation shielding at the moment
6	CDH	2	4	↓	M - The CDH subsystem shall be able to act autonomously when communication is not available. Communications receiver exceeds the LRO communication distance requirement by 16 times	Given potential communication failures, the rover could cease communication, disrupting mission control, and causing mission failure.	No updates
7	Thermal	1	4	→	M	Given the extreme lunar temperatures, the thermal system could fail to prevent overheating or freezing, leading to performance failures, and impeding the functionality of the rover.	Active
8	Thermal	2	5	→	M	Given the varying lunar environment, a thermal switch may fail to regulate temperatures, leading to system damage, impeding the rover, or causing failure.	Active

9	Instrumentation	2	4	→	M - The rover is capable of operating autonomously in the event of communication loss and the communication receiver can operate within 800 km of the LRO	Given the LRO's large flight path and the communication dependency of the rover, unexpected re-directions may occur, causing communication loss, and significantly delaying mission progress.	No updates
10	Program matics	3	3	→	M	Given team and objective needs, a new person may be needed to fill an administrative role, leading to mission delays, impacting the mission schedule.	Active
11	Program matics	3	3	→	M	Given the broad range of mission activities, an injury may occur to personnel, affecting availability and team dynamics, delaying mission activities, and affecting the mission schedule.	Active
12	Planetary Protection	3	3	→	M	Given the risk of contamination, the rover could contaminate the lunar surface, violating planetary protection protocols, compromising scientific integrity.	Active
13	Software Reliability	3	5	NEW	R	Given the complex nature of system software, critical bugs or failures may occur, leading to system malfunctions, affecting system performance and potential mission failure.	Active
14	Supply Chain	4	3	NEW	W	Given global supply chain issues, important components may be delayed or unavailable, delaying mission activities, affecting mission schedule, and increasing costs.	Active
15	Micromet	1	5	NEW	A	Given potential micrometeorite impacts, the rover	Active

	eorite Impacts					may be damaged, impacting functionality, compromising the mission.	
16	Political/R egulation Change	2	3	NEW	W	Given potential political and regulatory change, mission approval may be affected, causing delays, affecting mission schedule, and increasing costs.	Active
17	Data Integrity	2	5	NEW	A	Given cybersecurity threats, data may be corrupted or stolen, compromising the mission, causing delays, and affecting mission schedule	Active

L = Likelihood (1-5)		LxC Trend		Approach		Criticality
1 = not likely		↓ - Decreasing (improving)		A - accept		HIGH
5 = extremely likely		↑ - increasing (worsening)		M- mitigate		MED
C = Consequence (1-5)		→ - unchanged		W - watch		LOW
1 = low consequence		NEW - added this month		R - research		
5 = high consequence						

Figure 24. Risk Matrix

L I K E L I H O O D	5								
	4			14					
	3			10,11,12				2, 3, 5, 13	
	2		4	16	2, 9			1, 8, 17	
	1				7			16	

1 2 3 4 5

CONSEQUENCES

4.1.2. Failure Mode and Effect Analysis (FMEA)

The Failure Modes and Effects Analysis (FMEA) chart is the team's structured approach for addressing and preventing critically recognized risks. It breaks down each risk from its cause to its effect across the whole system and the mission along with specific prevention tactics and actions if the issue persists. Each failure mode is analyzed for severity, occurrence, and detection, leading to specific actions to mitigate risks, such as redundant systems, robust testing, and improved materials.

Table 21. FMEA Chart

Function	Failure Mode	Effects	Sev	Cause	Occ	Prevention	Det	RPN	Actions
Power	Solar cells fail to charge batteries.	System loss of power resulting in mission failure.	10	Timing the rover's charging times incorrectly could cause the rover to be stranded in a shadowed region without the ability to recharge using solar panels.	3	Keeping track of the battery drain from collecting & sending data, driving, and keeping the whole system powered. Planning driving routes with charging stops and time margin to reach a charging spot during the day.	9	270	If this failure continues, the number of science locations can decrease and increase the number of charging locations. Redundant batteries can be used, however, they carry extra mass.
Mechanical	Solar arrays fail to deploy.	System loss of power resulting in mission failure.	10	Onboard mechanical systems fail to deploy correctly causing the solar arrays to be unusable.	5	Testing deployment procedure in a Moon-like environment and ensuring robustness.	8	400	Implementing redundancy into the system to ensure the rover has some level of recharge capabilities.
CDH	Radiation exposure leading to data and equipment malfunction.	Could permanently damage some or all electronics.	8	Not testing the effectiveness of the radiation shielding could cause charged particles to disrupt the electronic system.	7	Use a test bed to ensure radiation shielding design prevents electronic damage.	4	224	Radiation is always present, so include redundant radiation shielding or a backup CPU.
	Radiation exposure leading to data malfunction.	Cause interference with data transfer, or could cause stored data loss before data transfer.	7	In the time between data collection and data transfer, radiation exposure that is damaging to electronics could also corrupt the data stored in the rover's drive.	5	Reduce the time between data collection and transfer. Add extra radiation protection around the rover's storage drive.	3	105	Back up data into a secondary drive. Add extra radiation shielding to the hard drive. Coordinate data collection timing with the satellite's data transfer range timing to reduce the time the data is stored.

Mechanical	Wheels of the rover wear over time in lunar terrain.	Science objective 3 cannot be accomplished if the rover is unable to travel to multiple locations.	8	Using a testing environment that does not accurately represent the terrain and environment of the Moon. Not being able to test the wheels for 3 years.	6	Use wheel design and material with heritage, especially 3+ years on the Moon.	9	432	Reduce the number of science locations. Increase the number of wheels (most likely not possible due to mass). Increase wheel durability.
Software	Critical software fails leading to system malfunction.	Could eliminate the use of critical components or the rover as a whole.	10	Software bugs or unexpected scenarios that were not properly accounted for.	4	Code redundancy to ensure robustness in onboard rover software. Broad testing scenarios to account for all possible situations.	8	320	Critical failure reset and bug awareness procedures.

4.1.3. Personnel Hazards and Mitigations

Personnel hazards in the workplace, particularly in high-risk environments such as those involving machinery and hazardous materials, require specialized care and awareness strategies to ensure safety. The Occupational Safety and Health Administration (OSHA) provides regulations and guidelines to manage these risks, while the Office of Safety and Mission Assurance (OSMA) ensures that safety protocols are adhered to. These administrations and mitigation techniques help the team to uphold high safety standards and protect the health of all parties involved with the mission.

Table 22. Personnel Hazards

Hazard	Mitigation
Debris/parts flying into eyes	Safety glasses with side shields must be worn
Clothing, jewelry, and hair can get stuck in machinery or parts	Do not wear loose clothing, loose neckwear, or exposed jewelry while operating machinery. Pull back and secure long hair
A serious injury occurs and there is nobody around to help	Do not work alone. Use the buddy system
Heavy object falls on foot	Wear strong close-toed shoes at all times
Sharp machinery, moving objects, and tools can cut and cause serious injury	Keep hands and other body parts away from moving machine parts, workpieces, and cutters that you are not using. Use tools for their designed purpose only
Machinery causes injury because safety mechanisms are not in place	If guards or safety mechanisms are present do not remove or disable them. Ensure all machinery has appropriate guards in place to prevent accidental contact. Regularly inspect guards for damage or wear
Inexperience with a machine poses a large risk	Do not operate any machine unless authorized, and do not set up or operate machinery if you are not trained and familiar with that setup
Attempts to clean and perform	Do not attempt to oil, clean, adjust, or

maintenance on machinery cause injury	repair any machine while it is running. Ensure the machine is in the "OFF" position. Use a brush to remove short, discontinuous types of chips--not hands, fingers, or rags
Injury due to working/moving equipment	Do not leave the machine until it has stopped running. Do not leave a machine until it has come to a complete stop
Emergency stop injury	Do not try to stop the machine with your hands or body. Press the emergency stop button and wait until the machine has come to a complete stop to proceed
Injury due to dull or broken tools	Check tools and machines before use to ensure they are safe to use. Always use correct speeds and feeds
Parts or tools become loose and become projectiles	Always check work and cutting tools on any machine are clamped securely
Injury due to carelessness	Concentrate on the work and do not talk unnecessarily while operating the machine. Make sure you are working with a calm and clear mind
Gloves get caught in rotating machinery and cause serious injury	Always remove gloves before turning on or operating any machine. If material is rough or sharp and gloves must be worn, place or handle material with the machine turned off. Never wear gloves or use rags to clean the workpiece or any part of a machine that is running. Rotating tools or parts can grab gloves and rags and pull you into the machine.
Cuts and injuries from chips and shavings	Never handle chips with your bare hands or fingers. Chips are extremely sharp and can easily cause cuts
Strain or injury from lifting heavy objects	Lift heavy objects with your legs and not your back. Additionally, ask for help
Hazardous walking environment caused by chips and debris	Floors, machines, and other surfaces should be kept free of wood and metal chips, dirt, and debris. Proper footwear

	should also be used
Flammable and hazardous materials	Flammable and combustible liquids such as oils, gasoline, cleaners, solvents, paints, and thinners should be stored in UL-approved liquid storage cabinets. Chemicals should be stored in cabinets approved for that use. Do not store incompatible chemicals together. Safety data sheets should be kept in the shop area for all chemicals used
Fire hazards from hot work areas	Hot work areas, such as welding and grinding areas, should be separated by at least 35 feet from flammable and combustible materials. Sparks should be controlled through the use of screens, if necessary
Risk due to improper signage	Emergency contact numbers are posted, hazard signs are posted and current, any additional warning signs are posted, and all containers are clearly labeled
Chemical or foreign body in eye	Emergency eyewash and shower are available and clearly labeled where chemicals are stored and used
Electrocution and electrical hazards	Ensure all electrical equipment is properly grounded. Inspect electrical cords and plugs for damage before use. Do not overload circuits. Use Ground Fault Circuit Interrupters (GFCIs) where necessary. Ensure that only qualified personnel perform electrical work
Excessive noise and hearing damage	Hearing protection is provided in high-noise areas.
Ergonomic hazards	Ergonomic workstations are provided to reduce repetitive strain injuries.
Chemical exposure	Proper ventilation systems are used to reduce inhalation hazards and personal protective equipment (PPE) is available in chemical areas

Radiation hazards	Proper shielding and dosimeters are used to monitor radiation exposure. Only trained personnel handle radioactive materials. Appropriate warning signs are posted in areas where radiation is present
Laser hazard	Appropriate eye protection is worn when operating lasers in controlled areas. Lasers have proper warning labels
Slips, trips, and falls	Walkways must be kept clear of obstacles and spills. Anti-slip mats are placed in areas prone to slip conditions. Proper footwear is used
Heat-related hazards and exhaustion	Adequate hydration areas and rest breaks for employees are provided, especially for those working in hot environments. Temperature and humidity levels are monitored. Use fans or air conditioning to cool work areas
Biological hazards	Proper sanitation and hygiene practices are used. Training is provided on handling biological materials. Availability of personal protective equipment such as gloves and masks is ensured
Fire and explosion hazards	Fire extinguishers are maintained and are easily accessible in all areas of the facility. Regular fire drills are conducted and employees are trained

5. Activity Plan

5.1. Project Management Approach

For all phases, management will include a program manager, deputy program manager of resources, chief scientist, and lead systems engineer. These will perform administrative tasks over their assigned purview over the duration of the mission. Similarly, each subteam will have a team lead assigned to overview all tasks assigned to that team. Each team will be allocated their own budget and purview, but direction will

be given for all subteams by the program manager, deputy program manager of resources, chief scientist, and lead systems engineer.

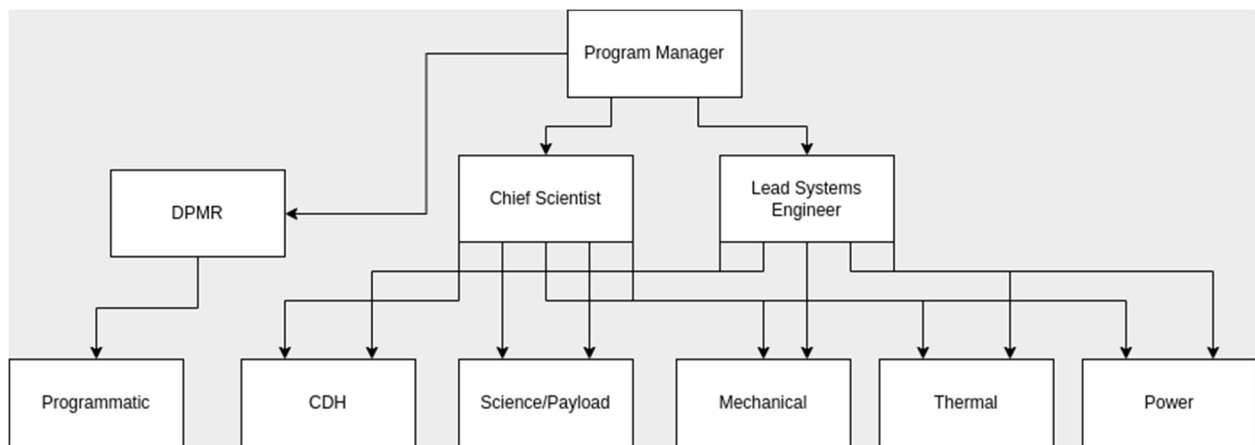


Figure 25. Full Team Organizational Chart

Phase C

The Command and Data Handling (CDH) team will comprise 6 engineers responsible for designing and implementing the crucial systems that manage data onboard the spacecraft, supported by 4 technicians who assist in hardware integration and testing, along with 2 scientists who provide scientific oversight to ensure data integrity and processing accuracy.

Simultaneously, the Science/Payload team, pivotal for the mission's scientific objectives, includes 4 engineers focusing on the integration and optimization of payload systems, 2 technicians responsible for hands-on assembly and testing, and 5 scientists who bring specialized knowledge in fields such as astrogeology and planetary science, contributing to the mission's scientific discoveries and data analysis capabilities.

The Mechanical team, which will focus on the spacecraft's structural and mechanical integrity, comprises 4 engineers who design and validate mechanical systems, 4 technicians who execute fabrication and assembly tasks with precision, and 2 scientists who ensure that mechanical designs meet stringent performance criteria and environmental conditions.

The thermal subteam will consist of 2 design engineers responsible for thermal system design and analysis, 2 assembly, integration & test engineers who oversee the integration of thermal hardware, 3 technicians who will conduct meticulous testing and calibration, and 1 scientist who provides expertise in thermal management and heat dissipation strategies.

Lastly for phase C, the power team will include 2 engineers specializing in power system design and optimization, supported by 2 technicians who handle power system integration and testing, ensuring the spacecraft's energy needs are met under varying operational conditions.

Thus, 45 people: 20 engineers, 15 technicians, and 10 scientists are needed for phase C of the mission.

Phase D

The CDH team will continue with 6 engineers overseeing the integration and testing of data handling systems. They are supported by 4 technicians who ensure seamless assembly and integration of CDH hardware components. Additionally, 2 scientists provide oversight to validate data processing algorithms and maintain scientific integrity.

Within the Science/Payload team, 4 engineers focus on the final assembly and ensuring scientific instruments meet performance standards for space operations. They collaborate closely with 2 technicians who facilitate testing. Furthermore, 5 scientists bring specialized expertise crucial for ensuring scientific objectives are met.

The Mechanical team will consist of 4 engineers who oversee the assembly and integration of the spacecraft's mechanical systems and structures, ensuring robustness and reliability in harsh space environments. They are supported by 4 technicians in assembly tasks and conducting thorough testing. Additionally, 2 scientists provide technical oversight.

In the Thermal subsystem, 2 engineers finalize the integration and testing of thermal control systems. They collaborate closely with 2 technicians who oversee the installation and verification of thermal hardware. Additionally, 1 scientist specializing in thermal management strategies ensures optimal thermal performance across all spacecraft components.

For the Power subsystem, 2 engineers oversee the integration and testing of power systems critical for sustaining spacecraft operations alongside 2 technicians who handle the installation and testing of power hardware components.

This means that a total of 42 people are needed for phase D: 18 engineers, 14 technicians, and 10 scientists.

Phase E

The CDH team transitions to an operational phase with 2 engineers who oversee real-time data management and system operations. They are supported by 1 technician who handles routine maintenance and troubleshooting of CDH hardware. Additionally, 2 scientists continue to validate data integrity and contribute to ongoing data analysis.

In the Science/Payload team, 3 engineers focus on optimizing payload operations and ensuring scientific instrument functionality assisted by 1 technician who supports instrument calibration and maintenance. Furthermore, 6 scientists derive scientific insights from the collected data and guide research efforts.

The Mechanical team includes 3 engineers who oversee spacecraft systems' mechanical operations and maintenance. 1 technician is required to ensure continuous functionality. Additionally, 2 scientists provide technical oversight to maintain mechanical systems' performance and reliability in lunar conditions.

Within the Thermal team, 2 engineers manage thermal control system operations and monitor temperature regulation on board the spacecraft. They are supported by 1 scientist who specializes in thermal management strategies.

For the Power subsystem, 2 engineers oversee power system operations and monitor energy consumption and distribution. They are assisted by 1 technician who handles routine checks and maintenance of power systems.

Thus 27 people are needed for phase E: 12 engineers, 4 technicians, and 11 scientists.

Phase F

The CDH team in Phase F consists of 1 engineer who oversees final data management tasks and is supported by 1 technician who assists in data retrieval and system shutdown procedures. Additionally, 1 scientist provides final validation of mission data.

For the Science/Payload team, 1 engineer alongside 1 technician focuses on finalizing payload operations and preparing instruments for shutdown or decommissioning. Furthermore, 1 scientist contributes to the final analysis and documentation of mission data and scientific outcomes.

In the Mechanical team, 1 engineer, assisted by 1 technician, oversees the final mechanical operations and ensures systems are prepared for decommissioning. Additionally, 1 scientist provides technical oversight for final mechanical assessments and documentation.

The Thermal subsystem includes 1 engineer who manages final thermal control system operations and ensures systems are safely shut down. They work with 1 scientist who verifies final thermal performance data and contributes to closing reports.

For the Power subsystem, 1 engineer, supported by 1 technician, oversees final power system operations and ensures systems are safely powered down. Additionally, 1 scientist provides assistance in power system performance documentation.

This means 14 people are required for phase F: 5 engineers, 4 technicians, and 5 scientists.

Phase C-F

The staff per phase trends down throughout the project. This is due to building over the rover requiring the most human resources.

Table 23: Subteam Work Overview by Phase

Subteam	Phase C	Phase D	Phase E	Phase F
CDH	6 Engineers 4 Technicians 2 Scientists	6 Engineers 4 Technicians 2 Scientists	2 Engineers 1 Technician 2 Scientists	1 Engineer 1 Technician 1 Scientist
Science / Payload	4 Engineers 2 Technicians 5 Scientists	4 Engineers 2 Technicians 5 Scientists	3 Engineers 1 Technician 6 Scientists	1 Engineer 1 Technician 1 Scientist
Mechanical	4 Engineers 4 Technicians 2 Scientists	4 Engineers 4 Technicians 2 Scientists	3 Engineers 1 Technician 2 Scientists	1 Engineer 1 Technician 1 Scientist
Thermal	4 Engineers 3 Technicians 1 Scientist	2 Engineers 2 Technicians 1 Scientist	2 Engineers 1 Scientist	1 Engineer 1 Scientist
Power	2 Engineers 2 Technicians	2 Engineers 2 Technicians	2 Engineers 1 Technician	1 Engineer 1 Technician 1 Scientist
Total	20 Engineers 15 Technicians 10 Scientists 45 total	18 Engineers 14 Technicians 10 Scientists 42 total	12 Engineers 4 Technicians 11 Scientists 27 total	5 Engineers 4 Technicians 5 Scientists 14 total

5.2. Mission Schedule

5.2.1. Schedule Basis of Estimate

The current mission schedule plans a launch date for September 1st, 2028 and an arrival date of September 5th, 2028, which exactly meets the L'Space mission constraint. However, it is likely that the team will be ready for launch well before the current given estimate, and further analogous comparison with similar lunar missions is needed for more accurate progress projection. Phases A and B are excluded from the timeline, and Phase C is assumed to directly start after the conclusion of the L'Space Program. Phases A and B have been excluded from the timeline. Phases C and D utilize data from the Psyche timeline, chosen for its similarities to the Ice Hunter mission in terms of program structure and scope. Phase E utilizes data from SpaceX and estimated timelines for the proposed rover at the Haworth landing site.

5.2.2. Mission Schedule

The major milestones for the mission are based around the following timeline:

Table 24. Table of Milestones

Milestone	Date
Full-scale assembly of rover instruments and subsystems complete	5/9/25
Completion of Initial Component Orders	7/16/25
Completion of Fabrication for Test	12/18/25
Systems Integration Review (SIR)	8/28/26
Critical Design Review (CDR)	9/30/26
Subsystem Testing Complete	6/11/27
Integrated Testing Complete	9/17/27
Launch Readiness Review (LRR)	12/17/27
Launch Date	9/1/28

1	Final Design Post PDR		0%	10/1/24	5/9/25	221	22	
1.1	Detailed design of Mechanical System	Mechanical	Not complete	10/1/24	2/7/25	130	13	
1.2	Detailed design of Electrical System	Electrical	Not complete	10/1/24	2/7/25	130	13	
1.3	Detailed design of Thermal System	Thermal	Not complete	10/1/24	2/7/25	130	13	
1.4	Detailed design of Computer/Hardware	Computer	Not complete	10/1/24	2/7/25	130	13	
1.5	Detailed design of Science Instrument Integrati	Science/Mechanical	Not complete	10/1/24	2/7/25	130	13	
1.6	Simulation Testing	All	Not complete	2/7/25	3/28/25	50	5	
1.7	Completed CAD of all components	All	Not complete	3/28/25	4/16/25	20	2	
1.8	Schedule Margin			4/17/25	5/8/25	22	3	
1.9	◆Full-scale CAD assembly of rover subsystems		Not complete	5/9/25	5/9/25	1		
2	Ordering and Procurement		0%	5/12/25	7/16/25	66	14	
2.1	Order Mechanical System Parts	Mechanical	Not complete	5/12/25	7/1/25	51	6	
2.2	Order Electrical System Parts	Electrical	Not complete	5/12/25	7/1/25	51	6	
2.3	Order Thermal System Parts	Thermal	Not complete	5/12/25	6/17/25	37	4	
2.4	Order Computer Hardware System Parts	Computer	Not complete	5/12/25	7/1/25	51	6	
2.5	Order NSS and NIRVSS	Science	Not complete	5/12/25	7/1/25	51	6	
2.6	Verify Test Facility Availability	Programmatics	Not complete	6/9/25	7/1/25	23	3	
2.7	Schedule Margin			7/2/25	7/15/25	14	2	
2.8	◆ Completion of Initial Component Orders		Not complete	7/16/25	7/16/25	1	1	
3	Manufacturing and Fabrication		0%	7/17/25	12/19/25	156	14	
3.1	Fabricating of all Subsystems	All	Not complete	7/17/25	11/14/25	121	13	
3.2	Fabricating electrical testbed	Electrical	Not complete	7/17/25	11/14/25	121	13	
3.3	Preliminary tests for integration	All	Not complete	11/14/25	12/5/25	22	3	
3.4	Schedule Margin			12/5/25	12/18/25	14	2	
3.5	◆ Completion of Fabrication for Test		Not complete	12/18/25	12/19/25	2	1	
4	Integration and Assembly		0%	12/22/25	8/28/26	250	22	
4.1	Assembly of each Subsystem	All	Not complete	12/22/25	6/30/26	191	20	
4.2	Integrating subsystems into test rover	All	Not complete	6/30/26	8/5/26	37	4	
4.3	Schedule Margin			8/6/26	8/27/26	22	3	
4.4	◆ Systems Integration Review (SIR)		Not complete	8/28/26	8/28/26	1	1	
4.5	◆ Critical Design Review (CDR)			9/30/26	9/30/26	1	1	
5	Subsystem Testing		0%	10/1/26	6/11/27	254	28	
5.1	Electrical Tests on Testbed	Electrical	Not complete	10/1/26	5/14/27	226	23	
5.2	Static and Dynamic Testing	Mechanical	Not complete	10/1/26	5/14/27	226	23	
5.3	TVAC testing of internal components	Thermal/Mechanical	Not complete	10/1/26	5/14/27	226	23	
5.4	Drill Tests	Mechanical	Not complete	10/1/26	5/14/27	226	23	
5.5	Communication Tests	Computer	Not complete	10/1/26	5/14/27	226	23	
5.6	Power cycle testing	Power	Not complete	10/1/26	5/14/27	226	23	
5.7	Mechanical wheel test in Moon Yard	Mechanical	Not complete	1/11/27	5/14/27	124	13	
5.8	Instrument Calibration Tests	Science/Mechanical	Not complete	10/1/26	5/14/27	226	23	
5.9	Schedule Margin			5/14/27	6/10/27	28	3	
5.1	◆ Subsystem Testing Complete		Not complete	6/11/27	6/11/27	1	1	
6	System-level testing		0%	6/11/27	9/17/27	99	14	
6.1	TVAC testing for complete rover	Thermal	Not complete	6/11/27	7/2/27	22	3	
6.2	Vibration Tests for complete rover	Mechanical	Not complete	7/2/27	7/16/27	15	2	
6.3	Simulated Emergency testing	All	Not complete	7/16/27	7/30/27	15	2	
6.4	Mock Mission Testing	All	Not complete	7/30/27	9/3/27	36	4	
6.5	Schedule Margin			9/3/27	9/16/27	14	2	
6.6	◆ Integrated Testing Complete		Not complete	9/17/27	9/17/27	1	1	
7	Launch Preparation		0%		9/20/27	12/17/27	89	15
7.1	Create final summary of risk analysis	Programmatics	Not complete		9/20/27	10/4/27	15	2
7.2	Verify mission operational status	Systems	Not complete		9/20/27	10/25/27	36	4
7.3	Prepare Launch Readiness Review	All	Not complete		9/20/27	12/3/27	75	8
7.4	Schedule Margin				12/3/27	12/17/27	15	2
7.5	◆ Launch Readiness Review (LRR)		Not complete		12/17/27	12/17/27	1	1
7.6	◆ Launch Date		Not complete		9/1/28	9/1/28	1	1

Figure 26. Gantt Chart

5.3. Budget

5.3.1. Budget Basis of Estimate

1. Complexity:

- a. The total mission is assumed to be of high complexity
 - i. It involves navigating the Moon's permanently shadowed regions, which lack heat and sunlight, bringing significant uncertainties to the mission environment
 - ii. Analogous missions (such as Curiosity) required significant margins to manage unexpected technical challenges or delays

2. Margins:

- a. The L'Space Project Management Handbook (which advises a 5% to 15% range for all budget margins) is assumed to be a reliable source for margin percentages. All budget estimates are within this range.
- b. The specific margin percentage for each portion of the budget is evaluated based on relative complexity, which each portion's section will elaborate on.

3. Travel rates:

- a. For lodging, meals, and incidental expenses, the General Services Administration-provided 2023 rates will be used. A list of each assumption is provided below:
 - i. Hotel (Lodging): \$182 per night
 - ii. Meals and Incidental Expenses (Per Diem): \$64/day
- b. For car rentals and flights, an AI-generated 2024 average cost is used:
 - i. Rental Car Estimate: \$60/day
 1. It is assumed that only 3-4 people can fit in one rental car (a very conservative estimate)
 - ii. Flight Estimate: \$500 per round-trip

4. Manufacturing and testing facility costs:

- a. It is assumed that all manufacturing and testing facilities will be hosted by JPL and all rates will be standard JPL rates
 - i. Manufacturing facility rates: \$200/hr
 - ii. Test facility rates: \$300/hr

5. Outreach

- a. It is assumed that Phase D will require the largest outreach budget since the public interest will likely be highest just before launch, and that Phase F will require the smallest budget since public interest will be the lowest after operations are concluded

- b. Outreach activities were assumed to be conducted primarily through digital platforms and local events to minimize costs, with historical outreach program costs and the increasing trend towards virtual engagement informing these estimates

6. Inflation

- a. The team assumed that the rate of inflation would remain constant throughout the course of the mission, based on historical inflation data and current economic forecasts, estimating a stable inflation rate of 2.6 percent per year
- b. Personnel costs were projected to increase at this rate, based on industry standards for aerospace and scientific research fields with the hope of ensuring competitive compensation would attract and retain top talent for the mission

7. Direct costs

- a. Direct costs associated with materials, equipment, testing, and facilities were assumed to remain stable with minor adjustments over the course of the development periods based on market analysis and publicly available information on supplier costs
- b. Facilities and Administrative (F&A) costs were projected to rise over the first half of the project and then fall off as project completion passes the halfway point
- c. Unless otherwise specified, the MCCET's Cost Estimation Relation (CER) equations were used to provide information on all direct costs
- d. For Phase D, the subteam testing is assumed to cost 30% of the total sub-instrument cost, as per the general MCCET guideline.

5.3.2. Total Mission Cost

Mission Phase	Phase C	Phase C	Phase C-D	Phase D	Phase E	Phase F	
Year	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Cumulative Total
FINAL COST CALCULATIONS							
Total F&A	\$ 1,850,000	\$ 2,643,502	\$ 1,434,747	\$ 1,881,839	\$ -	\$ -	\$ 7,810,088
Total Projected Cost	\$ 33,161,926	\$ 52,970,986	\$ 27,925,710	\$ 39,497,979	\$ 5,423,152	\$ 3,153,895	\$ 162,133,649
Total Cost Margin	\$ 2,707,472	\$ 4,316,644	\$ 2,141,690	\$ 3,005,913	\$ 330,177	\$ 181,969	\$ 12,683,865
	8.2%	8.1%	7.7%	7.6%	6.1%	5.8%	
Total Project Cost	\$ 33,161,926	\$ 52,970,986	\$ 27,925,710	\$ 39,497,979	\$ 5,423,152	\$ 3,153,895	\$ 162,133,649

Figure 27: Final Cost Calculations

Following the provided mission budget constraint of \$175,000,000, the Ice Hunters Budget mission has a project cost of \$162,133,649, which includes a total cost margin of \$12,686,162 (around 7.8%). The budget breaks down into the following categories: total personnel cost is projected to be \$30,205,108, total travel costs are \$816,534, total

outreach costs are \$1,683,186, and total direct costs are \$121,618,733. In viewing this data, most of the cost lies in the direct costs, with the personnel budget accounting for a lot of the rest of the budget.

5.3.3. Personnel Budget

Mission Phase	Phase C	Phase C	Phase C-D	Phase D	Phase E	Phase F	
Year	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Cumulative Total
PERSONNEL							
Science Personnel	\$ 800,000	\$ 820,800	\$ 841,600	\$ 862,400	\$ 971,520	\$ 452,000	\$ 4,748,320
Engineering Personnel	\$ 1,520,000	\$ 1,559,520	\$ 1,514,880	\$ 1,552,320	\$ 1,059,840	\$ 452,000	\$ 7,658,560
Technicians	\$ 900,000	\$ 923,400	\$ 883,680	\$ 905,520	\$ 132,480	\$ 135,600	\$ 3,880,680
Administration Personnel	\$ 120,000	\$ 123,120	\$ 126,240	\$ 129,360	\$ 132,480	\$ 135,600	\$ 766,800
Project Management	\$ 600,000	\$ 615,600	\$ 631,200	\$ 646,800	\$ 662,400	\$ 678,000	\$ 3,834,000
Total Salaries	\$ 3,940,000	\$ 4,042,440	\$ 3,997,600	\$ 4,096,400	\$ 2,958,720	\$ 1,853,200	\$ 20,888,360
Total ERE	\$ 1,099,654	\$ 1,128,245	\$ 1,115,730	\$ 1,143,305	\$ 825,779	\$ 517,228	\$ 5,829,941
Personnel Margin	\$ 352,776	\$ 361,948	\$ 357,933	\$ 366,779	\$ 264,915	\$ 165,930	\$ 1,870,281
TOTAL PERSONNEL	\$ 5,392,430	\$ 5,676,481	\$ 5,755,769	\$ 6,043,790	\$ 4,470,553	\$ 2,866,085	\$ 30,205,108

	Phase C	Phase C	Phase C-D	Phase D	Phase E	Phase F
# People on Team	FY 1	FY 2	FY 3	FY 4	FY 5	FY 6
Science Personnel:	10	10	10	10	11	5
Engineering Personnel:	19	19	18	18	12	5
Technicians:	15	15	14	14	2	2
Administration Personnel:	2	2	2	2	2	2
Management Personnel:	5	5	5	5	5	5

Figure 28: Personnel Budget

In each phase, there will be two administrative members, one serving as the team's project manager, and the other as the deputy manager of resources. Furthermore, there will be a manager for each subsystem, totalling 5. The total administration and management personnel (7) have been included in the total number of team members for each phase.

Phase C calls for 51 total team members. First, the team requires 19 engineers, as most of the design and simulation will occur in this phase. 6 engineers will be part of the CDH team since it requires the most precision, and the other subsystems will have 2-4 engineers on the team. Similarly, the team requires 15 technicians (of a similar distribution between subteams) since in-house manufacturing and assembly will occur during this period. 10 scientists, mostly part of the payload team, will advise this process and provide system requirements.

Phase D calls for 49 total team members. As the thermal subsystem is the least technically demanding, it will require one less engineer and technician than Phase C, and the rest of the team will remain, as they will be required to perform initial and final testing on their subsystem and complete system, and provide analysis and documentation on their results before launch in September 2028.

In the operational Phase E, 34 team members will remain. An additional scientist will be hired to aid the payload team in performing analysis of the data received by the

payload. 12 engineers will remain in order to advise the scientists, and a minimal amount of technicians will be on call in case the team faces any obstacles.

Lastly, Phase F requires only 19 team members as the project concludes. 5 scientists and engineers each will oversee the off-boarding process, while two technicians (one for the CDH team and one for the payload team) remain, as their subsystems are the most technically advanced and might require their knowledge.

The total margin for the personnel section is 7%. Each subteam has a liberal estimate for the number of team members needed: Phases C and D predict that over 45 team members are required, which is in the higher end of the suggested 25-50 team member range. Thus, the risk that more team members will need to be hired is quite low.

5.3.4. Travel Budget

TRAVEL							
Total Flights Cost	\$ 10,000	\$ 10,000	\$ 21,000	\$ 40,000	\$ 75,000	\$ 16,000	\$ 172,000
Total Hotel Cost	\$ 7,280	\$ 7,280	\$ 23,000	\$ 72,800	\$ 191,000	\$ 29,120	\$ 330,480
Total Transportation Cost	\$ 960	\$ 960	\$ 3,240	\$ 12,000	\$ 29,400	\$ 3,300	\$ 49,860
Total Per Diem Cost	\$ 2,560	\$ 2,560	\$ 8,064	\$ 25,600	\$ 67,200	\$ 10,240	\$ 116,224
Travel Margin	\$ 2,496	\$ 2,496	\$ 6,636	\$ 18,048	\$ 43,512	\$ 7,039	\$ 80,228
Total Travel Costs	\$ 23,296	\$ 23,902	\$ 65,161	\$ 181,587	\$ 448,348	\$ 74,240	\$ 816,534

Figure 29: Travel Budget

FY1 and FY2 require 4 trips each for design reviews and initial subteam meetings. These trips are small, creative sessions which will only require 5 team members each (one per subsystem). The cost breakdown is provided below:

- *Flights: 5 team members x \$500 per flight x 8 trips = \$20,000*
- *Hotels: 5 team members x \$182/night x 2 nights x 8 trips = \$14,560*
- *Transportation: 2 rental cars x \$60/day x 2 days x 8 trips = \$1,920*
- *Per Diem: 5 team members x \$64/day x 2 days x 8 trips = \$5,120*

FY3 requires 6 trips for component testing and fabrication visits to contractor facilities and JPL manufacturing and test facilities. These trips are slightly larger visits with management and a few subteam members (7 total members). The cost breakdown is provided below:

- *Flights: 7 team members x \$500 per flight x 6 trips = \$21,000*
- *Hotels: 7 team members x \$182/night x 3 nights x 6 trips = \$22,932*
- *Transportation: 3 rental cars x \$60/day x 3 days x 6 trips = \$3,240*
- *Per Diem: 7 team members x \$64/day x 3 days x 6 trips = \$8,064*

FY4 requires 8 trips, for peak fabrication and integration, to the team's manufacturing and testing facilities. These trips are large, comprehensive visits with most of the subteam included (10 total members). The cost breakdown is provided below:

- *Flights: 10 team members x \$500 per flight x 8 trips = \$40,000*
- *Hotels: 10 team members x \$182/night x 5 nights x 8 trips = \$72,800*

- *Transportation: 5 rental cars x \$60/day x 5 days x 8 trips = \$12,000*
- *Per Diem: 10 team members x \$64/day x 5 days x 8 trips = \$25,600*

FY5 requires 1 trip to Cape Canaveral for launch. All team members in Phase E (32 people) will travel to conduct pre-launch procedures and testing, and witness the eventual launch. The cost breakdown is provided below:

- *Flights: 32 team members x \$500 per flight x 1 trip = \$16,000*
- *Hotels: 32 team members x \$182/night x 5 nights x 1 trip = \$29,120*
- *Transportation: 11 rental cars x \$60/day x 5 days x 1 trip = \$3,300*
- *Per Diem: 32 team members x \$64/day x 5 days x 1 trip = \$10,240*

FY6 will require 5 trips for post-launch operations across NASA centers and U.S universities, for 10 team members each. These trips are open-ended to account for the possibility of any previously unaccounted necessity to travel:

- *Flights: 10 team members x \$500 per flight x 5 trips = \$24,000*
- *Hotels: 10 team members x \$182/night x 4 nights x 5 trips = \$36,400*
- *Transportation: 3 rental cars x \$60/day x 4 days x 5 trips = \$3,600*
- *Per Diem: 10 team members x \$64/day x 4 days x 5 trips = \$12,800*

As travel can be unpredictable and there likely will be additional necessity for travel (such as additional visits to contractor sites or testing facilities), a margin of 12% is provided within the travel budget.

5.3.5. Outreach Budget

OUTREACH							
Total Outreach Materials	\$ 15,000	\$ 15,000	\$ 30,000	\$ 100,000	\$ 75,000	\$ 50,000	\$ 285,000
Total Outreach Venue Costs	\$ 3,000	\$ 3,000	\$ 30,000	\$ 150,000	\$ 100,000	\$ 40,000	\$ 326,000
Total Outreach Travel Costs	\$ 3,000	\$ 3,000	\$ 30,000	\$ 150,000	\$ 100,000	\$ 40,000	\$ 326,000
Total Outreach Services Costs	\$ 8,000	\$ 8,000	\$ 50,000	\$ 100,000	\$ 100,000	\$ 25,000	\$ 291,000
Total Outreach Personnel Costs	\$ 15,000	\$ 15,000	\$ 60,000	\$ 75,000	\$ 60,000	\$ 25,000	\$ 250,000
Outreach Margin	\$ 2,200	\$ 2,200	\$ 10,000	\$ 28,750	\$ 21,750	\$ 9,000	\$ 73,900
Total Outreach Costs	\$ 46,200	\$ 47,401	\$ 220,920	\$ 650,843	\$ 504,252	\$ 213,570	\$ 1,683,186

Figure 30: Outreach Budget

For the allocation of the outreach budget, a percentage breakdown for each phase is provided:

For FY1 and FY2, 10% of the budget is allocated, as activities are minimal, indicating low travel and venue costs. The focus is on initial planning and material creation for future years.

For FY3, 20% is allocated as outreach efforts for public engagement ramp up. For FY4, 30% is allocated since the period directly before a mission launch is typically when the public interest and engagement is highest. There is significant investment in materials, travel, and events.

For FY5, 20% is allocated as outreach programs continue but are scaled down.

For FY6, 10% is allocated, as the goal is education efforts on mission outcomes rather than active engagement.

As the allocated outreach budget is above 1%, a 5% outreach margin is provided (a relatively low value).

News Outreach

This includes costs associated with outreach associated with reaching out to traditional and nontraditional media outlets. This includes compensation for creative works, especially podcasts and blogs commissioned to report on events. This will be allocated a budget of \$25,000 as this is expected to take up a small section of the outreach efforts. There will also be \$5,000 of margin for this outreach budget category to ensure necessary expenditures are met.

Digital Media Outreach

This includes costs associated with social media, digital content creation, and creating lesson plans and citizen science projects. The total budget for this section will be allocated \$520,000.

Social media efforts will be allocated \$75,000 for the purpose of creating and distributing posts on social media platforms. This includes the creation of graphics and writing the body of the post as well as moderating the posts after publishing.

\$300,000 will be allocated for video production with \$100,000 of that being allocated to short form video production and \$200,000 being allocated for long form video production. Video production funds will go towards writing, creating sets, recording, editing, and publishing these videos.

\$50,000 will be allocated to creating materials to provide materials and resources to parents and teachers such as K-12 lessons with activities and experiments, coloring pages, and home experiments and activities

\$25,000 will be allocated to maintaining and moderating citizen science projects related to the mission. And \$70,000 will be allocated for margin in this category.

In person outreach

In person outreach includes expenses related to in person demonstrations as well as internship expenses. \$520,000 will be allocated to these expenses.

\$200,000 will be allocated towards in person outreach demonstrations such as creating models of the rover, posters, stickers, flyers, as well as any other needed materials for demonstration purposes and finally, \$250,000 will be allocated towards internships.

5.3.6. Direct Costs

DIRECT COSTS							
Mechanical Subsystem	\$ 5,000,000	\$ 9,000,000	\$ 6,000,000	\$ 6,000,000	\$ -	\$ -	\$ 26,000,000
Power Subsystem	\$ 500,000	\$ 700,000	\$ 824,000	\$ 607,200	\$ -	\$ -	\$ 2,631,200
Thermal Control Subsystem	\$ 1,000,000	\$ 1,000,000	\$ 747,200	\$ 824,160	\$ -	\$ -	\$ 3,571,360
Comms & Data Handling Subsystem	\$ 6,000,000	\$ 9,000,000	\$ 3,000,000	\$ 5,400,000	\$ -	\$ -	\$ 23,400,000
Science Instrumentation	\$ 6,000,000	\$ 6,000,000	\$ 3,000,000	\$ 4,500,000	\$ -	\$ -	\$ 19,500,000
Spacecraft Cost Margin	\$ 1,850,000	\$ 2,570,000	\$ 1,357,120	\$ 1,733,136	\$ -	\$ -	\$ 7,510,256
Total Spacecraft Direct Costs	\$ 20,350,000	\$ 29,005,020	\$ 15,704,593	\$ 20,551,527	\$ -	\$ -	\$ 85,611,139
Manufacturing Facility Cost	\$ 5,000,000	\$ 13,800,000	\$ 2,600,000	\$ -	\$ -	\$ -	\$ 21,400,000
Test Facility Cost	\$ -	\$ -	\$ 1,500,000	\$ 8,592,000	\$ -	\$ -	\$ 10,092,000
Facility Cost Margin	\$ 500,000	\$ 1,380,000	\$ 410,000	\$ 859,200	\$ -	\$ -	\$ 3,149,200
Total Facilities Costs	\$ 5,500,000	\$ 15,574,680	\$ 4,744,520	\$ 10,188,394	\$ -	\$ -	\$ 36,007,594

Figure 31: Direct Costs

For the below cost breakdown based on the subsystem, the provided costs (such as the estimated mechanical costs) might include testing costs. Thus,

Furthermore, most of the in-house manufacturing occurs in Phase C (mostly in FY1 and FY2), so the costs for manufacturing facilities are allocated accordingly. The reverse is true for testing facilities, which exclusively occurs in Phase D, and mostly later (FY4).

The direct costs associated with the mechanical subsystem is estimated to be a total of \$44,000,000 including all components (wheels, suspension, chassis, drill), materials, testing and personnel. This section will go over all direct costs and quantify each amount.

The chassis and suspension will be constructed in-house of aluminum 6061, a widely used material in different space applications. This material will provide all the necessary properties to perform effectively. The size of this component will be the largest of the three main components with dimensions 1 x .5 x .25 m. The cost of this material can be purchased in bulk sheets or solid blocks and tubing for machining at ~5 dollars per pound. Manufacturing costs and fabrication of the chassis and suspension are estimated to be approximately \$28,000,000. This includes manufacturing, facility costs, materials, and personnel services.

The wheels are the final component to be considered will also be constructed through a NASA facility, with each wheel being constructed from a single block of aluminum and the estimated cost for each wheel is estimated to be ~166,000 dollars each or ~1 million dollars total for all 6 wheels, this would also include manufacturing, facility costs, and materials.

The TRIDENT drill is the final component of the mechanical system and can be contracted through a third party company Honeybee Robotics, this product has a mass of 20 kg and draws 5 W of power, this product is estimated to cost \$ ~15,000,000 for manufacturing and testing. All of these costs were derived using the Mechanical CER formula from the MCCET.

Power

The direct costs of the power subsystem involve the parts of the system: the solar panel, battery pack, and combined solar charge converter / power supply, As well as the personnel required to assemble the system. In this section of the direct costs. the costs of each component of the subsystem and the cost to construct and assemble the subsystem.

Firstly, the largest direct cost of the system is the ZTJ solar panel which will cost approximately 1.5 million dollars. This was found by looking at previous missions that have used the ZTJ solar cell and also via indirect contact with the company. After the solar cell, the next most expensive item of the system is the battery pack; this is estimated at a total of 400000 dollars for 7 batteries as of now but may increase in cost and number because of expanding scope. After this the final piece of the power subsystem is the power supply; this component performs several roles such as the solar charge controller and converter from low voltage from the solar panels to high voltage that the dedicated portion of the PSU uses, however, it will likely cost the least with an estimated cost of 100000 dollars given by previous missions using the STARBUCK-MINI such as the Intuitive Machines 2024 lunar mission. The estimated total for this portion of the dedicated costs comes to around 2 million dollars and makes up the larger portion of the dedicated costs between components and personnel costs.

After calculating all of the costs, the cost of the power subsystem should come out to 2.6 million dollars for the components needed to manufacture the system.

Communications and Data Handling

The single-board computer, the BAE Systems RAD5545 will be supplied through a contractor. The estimated total cost is \$1,900,000.00. The estimated total manufacturing cost is \$1,400,000.00, and the estimated total testing facility cost is \$500,000.00.

The SpaceWire data bus's supplier is STAR-Dundee. The mass/cost for the item is variable depending on the length of cables used, thus the maximum length of 10 meters was picked in order to have the safest margin. The estimated total cost is \$2,200,000.00. The estimated total manufacturing cost is \$1,700,000.00, and the estimated total testing facility cost is \$500,000.00.

The antenna (transmitter+receiver), the BAE Systems Ka-band multi-orbit LEO optimized antenna, will be supplied through ViaSat. The estimated total cost will be \$8,300,000.00. The estimated total manufacturing cost is \$6,400,000.00, and the estimated total testing facility cost is \$1,900,000.00.

The 2GB of radiation-hardened flash storage will be supplied by Philson USA. The estimated total cost will be \$500,000.00. The estimated total manufacturing cost is \$400,000.00, and the estimated total testing facility cost is \$100,000.00.

The total subsystem cost is \$12,900,000.00, with \$9,900,000.00 coming from manufacturing and \$3,000,000.00 coming from testing.

All of these costs were derived using the CER formula for Electronics and Software from the MCCET.

Thermal

The heater, Minco Polyimide Thermofoil HK Series, will be supplied by their contractor. The estimated total cost is \$500,000. Including manufacturing and labor cost can be estimated up to \$300,000 and testing facility cost can be estimated \$200,000.

The radiator, Redwire Space: Q-Rad will be manufactured by the main supplier. The estimated total cost for the radiator is \$500,000. The estimated total manufacturing and labor cost will be \$350,000 and the total testing facility cost can range from \$100,000-250,000. This equation is taken into account with the four main factors of Total Cost = Material Cost + Labor Cost + Overhead + Testing and QA Cost.

The thermal switch Diaphragm Thin Plate will be manufactured and tested by Sierra Space. The estimated total cost for the thermal switch is \$100,000. The estimated manufacturing and labor cost will be \$80,000 and the testing cost will be \$40,000. This equation is taken into account with factors of the product material, labor and facility testing within Sierra Space.

The sprayable thermal paint, Socomore A276 Polyurethane will be purchased from Socomore. One unit of a bucket of paint will cost \$179.73. Labor and testing costs could be estimated to be up to \$3,000-5,000.

Science

The two science instruments that make up the direct costs will be contracted through the same company, Astrobotic. The total direct costs for the following instruments are approximately \$17,200,000.

The first instrument is the Neutron Spectrometer system (NSS), a COTS part to be developed through the selected contractor. This part has a mass of 1.6 kg and draws 1.5 W of power to operate. The total for this instrument is estimated at \$3,700,000. The costs associated with this part would also require testing which can be estimated to be up to 50K.

The second instrument is the Neutron Infrared Volatile Spectrometer system (NIRVSS). This part will also be contracted by Astrobotic and has a mass of 3.6 kg and draws 30 W of power. The total for this instrument is estimated at \$13,400,000. This would include the manufacturing and development through the mentioned contractor. Testing for this part can add up to an additional 50K. All of these costs were derived using the CER formula from the MCCET.

5.4. Scope Management

5.4.1. Change Control Management

Our team's process for change control is to, first and foremost, communicate any changes in a way that everyone can be aware of. These methods may include email, personal messages, and verbal communication through meetings. If there are any updates to a change, then these updates will be brought up as soon as possible. Any miscommunications must be avoided as they will hinder the team in the future.

Our plans for implementing customer-requested changes are as follows: After receiving an RFA or ADV, these changes will be tracked in a spreadsheet to track document changes. After an update, a document will gain a distinguisher marking, such as A, B, C, etc. These will be incremented through the alphabet when a change has occurred. These changes will be communicated throughout the team via the spreadsheet tracking different versions of the documents and through Discord communication channels.

For internal requested changes, the implementation plans are as follows: Feedback will always be encouraged from all team members. Any feedback necessitating filling out an RFA or ADV will be brought to the mission administration for review. Upon review, an

RFA or ADV will be filled out and these changes will be tracked in the spreadsheet mentioned above. The process will continue as described in the previous paragraph.

5.4.2. Scope Control Management

Several methods will be implemented if the team must downslope to correct overrun costs or schedules. First, the team will conduct subsystem evaluation meetings, similar to the previous process, to reduce the mission budget from \$225 million to \$175 million. These meetings will involve key stakeholders to assess the importance of specific components and alternative approaches.

Manufacturing and procurement methods will be reviewed to identify cost-saving opportunities while keeping quality high. This might include switching to more cost-effective suppliers, changing manufacturing methods, changing materials, or changing physical characteristics.

Non-essential activities such as travel, outreach, and logistics will be evaluated for budget reductions, with funds redirected to more critical mission components to ensure the mission's success. As a last resort, specific science instruments may be removed to prioritize instruments with the highest scientific value.

Conversely, if the team has an additional margin for budget, funds will be used to enhance existing capabilities and subsystem performance. Additional budget may be used to add new science instruments, expand outreach and public engagement activities, improve redundancy and reliability, and enhance data analysis and processing capabilities.

The approach to scope control will vary depending on the mission life cycle phase. In the early stages, thorough planning and design reviews will identify potential risks and cost-saving opportunities, while mid-mission phases will focus on monitoring progress and implementing corrective actions. In the late phases, priority will be given to testing and validation to ensure mission readiness and address any last-minute issues or changes.

5.5. Outreach Summary

The purpose of the outreach directive of the mission is to inform the taxpayer of the mission as well as the science obtained from the mission.

News Outreach Efforts

The first aspect of outreach to consider is to ensure positive publicity in traditional and nontraditional news organizations when approaching key mission points or discoveries

such as launch or critical science discoveries. These efforts will target mainly adults and have two categories of audiences which are general audiences and technically inclined audience members with interest in space exploration. Traditional news will target a mainstream, general audience. In this context, traditional media includes print, broadcast, radio, etc. Reaching out to national and local media distributors and journalists will ensure people of a range of backgrounds are aware of the mission.

Non-traditional media includes space themed news podcasts, blogs, etc. These outlets are geared towards people already interested in space news and information. By spreading information through these avenues, the team will engage with communities that are passionate about the mission who are likely to share technological successes as well as scientific discoveries with friends and family.

The success of these efforts can be measured by the exposure metrics recorded from these outlets. A goal for total exposure can be set at 100 million interactions.

Digital Media Outreach Efforts

In addition, including digital media efforts in outreach will be critical to interacting with and informing a broad audience. Managing social media pages on mainstream sites and posting about mission updates as well as science discoveries made in a timely fashion enhances the ability to communicate with the taxpayer and allows for engagement and involvement.

This effort would target social media users, which are aged 13 or older. These people may or may not have a technical or scientific background so communicating in an engaging and easy to understand manner will be crucial. Posts will include mission updates and will usually be accompanied by a graphic or video relating to the topic being discussed in the body of the post. Platforms utilized can include Facebook, X, Instagram. Others may be added to this list as necessitated or decided by the team. YouTube will be discussed further in this section. Making specific content for national holidays and special events will keep content engaging and relevant to the mission as well as the general public. Special efforts can be put into making images operable with screen readers to ensure accessibility. Efforts can also be made to create posts that are available in languages such as Spanish. Success on this platform can be measured by interactions and engagement. A quantitative goal on this front would be to aim for 1-2 million followers on all platforms in total.

It is important to utilize existing NASA social media presence in the outreach efforts as applicable, such as the official NASA page, NASA Center pages, and any other page deemed to be appropriate.

Video presence will be important for outreach purposes of the mission. Videos can be addressed to adults as well as K-12 students. There are two types of videos under consideration. The first, short form videos, in this context a video less than 10 minutes. The second is long form videos, in this context a video with a length that is greater than 10 minutes. All forms of videos can include closed captioning as well as audio description to increase audience breadth and reach. Languages such as Spanish can be included as an audio track and captions as well.



Figure 32. Example Social Media Post

Short form videos will be focused on engineering or science topics related to the mission. They will be designed to be approachable by children or anyone with rudimentary technical knowledge. The aim of these videos will be to increase general scientific literacy, while still focusing on aspects specifically related to the mission. These videos will make up a series of videos. These videos will be accessible on

YouTube, the NASA website, as well as on NASA+. There is also the possibility of creating short videos related to special events for the purpose of increasing visibility such as videos relating to launch or landing on the Moon.

Long form videos will be focused on the design and operation of the rover as well as scientific discoveries derived from received data. The aim of these videos will be to inform the general public of the work being done within the team as well as inspire the next generation of scientists and engineers. These videos will result in the creation of a documentary series available on YouTube, the NASA website, as well as NASA+. This series will be released at or some time after the rover arrives on the lunar surface.

For both video formats, success can be measured in views or downloads. A goal of 10 million interactions will be the target goal for this media.

Creating classroom aids for teachers as well as parents will be critical to reaching K-12 students. A curation of classroom lessons and activities related to the mission will be made available to teachers for the purpose of distributing and instructing students about science and engineering principles related to the mission. These lessons and activities will include powerpoint presentations as well as interactive crafts or experiments to perform. There will also be a science breakdown instructing teachers on the basic science principles of the phenomena described in the lesson, experiment, or craft provided. While each state may have a different science curriculum for students, keeping education standards in the creation of these resources will be imperative to adding as much value to teachers and students as possible.

At-home activities such as coloring books, science experiments, and crafts will make the mission accessible to parents and children. These resources will include instructions, explanations about the relevant science or engineering topics related to the mission, as well as further resources for inquisitive children or parents interested in learning more. The coloring pages will feature the rover and its various subsystems with brief descriptions of the system(s) depicted as well as highlighting the science objectives of the mission. The science experiments and crafts will demonstrate relevant science and engineering principles related to the rover and the science objectives of the mission. This will provide children and parents with interactive and inexpensive means of engaging with the mission and the science objectives.

For the previous two resources, success can be measured through page visits and downloads. Expected resource downloads will aim to hit at least 500 thousand downloads total.

Citizen science will be an interesting and engaging way of enabling volunteers, particularly highschool and college students, to interact directly with the mission. Presenting data from the rover in a way that ordinary people can contribute to the science will provide meaningful experiences for regular people to engage with research directly related to the mission. Zooniverse would be a great resource to host this citizen science opportunity. Success in this will be measured in number of volunteers and in the data processed. If the data released for processing is completed, that will equate to a success. Similarly, if volunteers equal 100 thousand or more, the project will be considered a success.

In Person Outreach Efforts

Having in person outreach efforts, specifically near key mission milestones, such as launch or landing on the lunar surface as these in person efforts will be most impactful adjacent to events such as launch or landing on the lunar surface. Large events will ideally feature small scale models of the rover, informational posters, a video presentation, small samples of lunar regolith for view, one or two interactive demonstrations, and items to distribute such as stickers or flyers. Small events can feature as little as informational posters, one or two interactive demonstrations, and items to distribute such as stickers or flyers.

Large events will be performed directly by the team and will take place near key mission milestones. Other demonstrations will be facilitated by the contracted partners as well as volunteers from the community. Large events will take place in public places such as NASA centers or planetariums. Smaller events can take place at STEM community nights for universities or schools as well as other events as deemed appropriate by the team.

The team will also offer internships and opportunities for college or highschool students to participate in the work being performed on the mission. This will engage a variety of different individuals and develop the incoming STEM workforce. Engaging in this way will provide hands-on opportunities for interested emerging workers to engage with their specific field of study as well as introduce the work environment in a technical field and specifically at NASA.

Success for in person outreach will be met if at least 100 hours of in person public outreach is carried out.

Partner Outreach

The team will utilize mission partners in outreach efforts. Ensuring that partners share this team’s outreach materials as well as enabling them to share resources related to their own contributions to the mission will increase the efficacy of the mission’s outreach efforts as well as reach individuals specific to regions where mission partners are located.

Table 25. Example Outreach schedule

Social Media Posting Schedule		
Platform	Frequency	Media Type
Facebook	Weekly	Photos, Infographics, Short Videos
Instagram	Weekly	Photos, Infographics, Short Videos
X	Weekly	Photos, Infographics, Short Videos
Youtube	Bi-Monthly	Short Videos, Long videos

6. Conclusion

The Ice Hunters team showed a plan to conduct essential surface measurements and mapping to advance understanding of lunar water distribution and abundance on the PSR of the lunar poles. This mission will travel to Haworth and use the NSS and NIRVSS instruments in order to determine the availability of hydrogen in Haworth's PSR. To accomplish this goal, each subteam has determined the necessary requirements and components in order to accomplish their portion of the rover. This mission schedule and its accompanying budget will fall within the restraints given, with total cost being \$162,133,649, which includes a total cost margin of \$12,686,162 (around 7.8%). A clear team structure has been laid out, with this structure defining conflict management and how changes will be defined.

If the team had more time, the team would focus on improving certain aspects of the PDR. Unfortunately, not all members contributed equally and with an already stretched thin team, the Project Manager made the decision to assign one person as the lead of each part. The reason this decision was made is adding more than one person is that certain individuals would have to do even more work to cover others.

The team plans to practice for the CDR, which will require finding a time everyone is available. This will allow for everyone to get comfortable with the content that will be presented. Furthermore, clear instructions for what everyone will contribute to the CDR will be given out by the Project Manager. Upon approval, the Ice Hunters will move into the next phase. This will allow the team to begin to build the rover. Furthermore, following the guidelines laid out in the budget category, the team will expand in order to accommodate the increase in labor.

Statement of Generative AI

Generative AI (ChatGPT, Meta Llama 3.1) was used in the CDH subsection as a tool to find trustworthy resources in which to obtain information regarding criteria such as component lead times, suppliers, and software implementation. It was also used in the budgeting section in order to provide cost estimates and averages in multiple areas.

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Appendix

Time when change was approved	RFA's	ADV's	CCB's
Post MDR	-	-	7kg change from 63 to 70 kg total craft weight from increasing the number of batteries in the Power System from 7 to 12

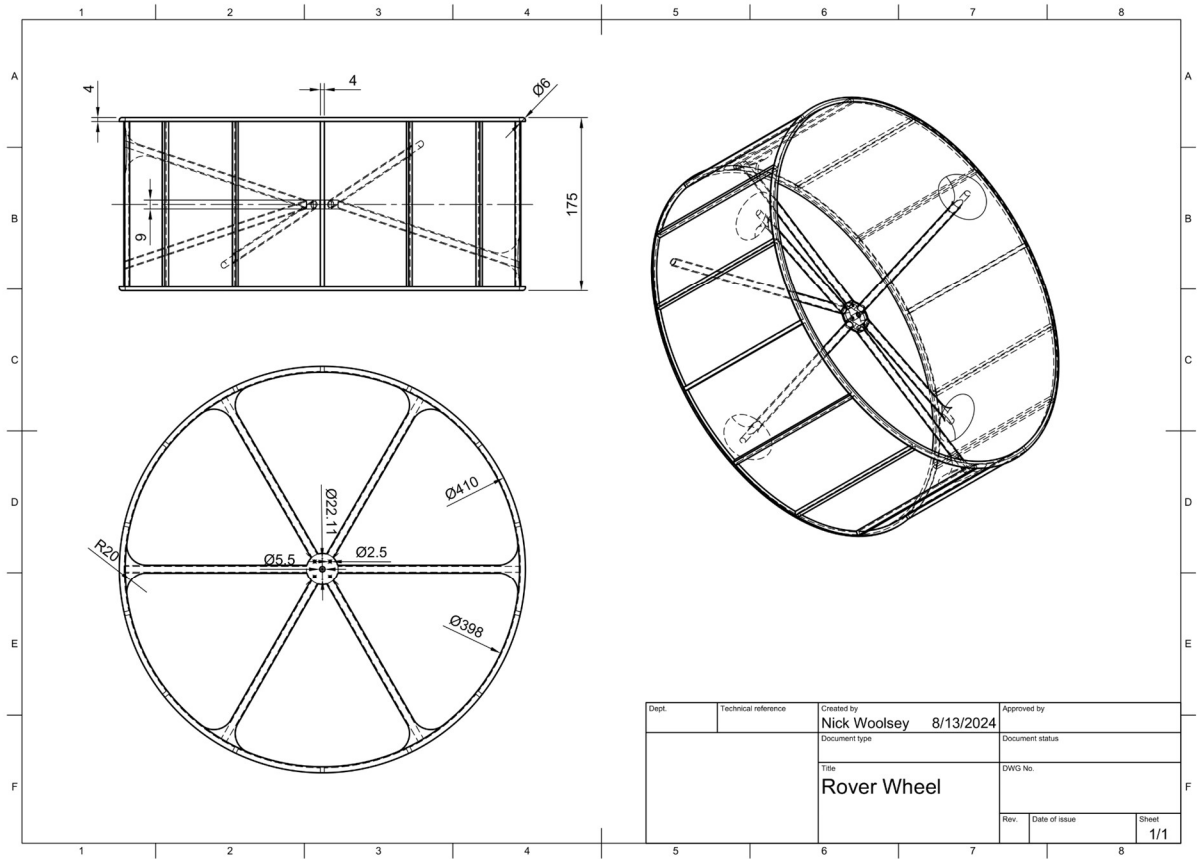


Figure 33: Rover Wheel

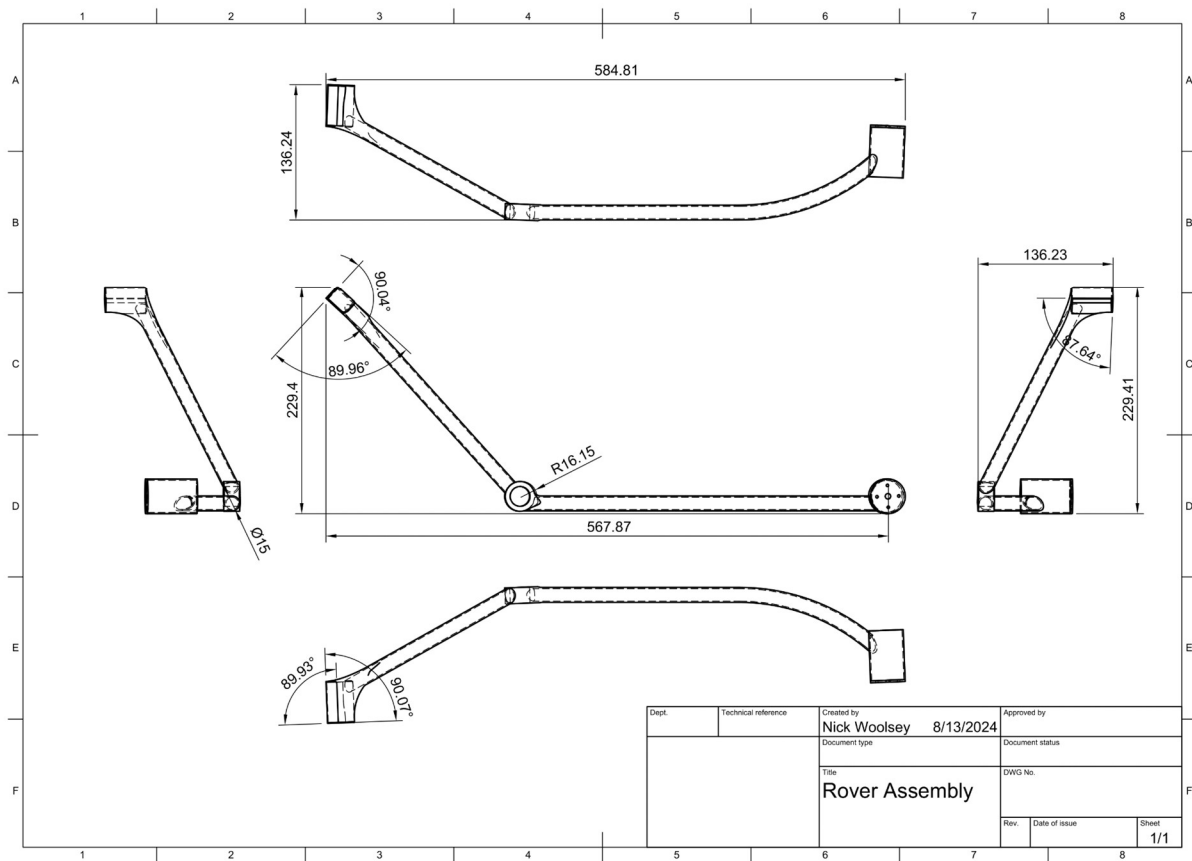


Figure 34: Rover Assembly 1

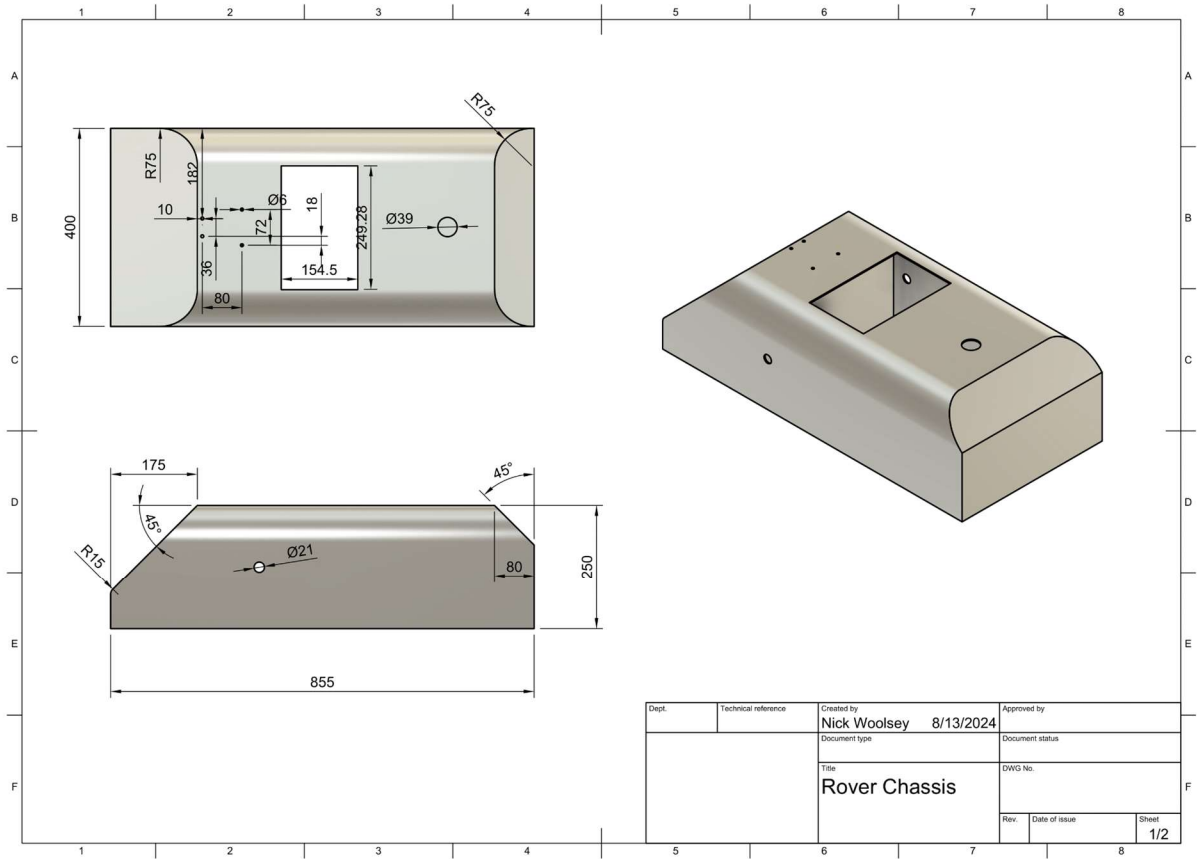


Figure 35: Rover Chassis

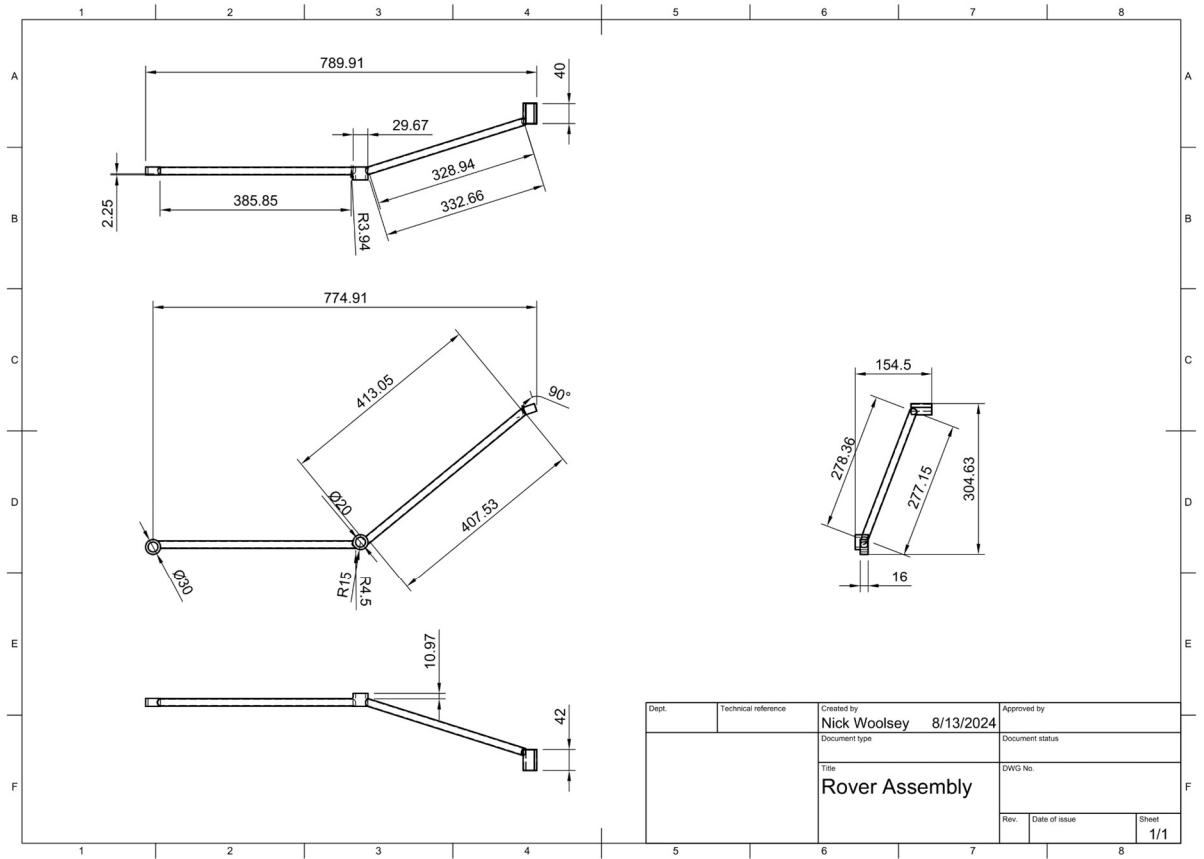


Figure 36: Rover Assembly 2

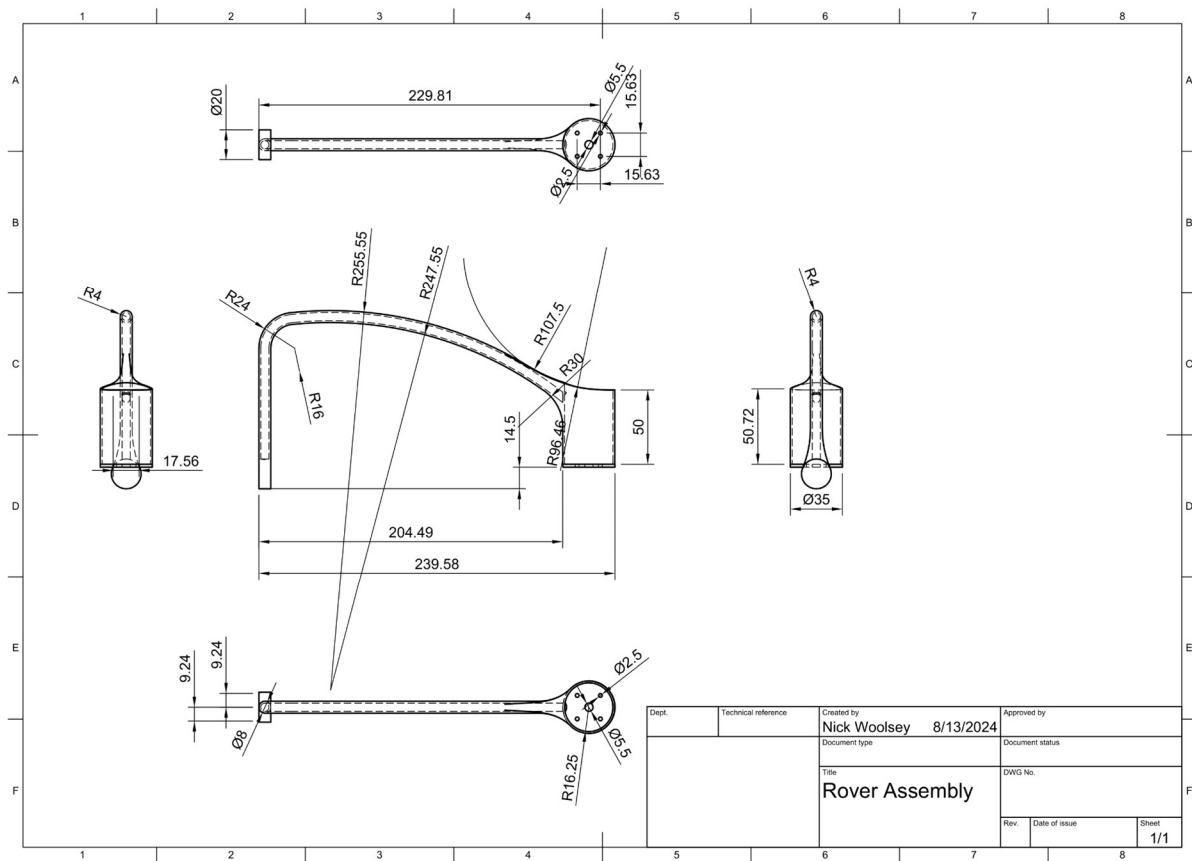


Figure 37: Rover Assembly 3